

Backup · Z4 full results gallery

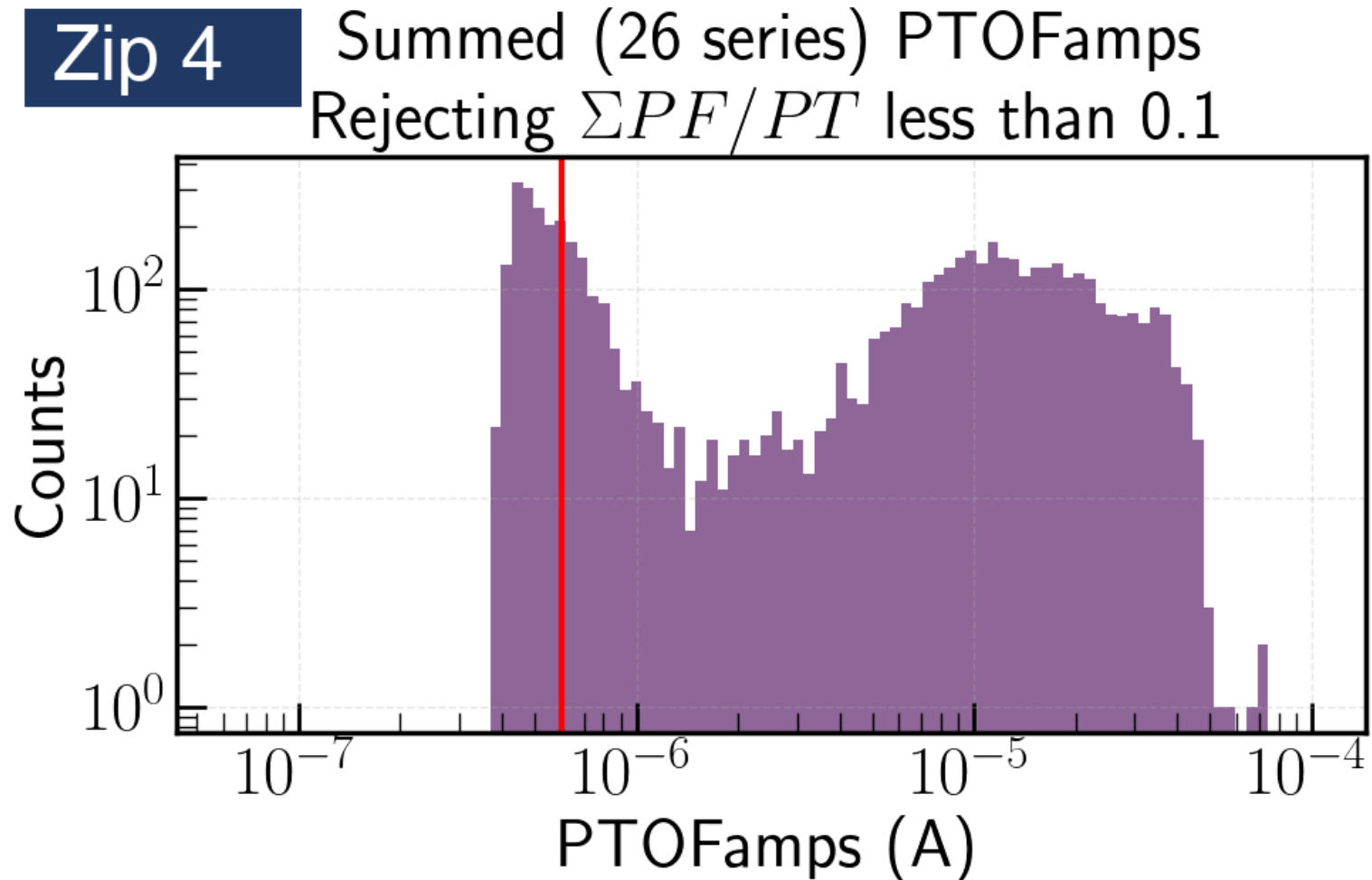
every diagnostic and result figure of the pipeline, in processing order

- Dataset: the K-line selection window
- Example grids: LP + fit and raw vs fit, event × channel
- Fit diagnostics: NRMSE, fitted pretrigger, τ_{rise} / τ_{fall} distributions
- Aligned fans: no cut $\rightarrow$ $\text{NRMSE} \leq 0.4 \rightarrow + \tau_{\text{rise}} \leq 0.3$ ms; kept vs rejected; measured traces
- Special populations: NRMSE-rejected events, well-fit slow-rise events
- Templates: PCA nxm0–4 per channel, summed PT / PS1 / PS2

Z4 · K-line selection window

figure: kline_zip4.png

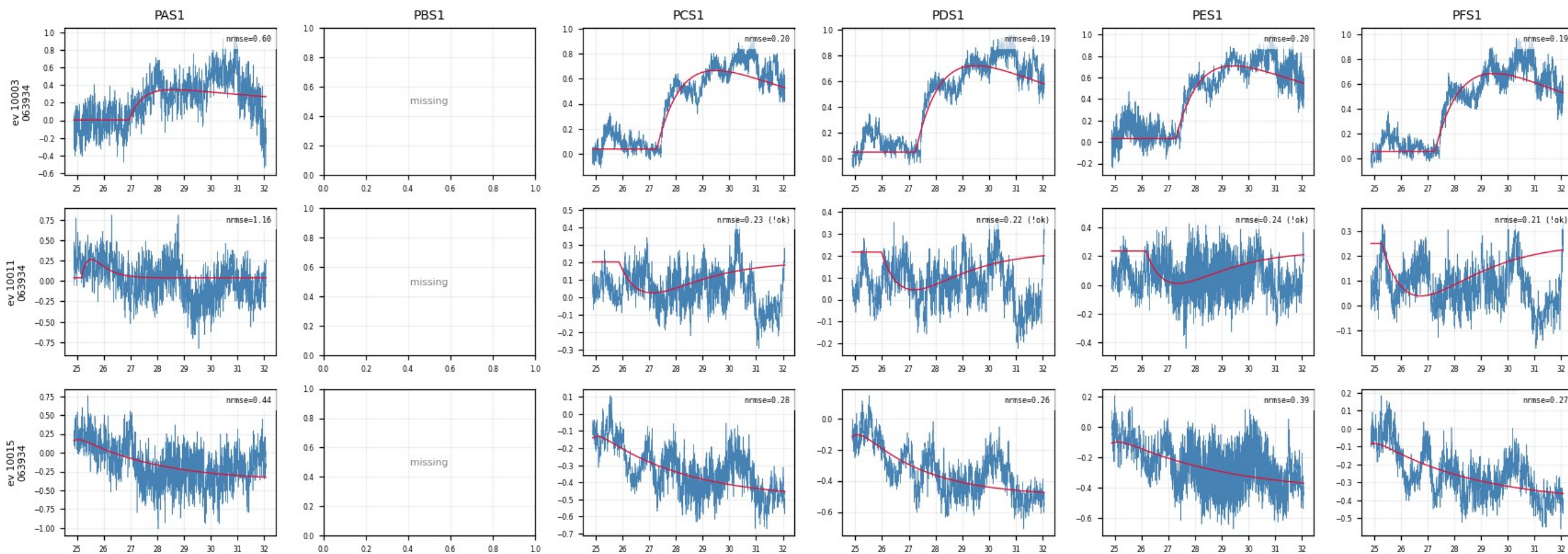
Summed PTOFamps spectrum of Z4; red line = Prof. Saab's 10.37 keV K-line position; the $\times/\div 1.35$ window around it is the event selection; everything inside is cached raw.



Z4 · fit_examples: LP trace vs 2-exp fit (1/10)

figure: zip4_fit_examples.png, event rows 1–3 of 15, left half of the channels

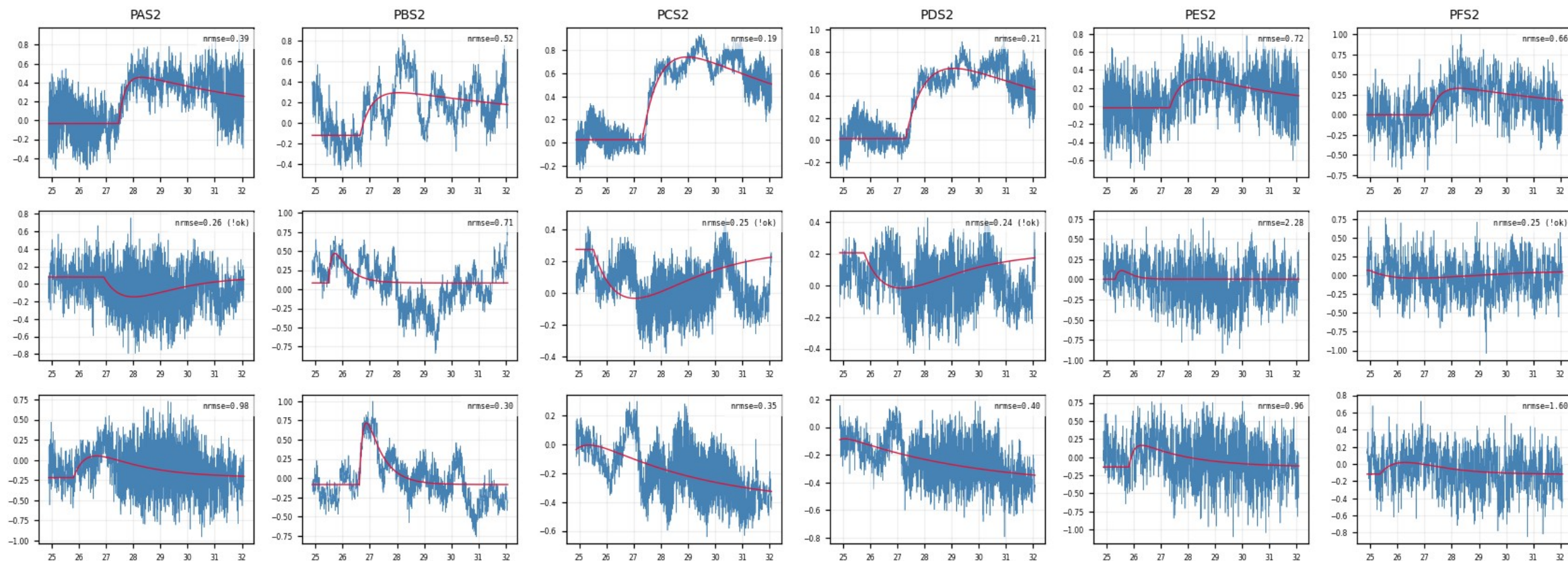
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



Z4 · fit_examples: LP trace vs 2-exp fit (2/10)

figure: zip4_fit_examples.png, event rows 1–3 of 15, right half of the channels

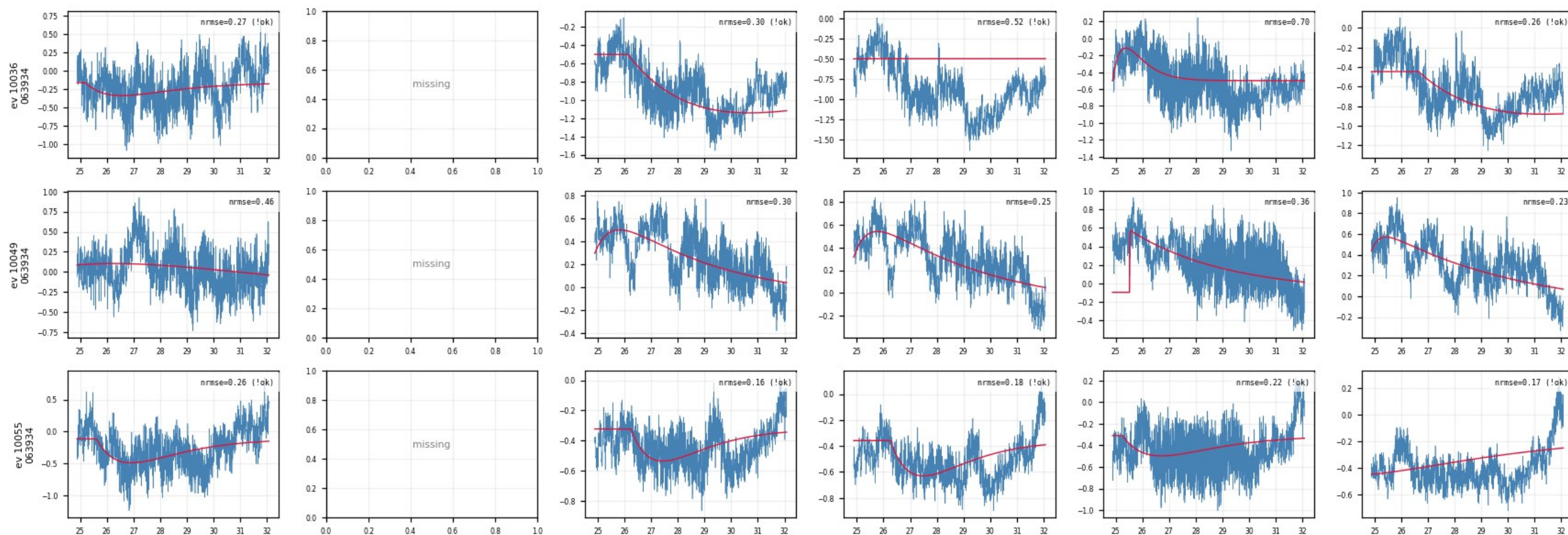
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



Z4 · fit_examples: LP trace vs 2-exp fit (3/10)

figure: zip4_fit_examples.png, event rows 4–6 of 15, left half of the channels

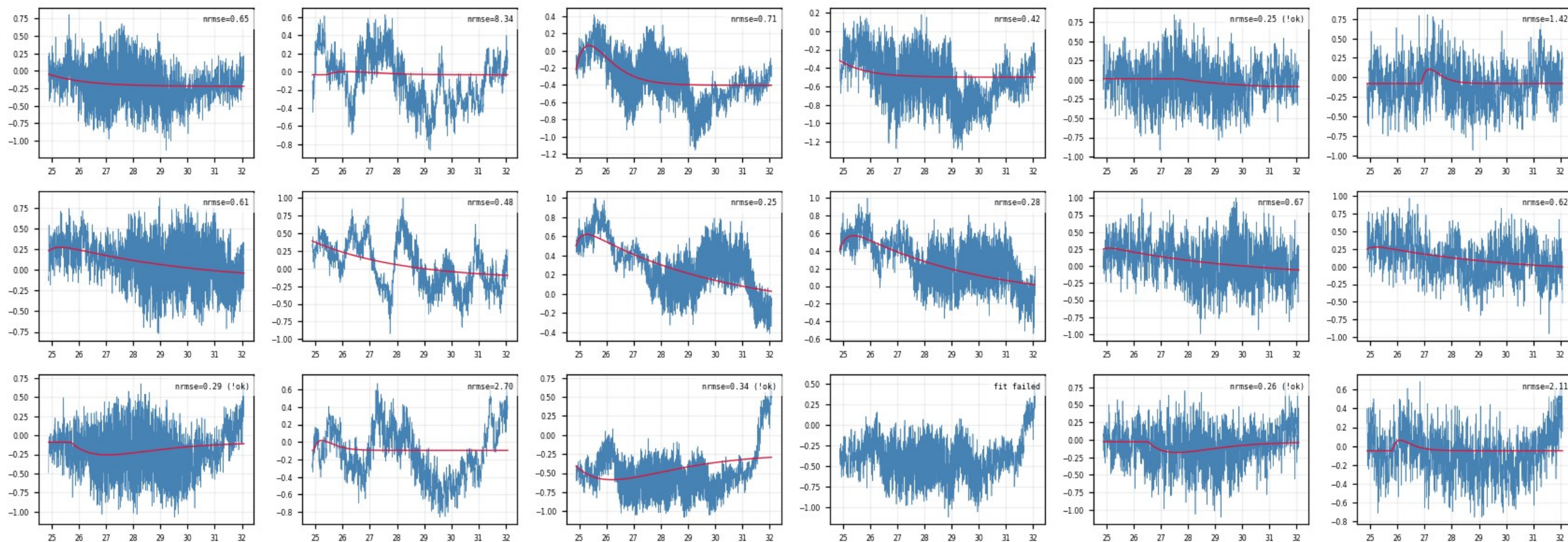
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



Z4 · fit_examples: LP trace vs 2-exp fit (4/10)

figure: zip4_fit_examples.png, event rows 4–6 of 15, right half of the channels

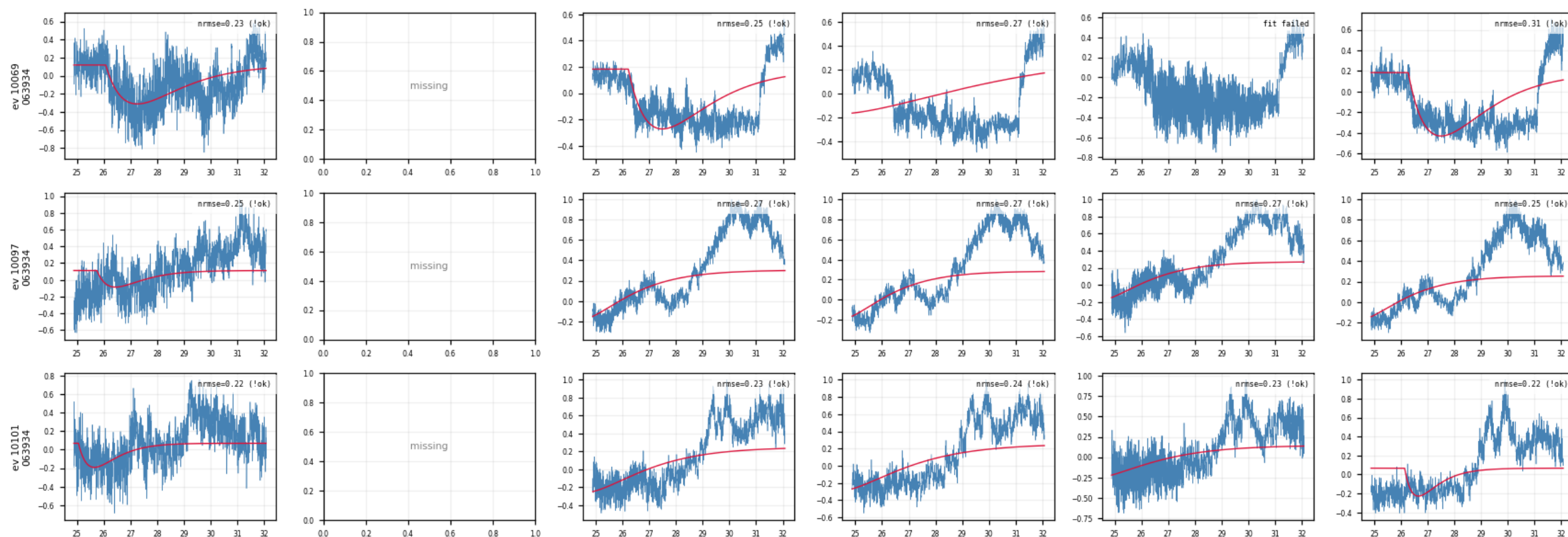
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



Z4 · fit_examples: LP trace vs 2-exp fit (5/10)

figure: zip4_fit_examples.png, event rows 7–9 of 15, left half of the channels

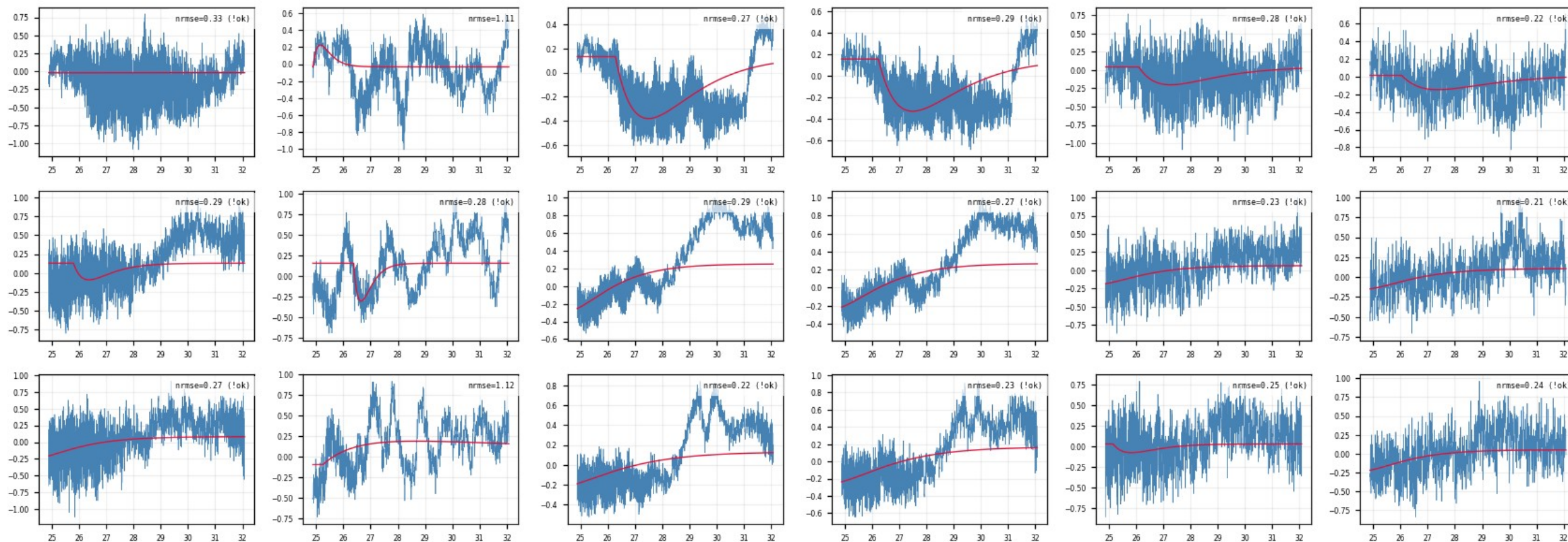
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



Z4 · fit_examples: LP trace vs 2-exp fit (6/10)

figure: zip4_fit_examples.png, event rows 7–9 of 15, right half of the channels

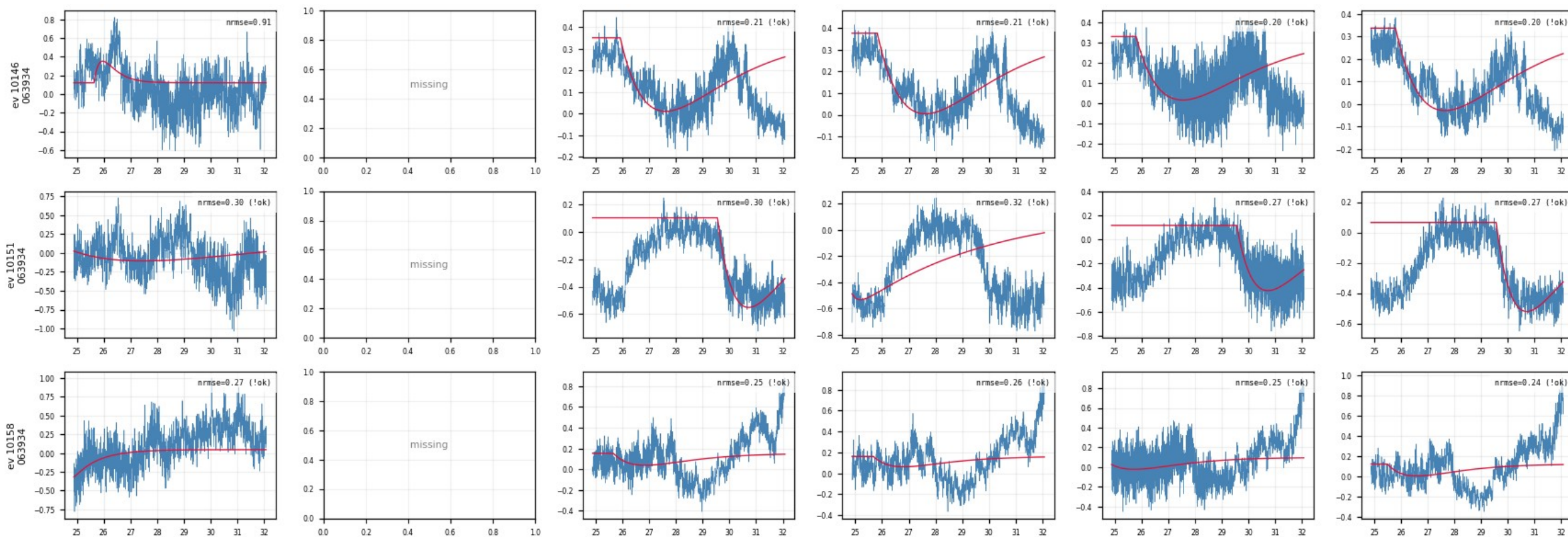
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



Z4 · fit_examples: LP trace vs 2-exp fit (7/10)

figure: zip4_fit_examples.png, event rows 10–12 of 15, left half of the channels

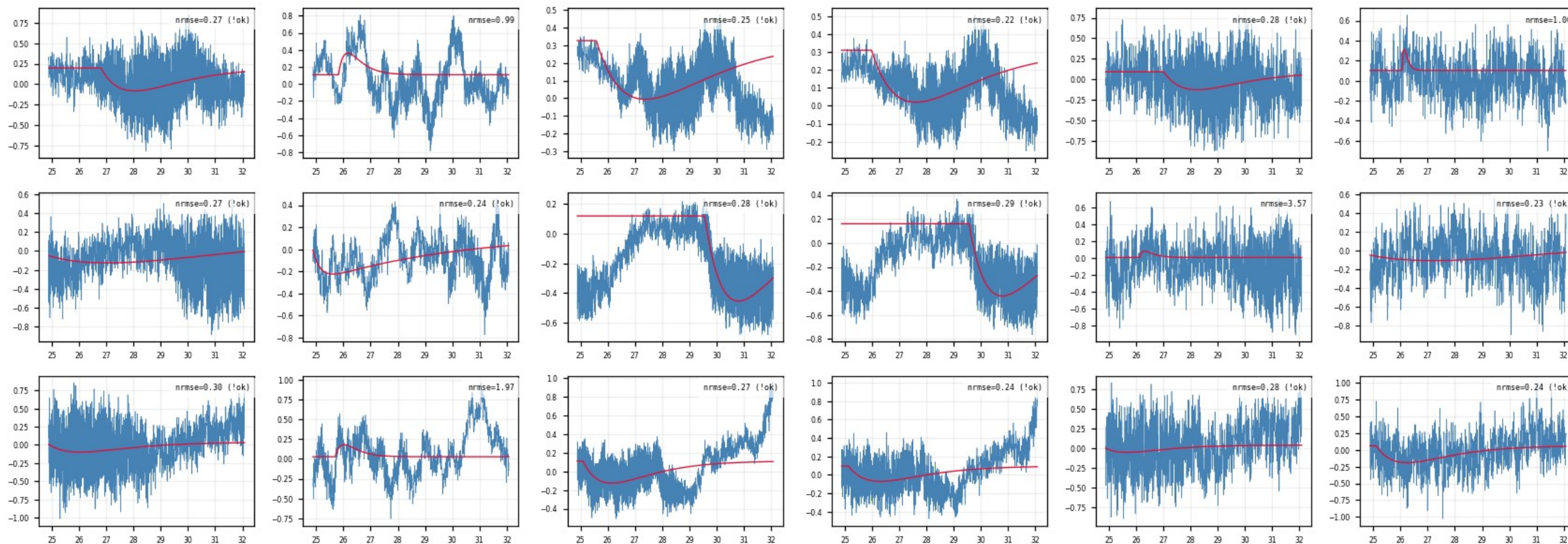
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



Z4 · fit_examples: LP trace vs 2-exp fit (8/10)

figure: zip4_fit_examples.png, event rows 10–12 of 15, right half of the channels

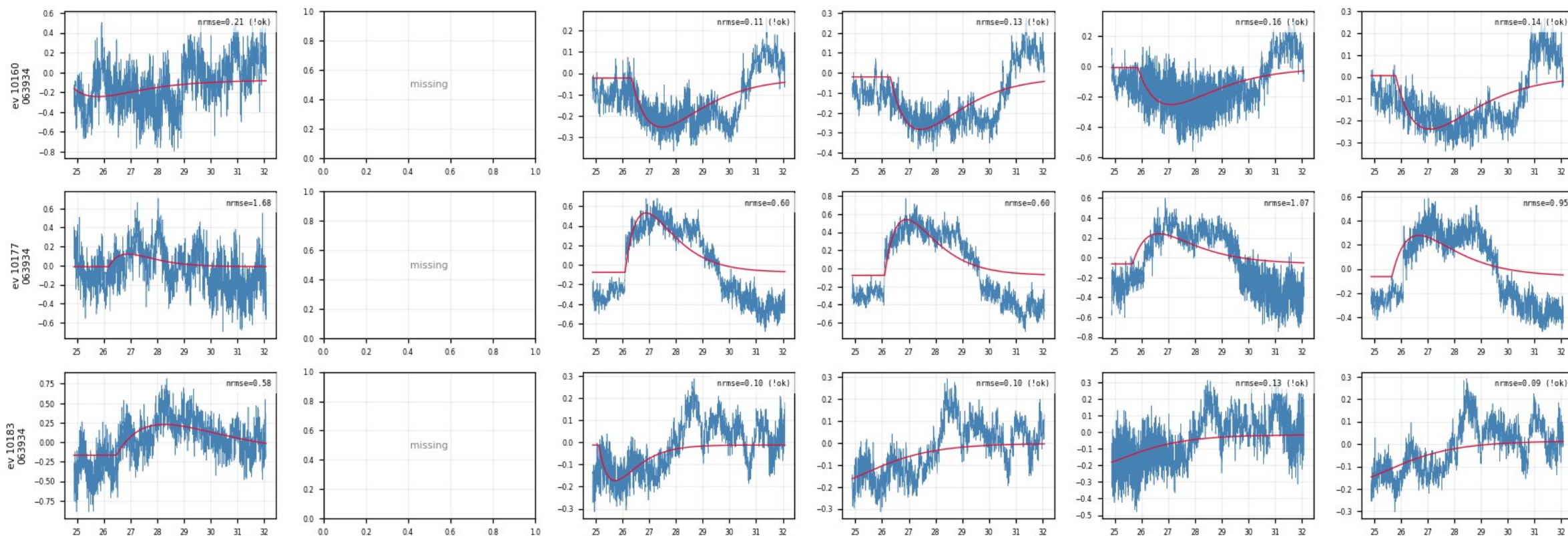
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



Z4 · fit_examples: LP trace vs 2-exp fit (9/10)

figure: zip4_fit_examples.png, event rows 13–15 of 15, left half of the channels

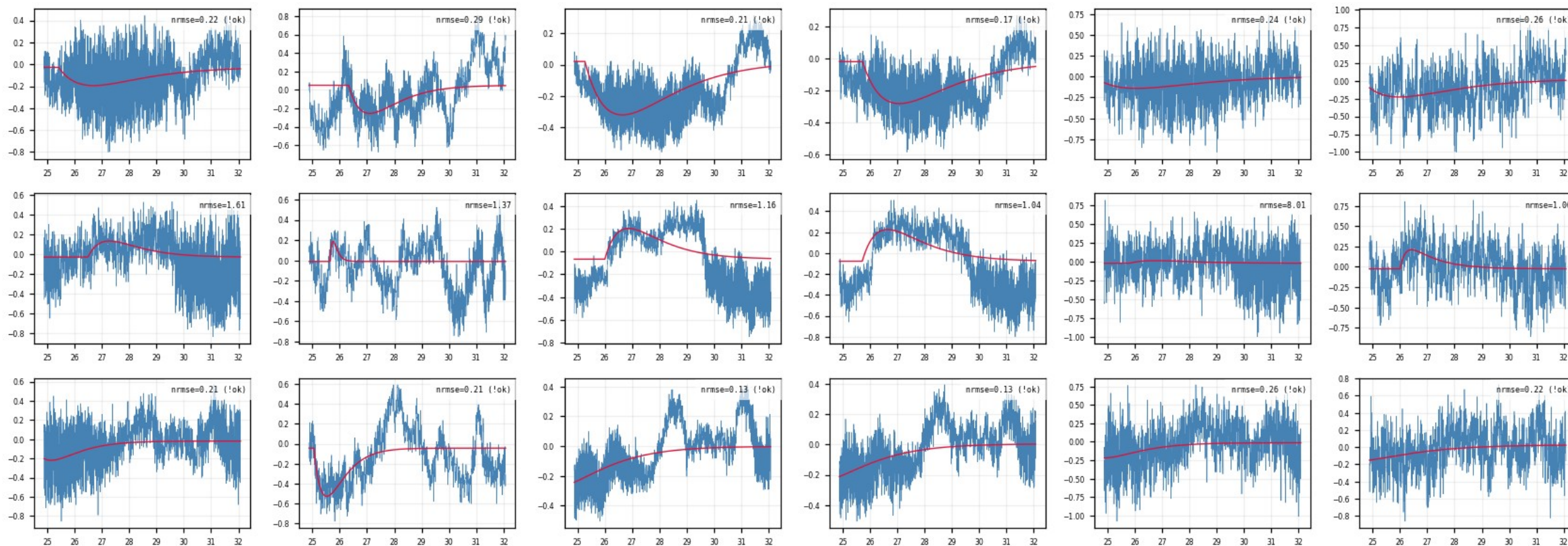
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



Z4 · fit_examples: LP trace vs 2-exp fit (10/10)

figure: zip4_fit_examples.png, event rows 13–15 of 15, right half of the channels

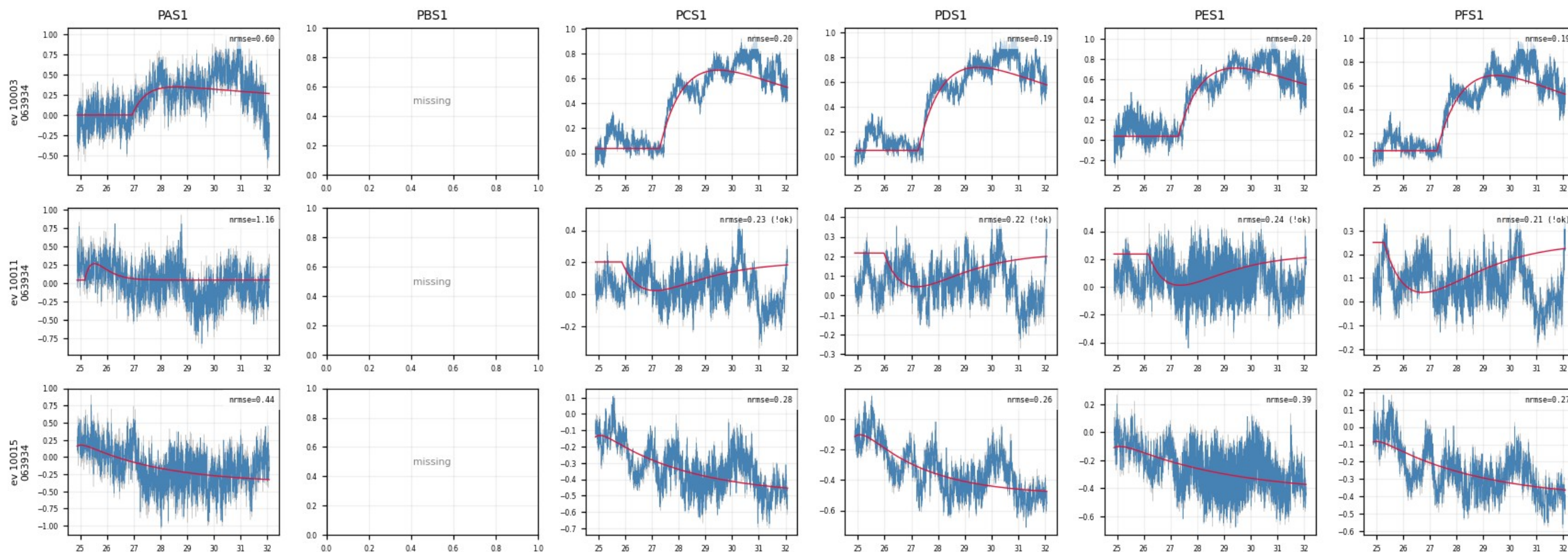
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



Z4 · raw_vs_fit_examples: raw trace vs fit (1/10)

figure: zip4_raw_vs_fit_examples.png, event rows 1–3 of 15, left half of the channels

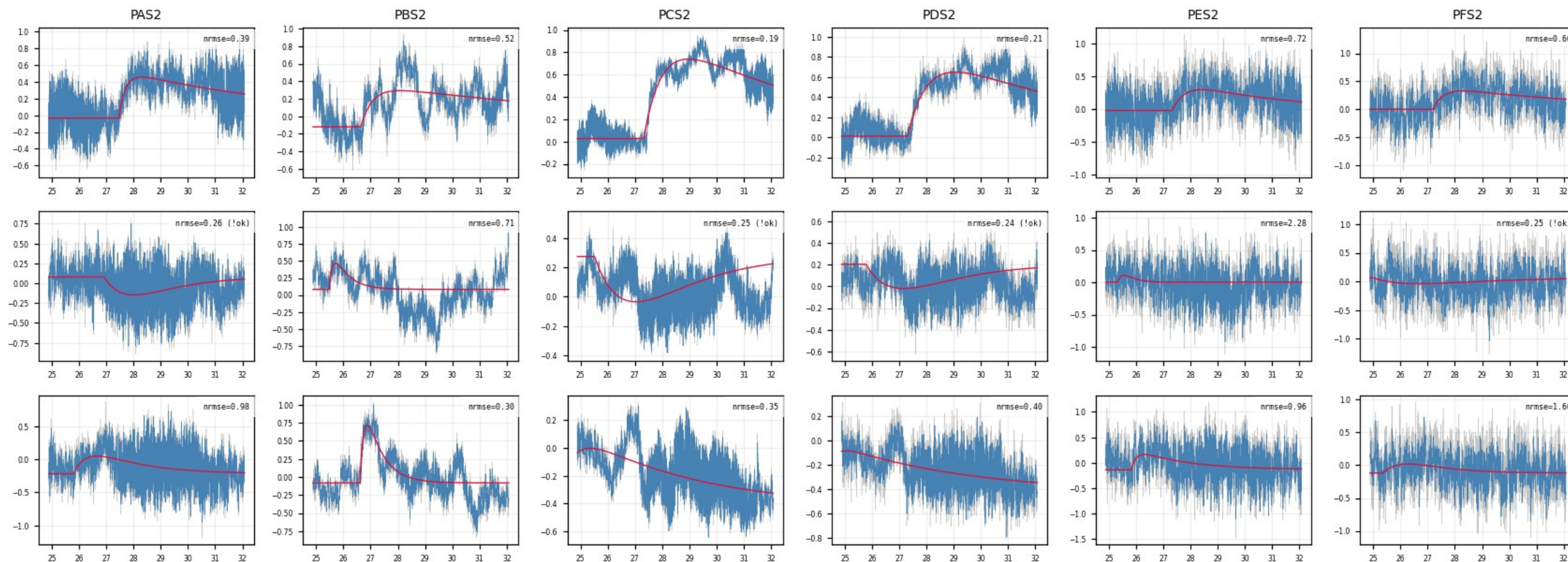
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Z4 · raw_vs_fit_examples: raw trace vs fit (2/10)

figure: zip4_raw_vs_fit_examples.png, event rows 1–3 of 15, right half of the channels

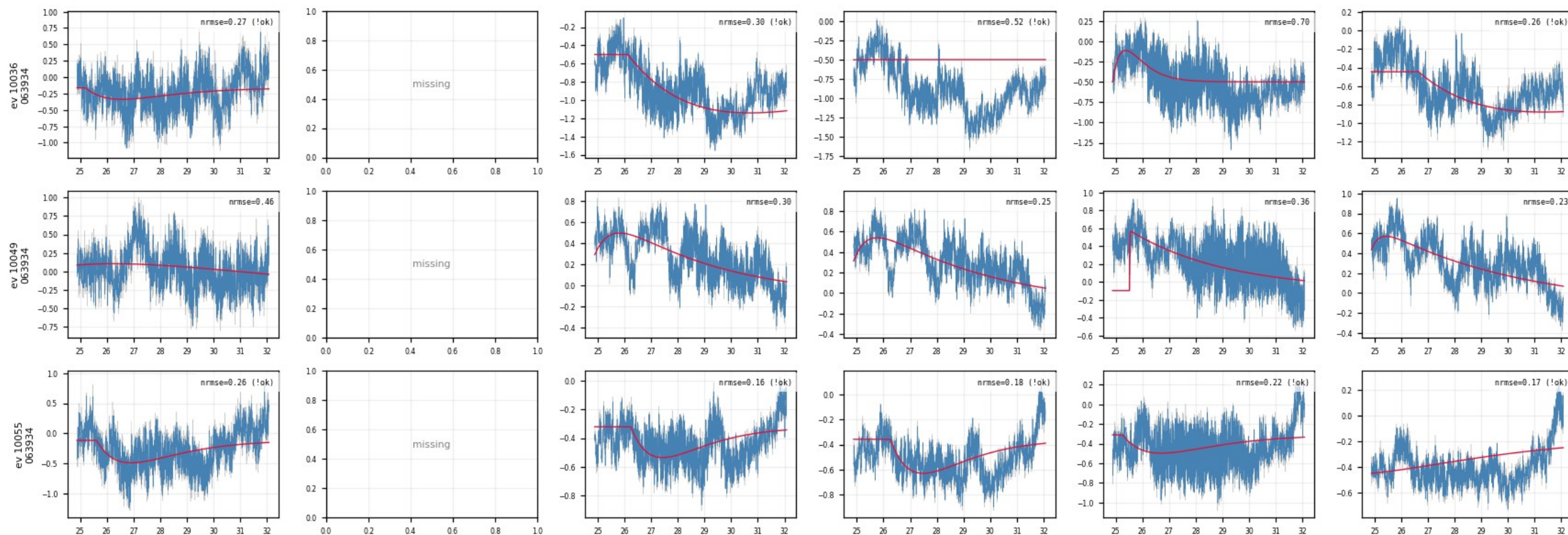
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Z4 · raw_vs_fit_examples: raw trace vs fit (3/10)

figure: zip4_raw_vs_fit_examples.png, event rows 4–6 of 15, left half of the channels

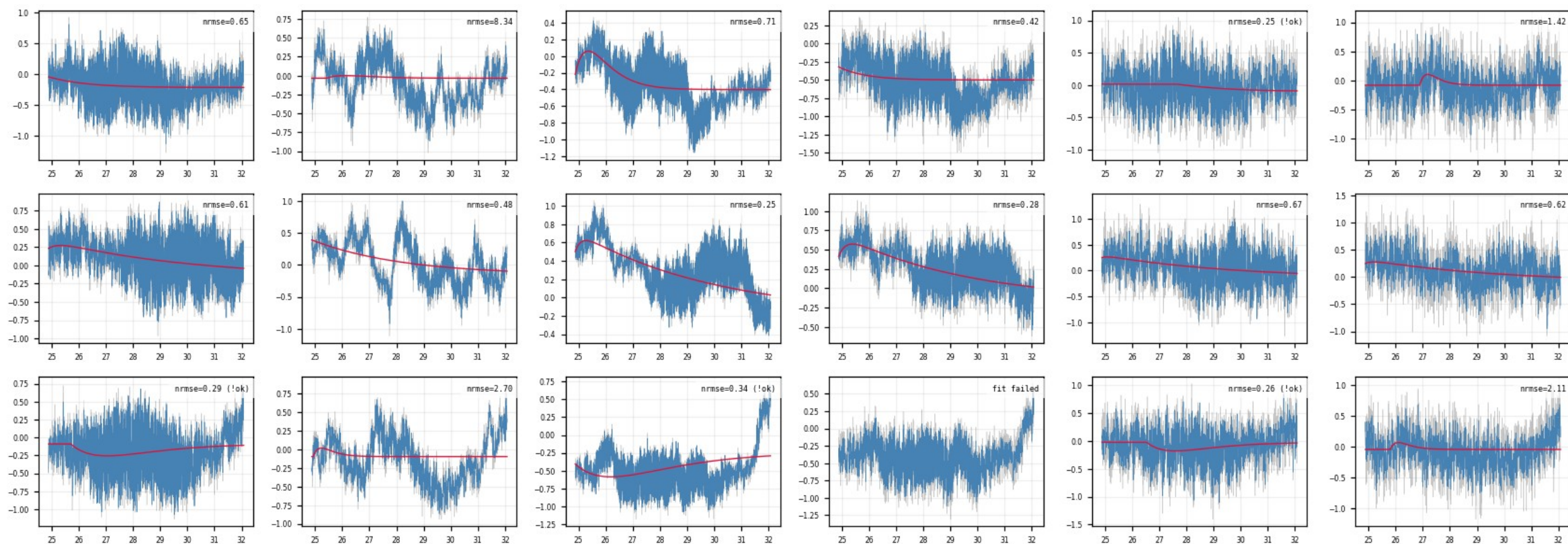
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Z4 · raw_vs_fit_examples: raw trace vs fit (4/10)

figure: zip4_raw_vs_fit_examples.png, event rows 4–6 of 15, right half of the channels

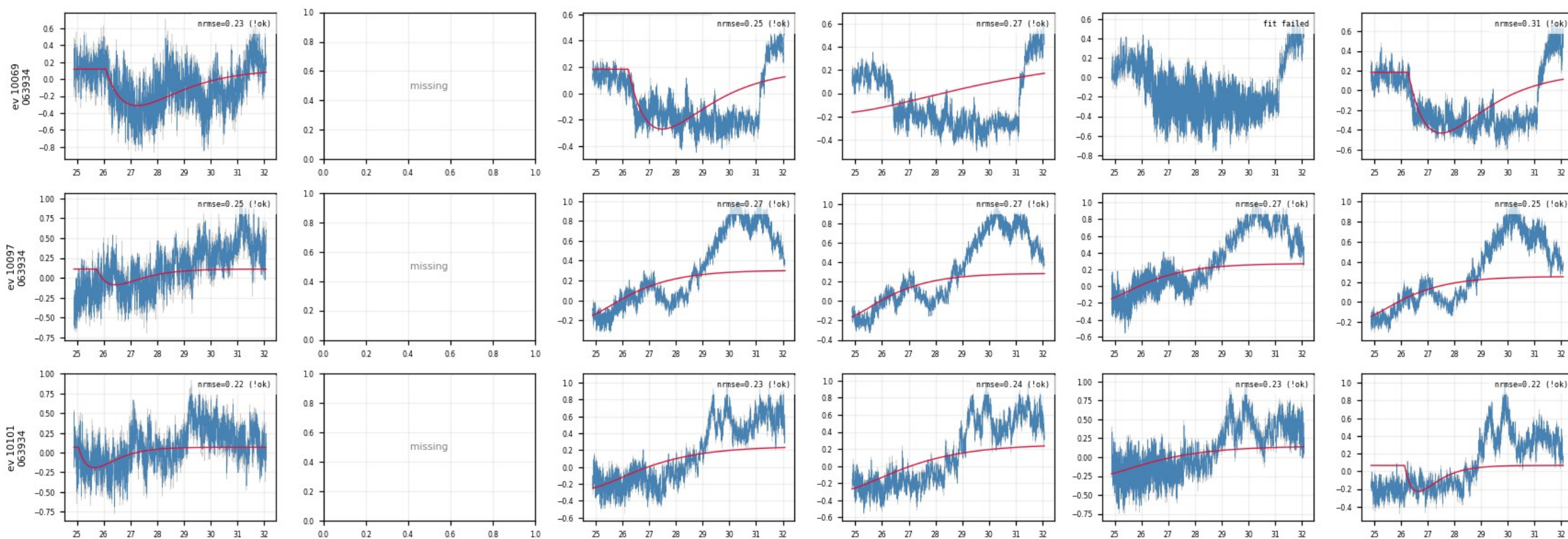
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Z4 · raw_vs_fit_examples: raw trace vs fit (5/10)

figure: zip4_raw_vs_fit_examples.png, event rows 7–9 of 15, left half of the channels

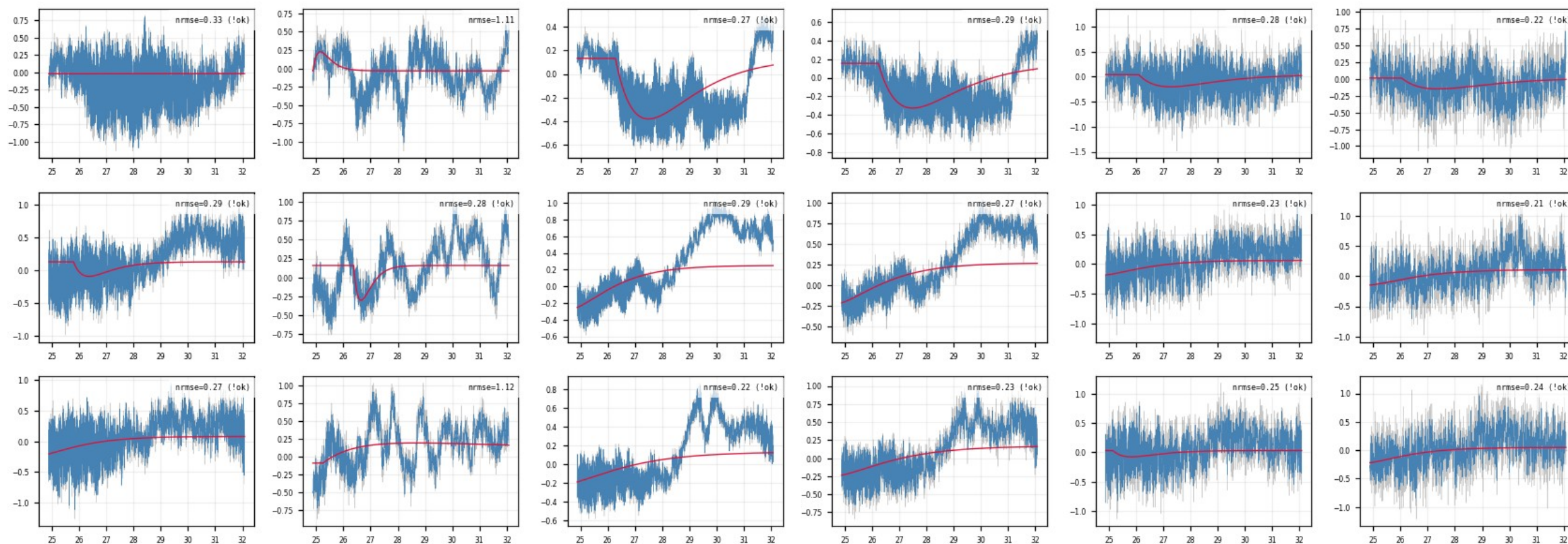
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Z4 · raw_vs_fit_examples: raw trace vs fit (6/10)

figure: zip4_raw_vs_fit_examples.png, event rows 7–9 of 15, right half of the channels

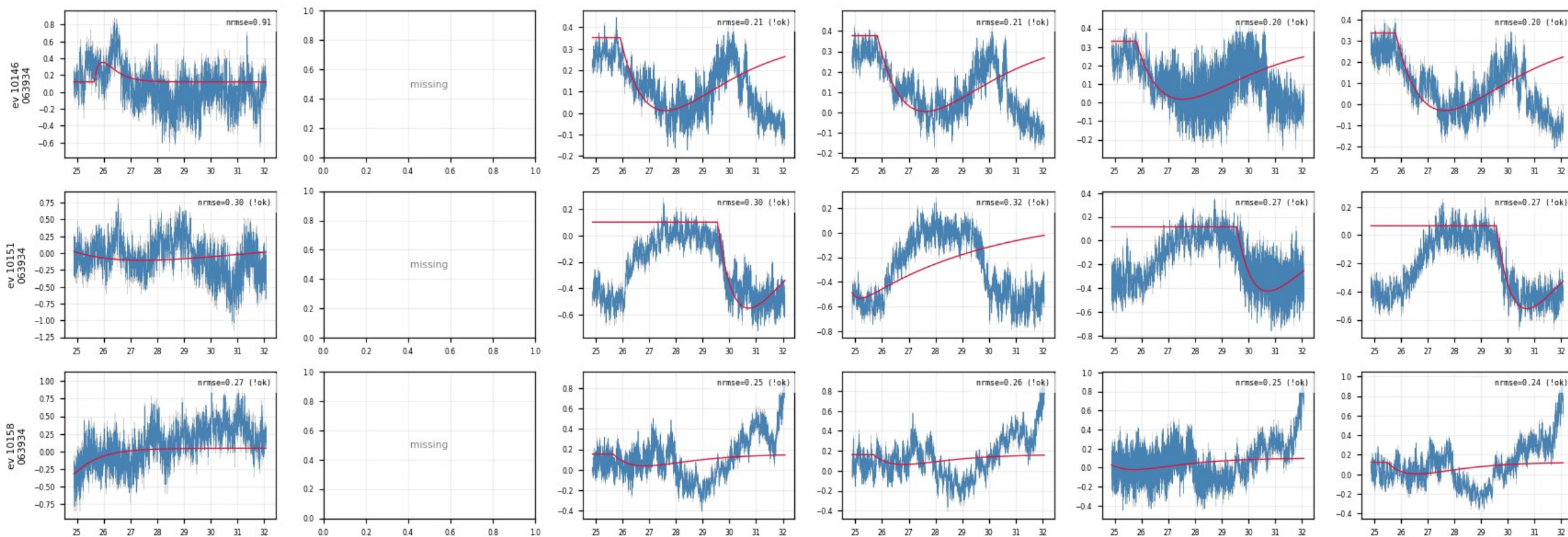
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Z4 · raw_vs_fit_examples: raw trace vs fit (7/10)

figure: zip4_raw_vs_fit_examples.png, event rows 10–12 of 15, left half of the channels

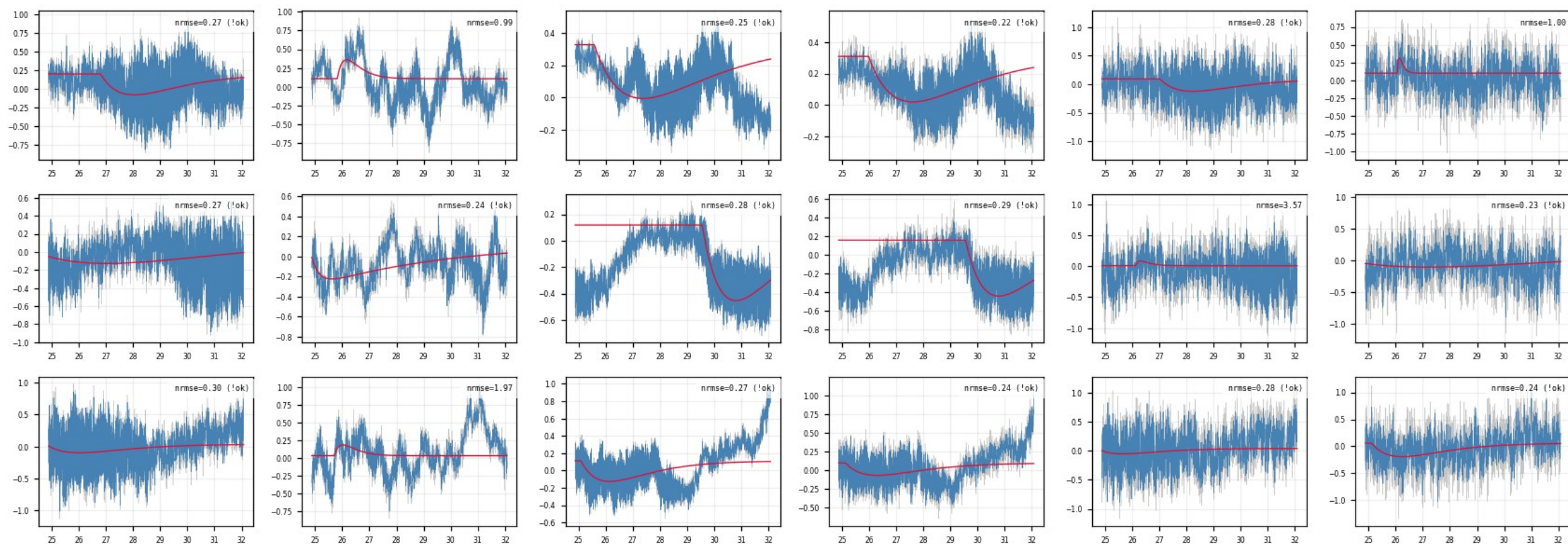
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Z4 · raw_vs_fit_examples: raw trace vs fit (8/10)

figure: zip4_raw_vs_fit_examples.png, event rows 10–12 of 15, right half of the channels

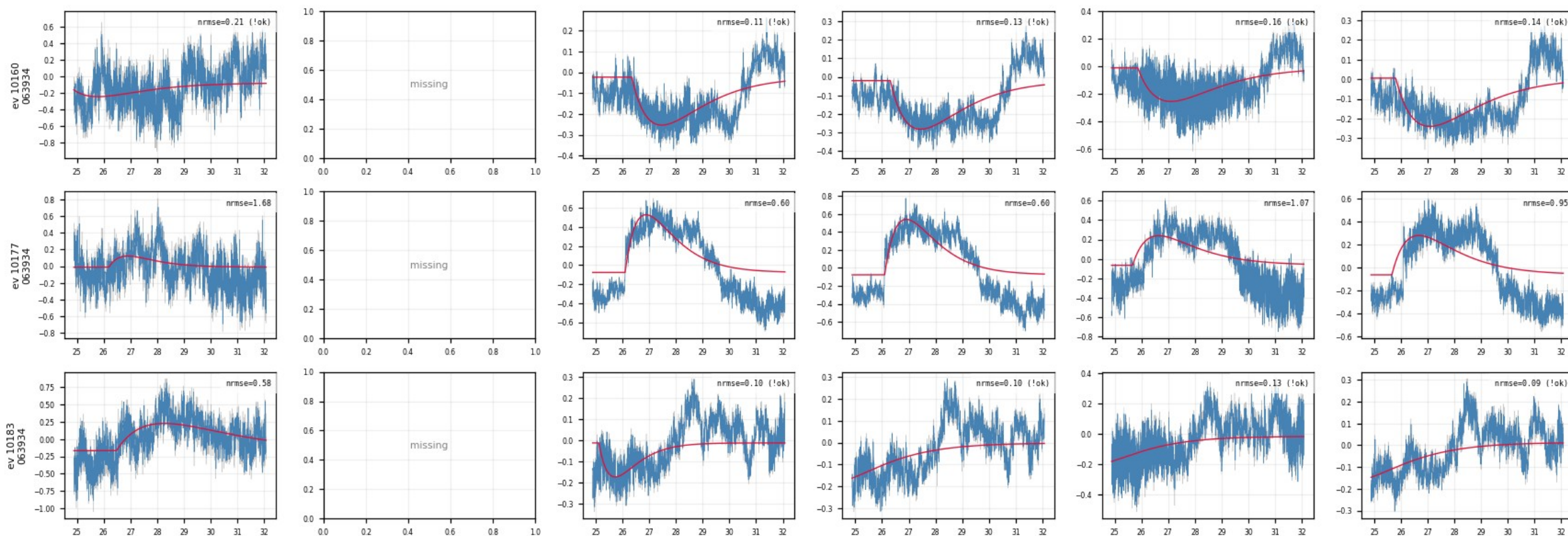
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Z4 · raw_vs_fit_examples: raw trace vs fit (9/10)

figure: zip4_raw_vs_fit_examples.png, event rows 13–15 of 15, left half of the channels

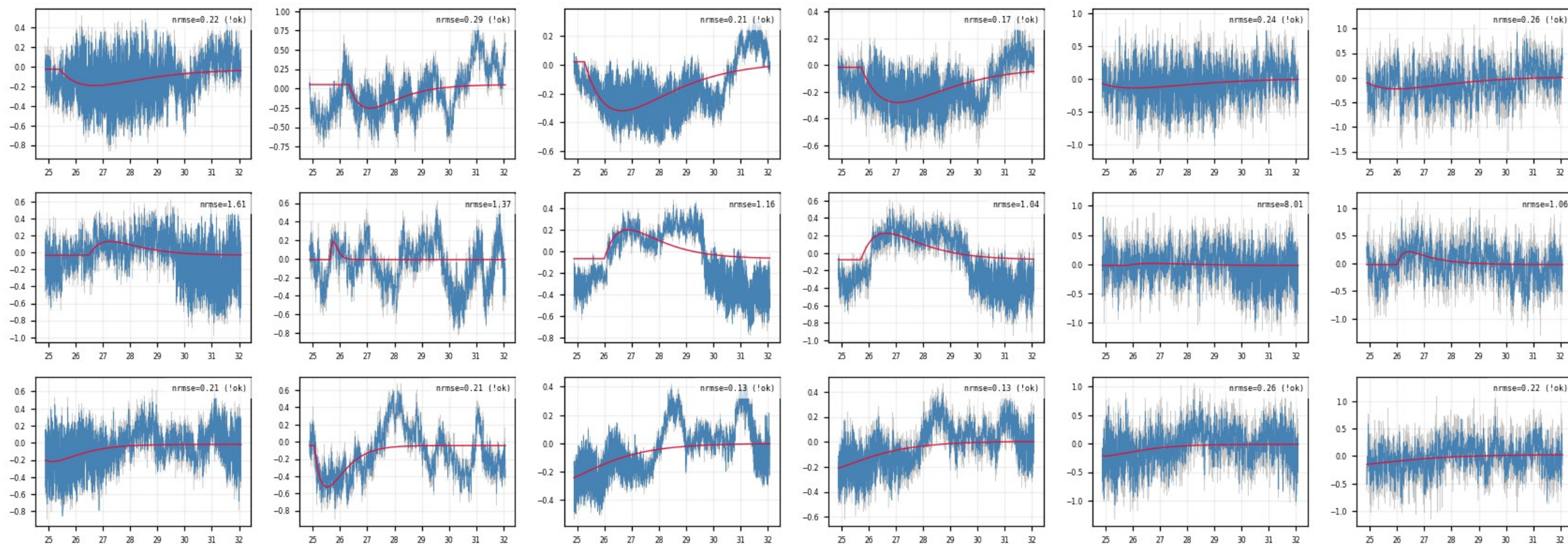
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Z4 · raw_vs_fit_examples: raw trace vs fit (10/10)

figure: zip4_raw_vs_fit_examples.png, event rows 13–15 of 15, right half of the channels

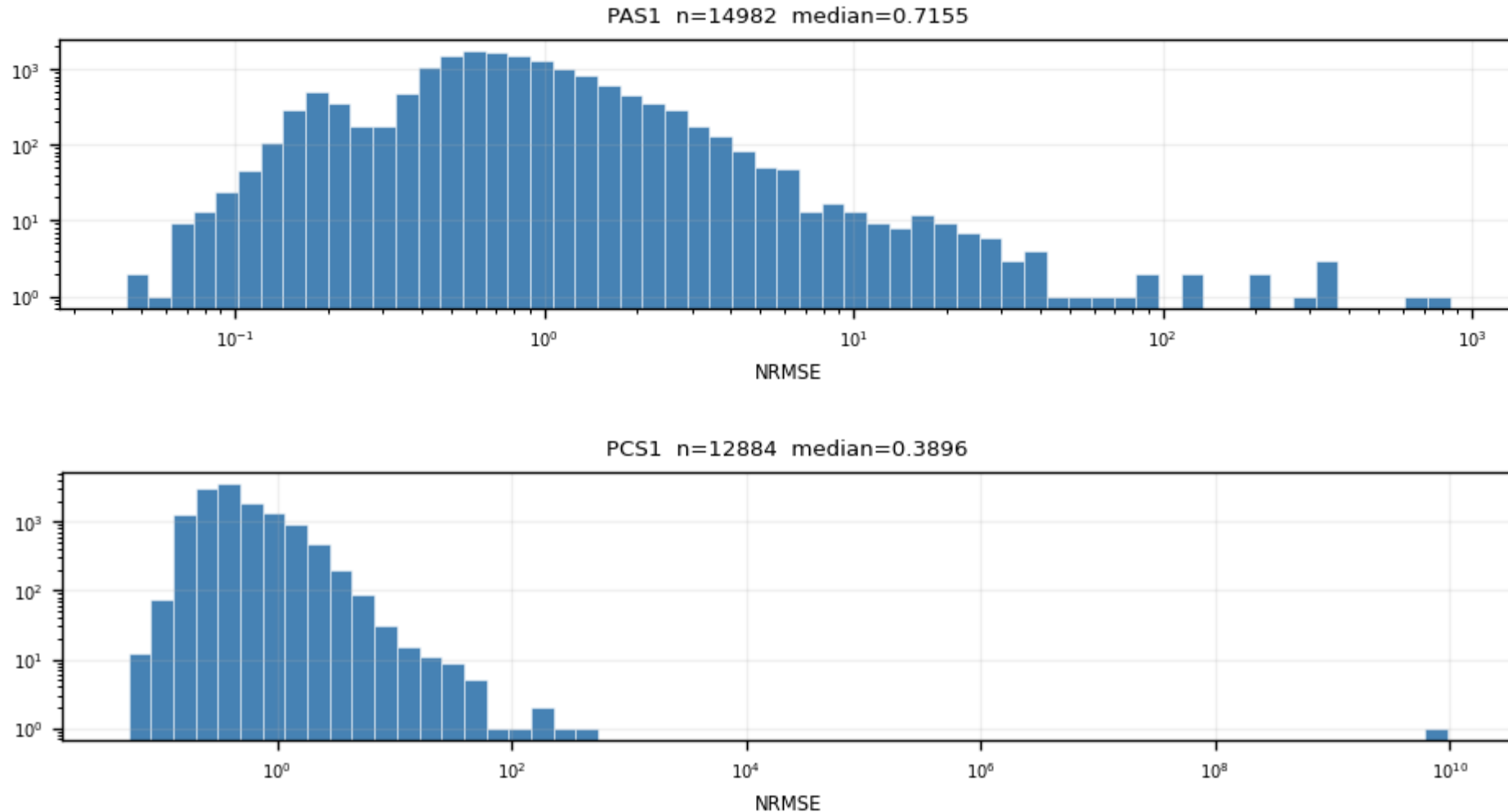
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Z4 · nrmse: NRMSE distribution per channel (1/6)

figure: zip4_nrmse.png, channel panels 1–2 of 11, top to bottom

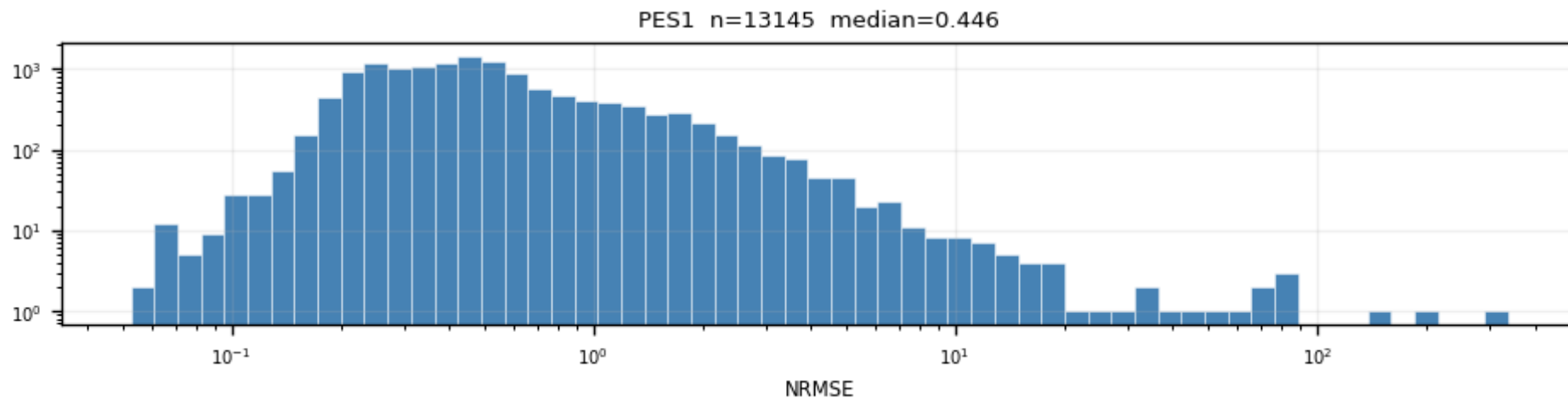
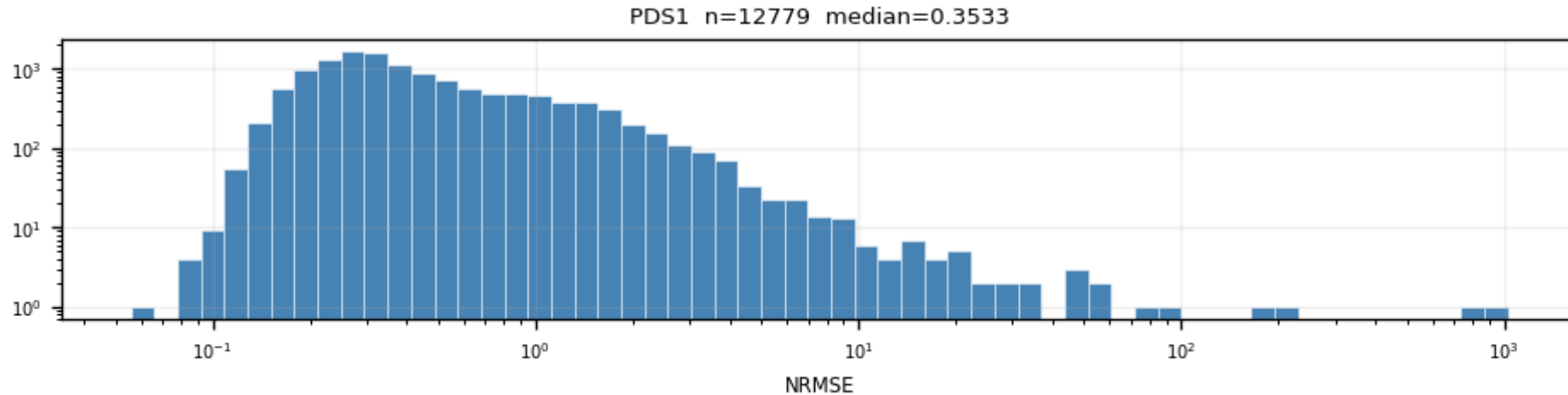
NRMSE = $\text{RMS}(\text{fit residual}) / \text{fitted peak}$, log-log. Bimodal: good fits vs noise triggers; the valley at ≈ 0.4 sets quality cut 1.



Z4 · nrmse: NRMSE distribution per channel (2/6)

figure: zip4_nrmse.png, channel panels 3–4 of 11, top to bottom

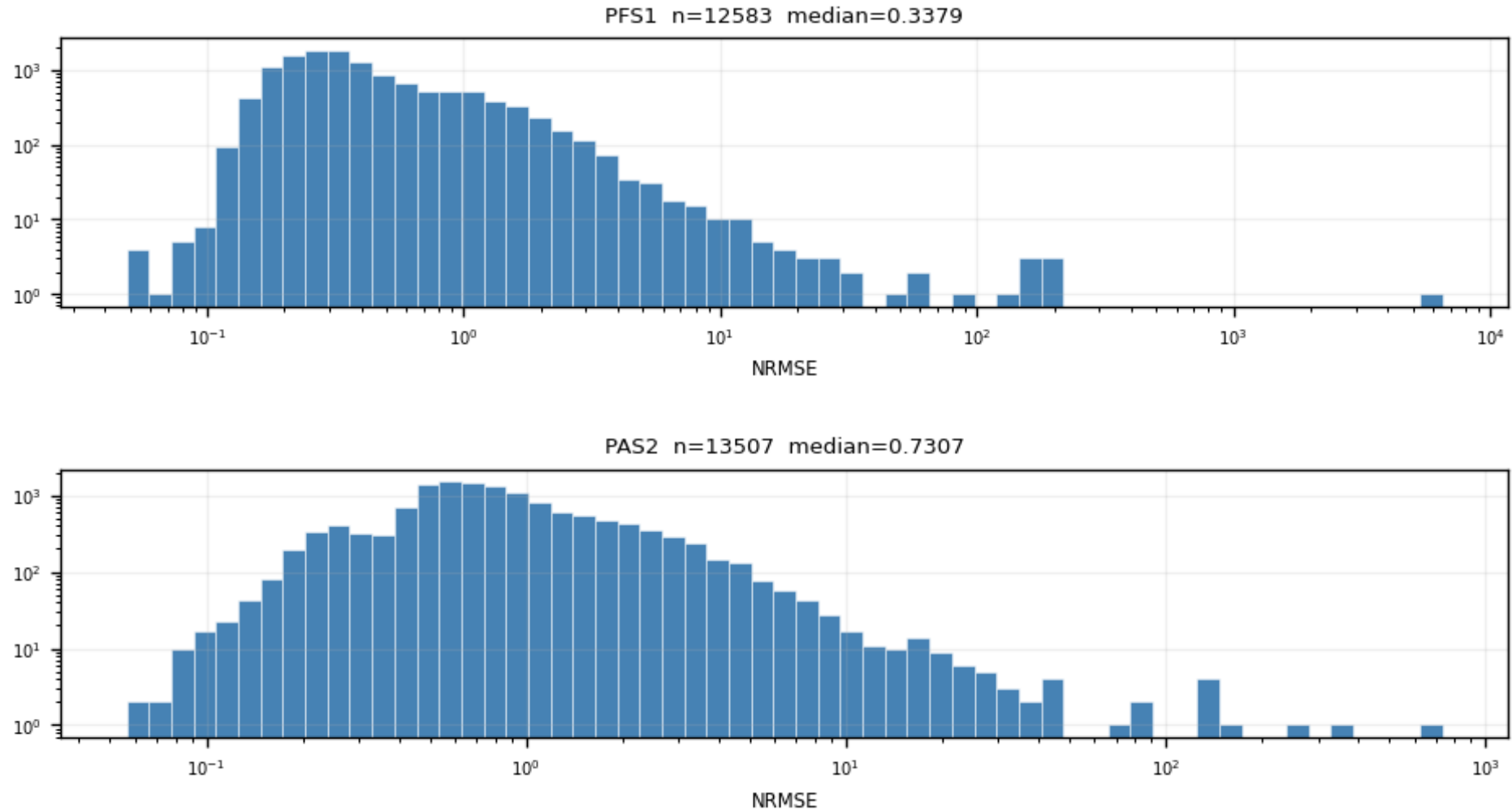
NRMSE = $\text{RMS}(\text{fit residual}) / \text{fitted peak}$, log-log. Bimodal: good fits vs noise triggers; the valley at ≈ 0.4 sets quality cut 1.



Z4 · nrmse: NRMSE distribution per channel (3/6)

figure: zip4_nrmse.png, channel panels 5–6 of 11, top to bottom

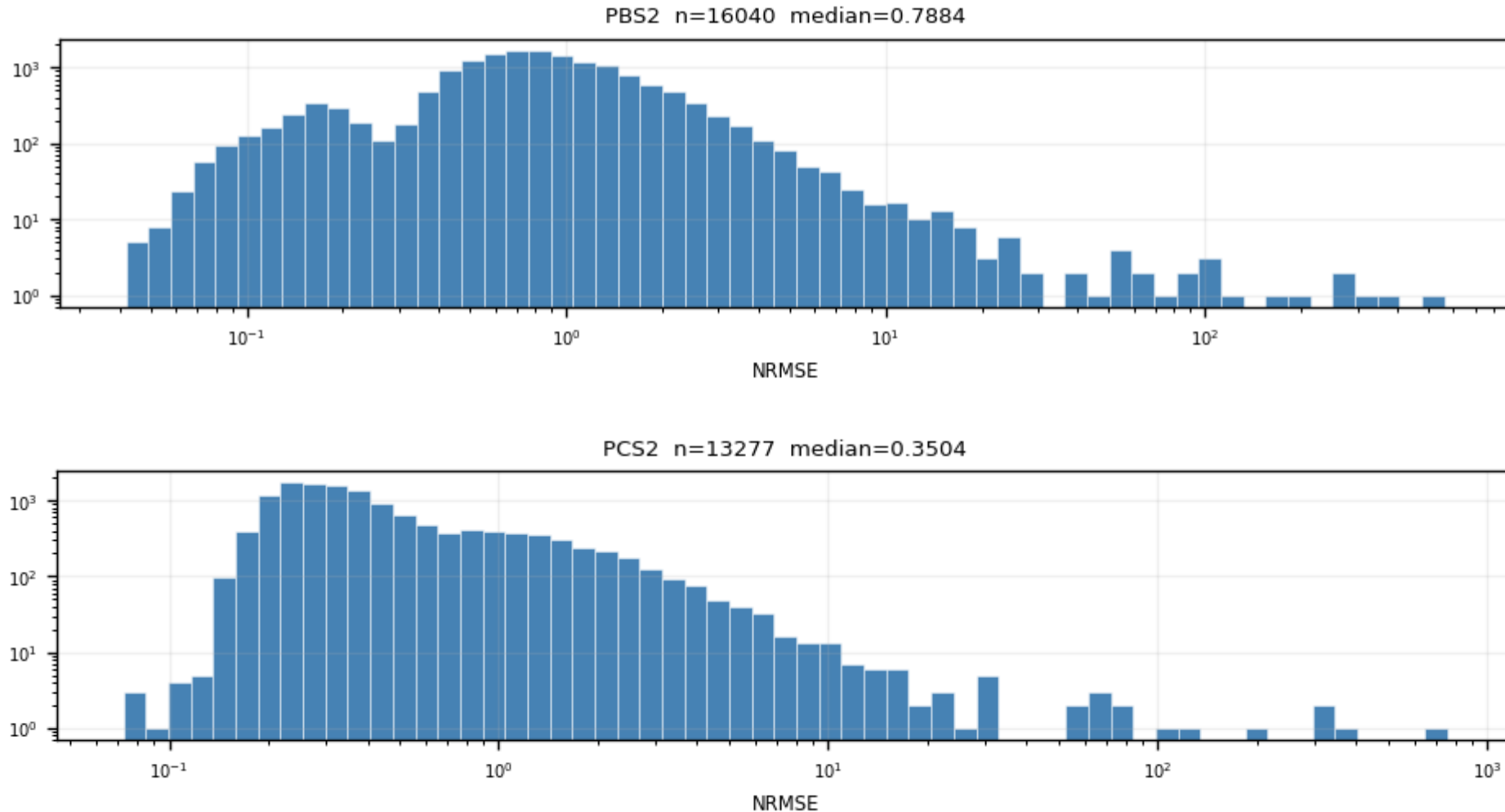
NRMSE = $\text{RMS}(\text{fit residual}) / \text{fitted peak}$, log-log. Bimodal: good fits vs noise triggers; the valley at ≈ 0.4 sets quality cut 1.



Z4 · nrmse: NRMSE distribution per channel (4/6)

figure: zip4_nrmse.png, channel panels 7–8 of 11, top to bottom

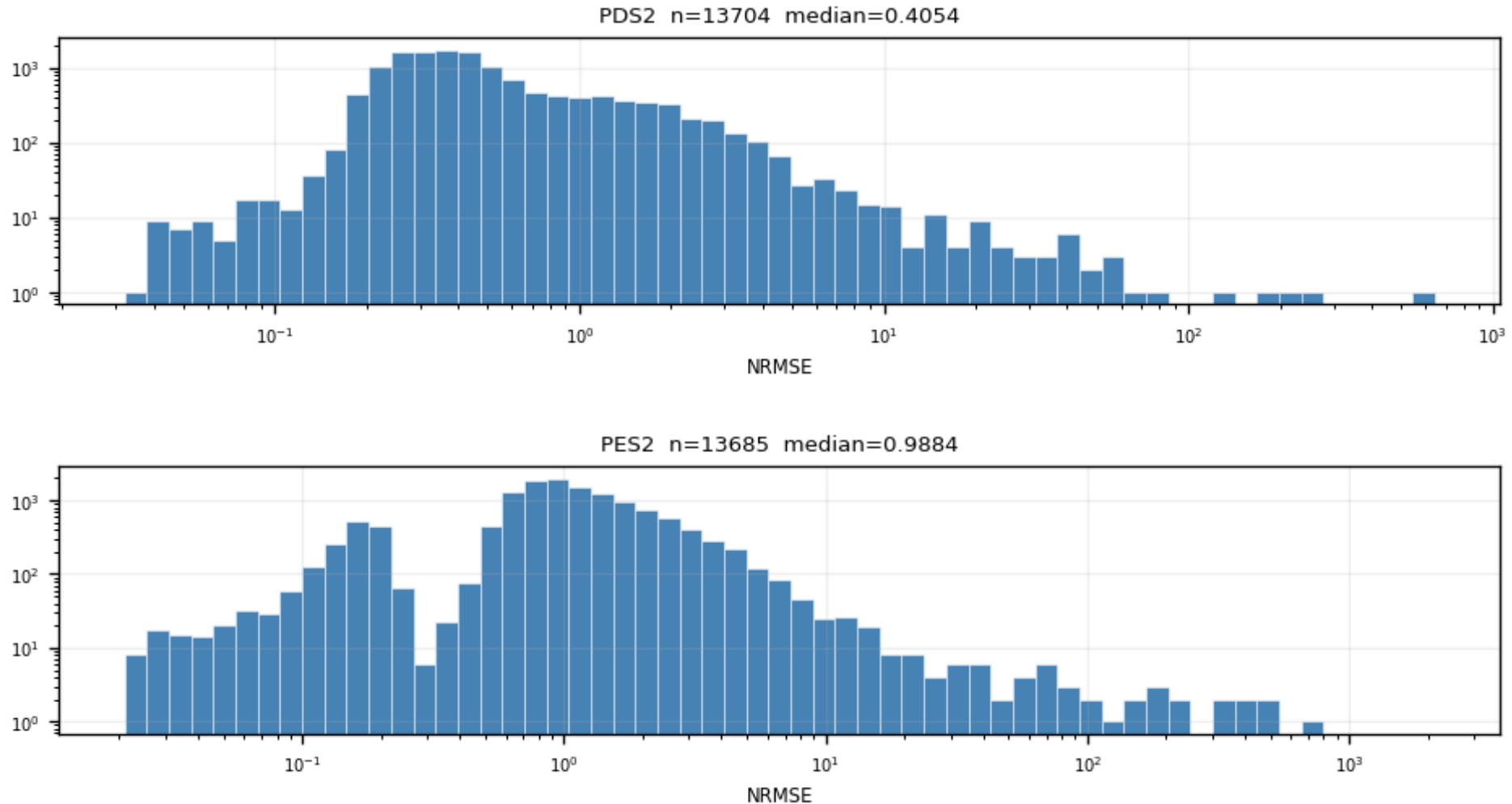
NRMSE = $\text{RMS}(\text{fit residual}) / \text{fitted peak}$, log-log. Bimodal: good fits vs noise triggers; the valley at ≈ 0.4 sets quality cut 1.



Z4 · nrmse: NRMSE distribution per channel (5/6)

figure: zip4_nrmse.png, channel panels 9–10 of 11, top to bottom

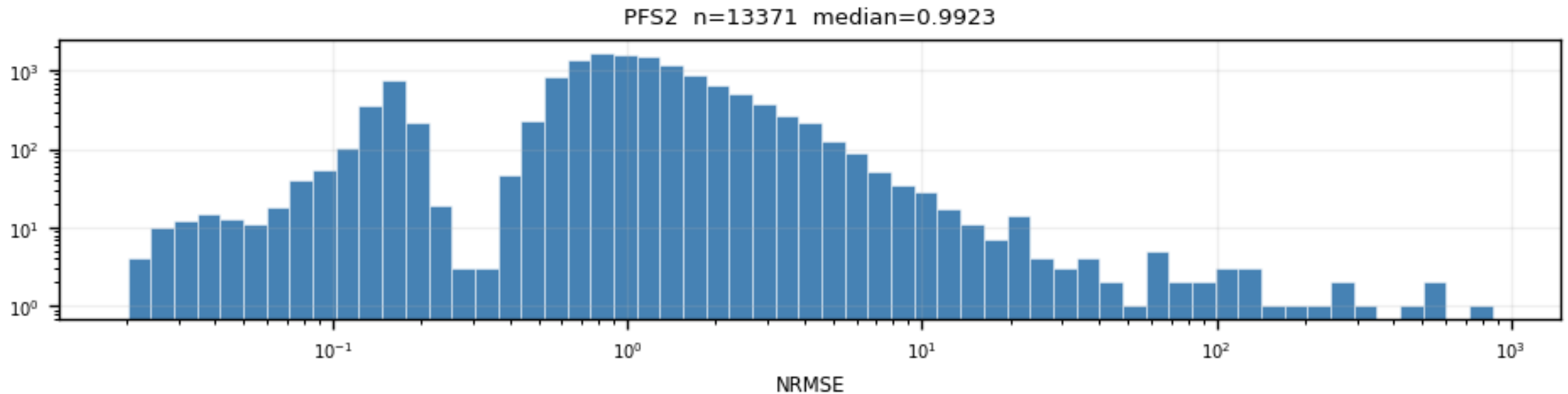
NRMSE = $\text{RMS}(\text{fit residual}) / \text{fitted peak}$, log-log. Bimodal: good fits vs noise triggers; the valley at ≈ 0.4 sets quality cut 1.



Z4 · nrmse: NRMSE distribution per channel (6/6)

figure: zip4_nrmse.png, channel panels 11–11 of 11, top to bottom

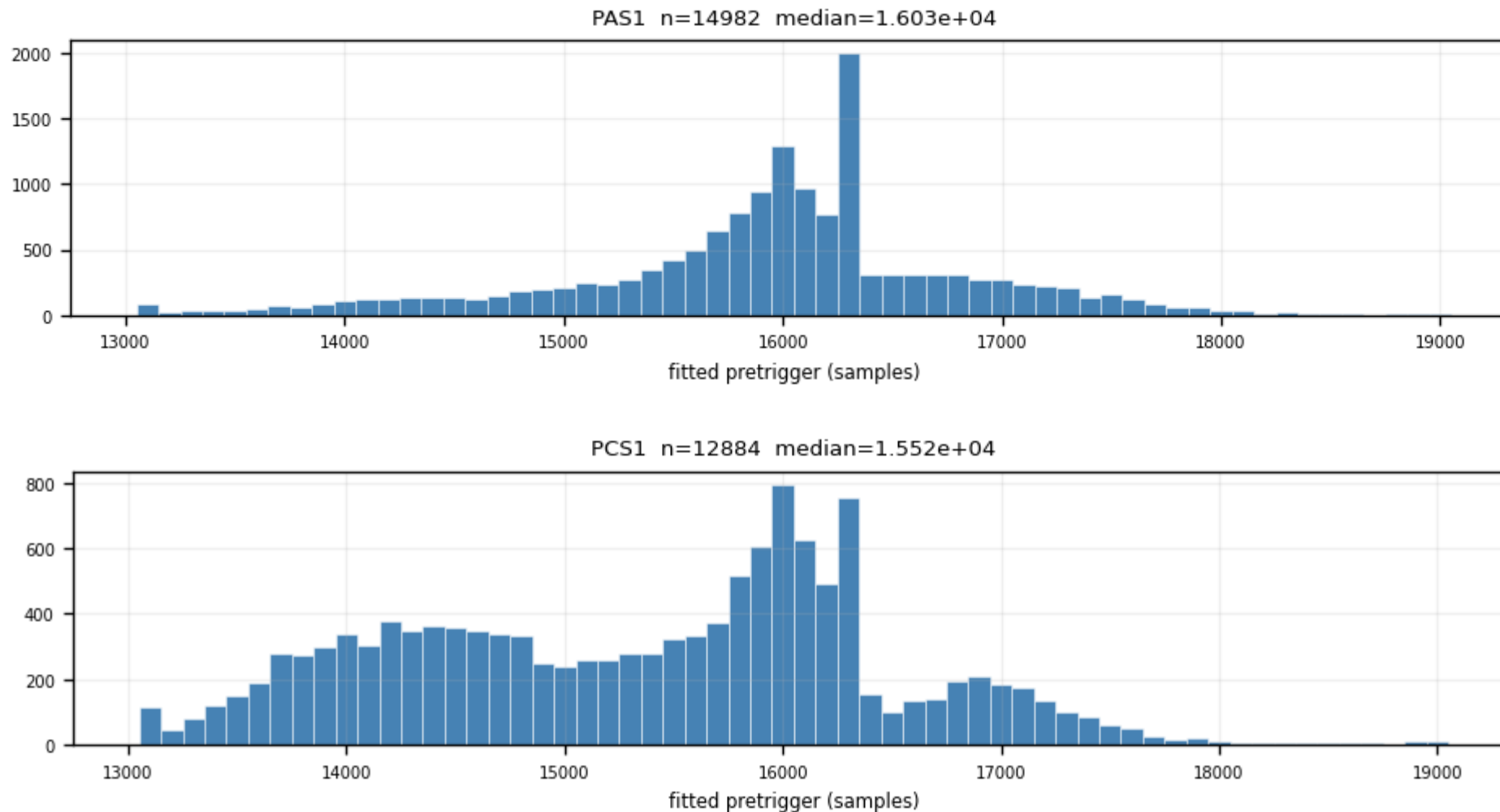
NRMSE = $\text{RMS}(\text{fit residual}) / \text{fitted peak}$, log-log. Bimodal: good fits vs noise triggers; the valley at ≈ 0.4 sets quality cut 1.



Z4 · pretrigger: fitted onset per channel (1/6)

figure: zip4_pretrigger.png, channel panels 1–2 of 11, top to bottom

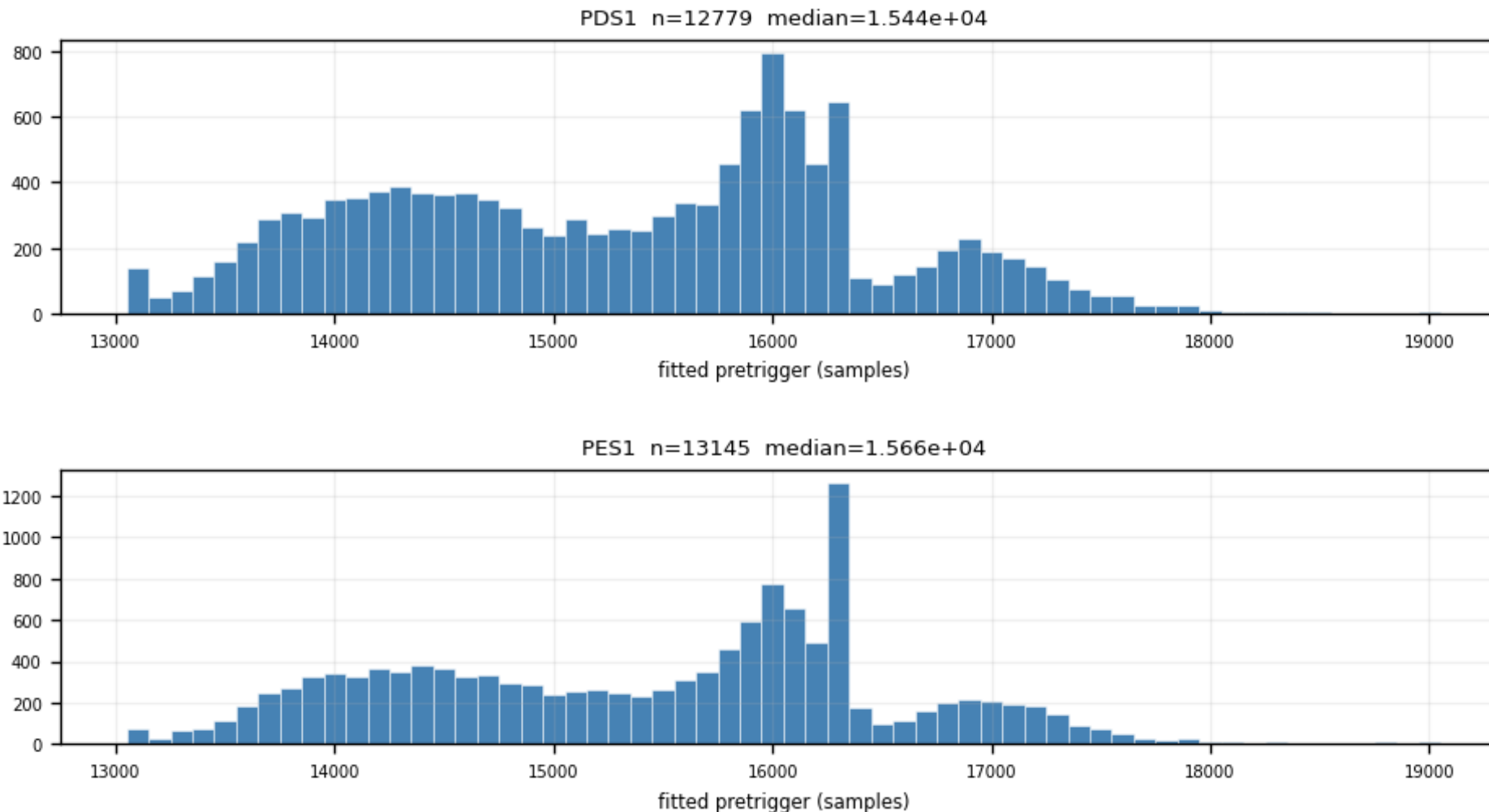
The onset is a FREE fit parameter; pinning it inside the fit would bias every other parameter.



Z4 · pretrigger: fitted onset per channel (2/6)

figure: zip4_pretrigger.png, channel panels 3–4 of 11, top to bottom

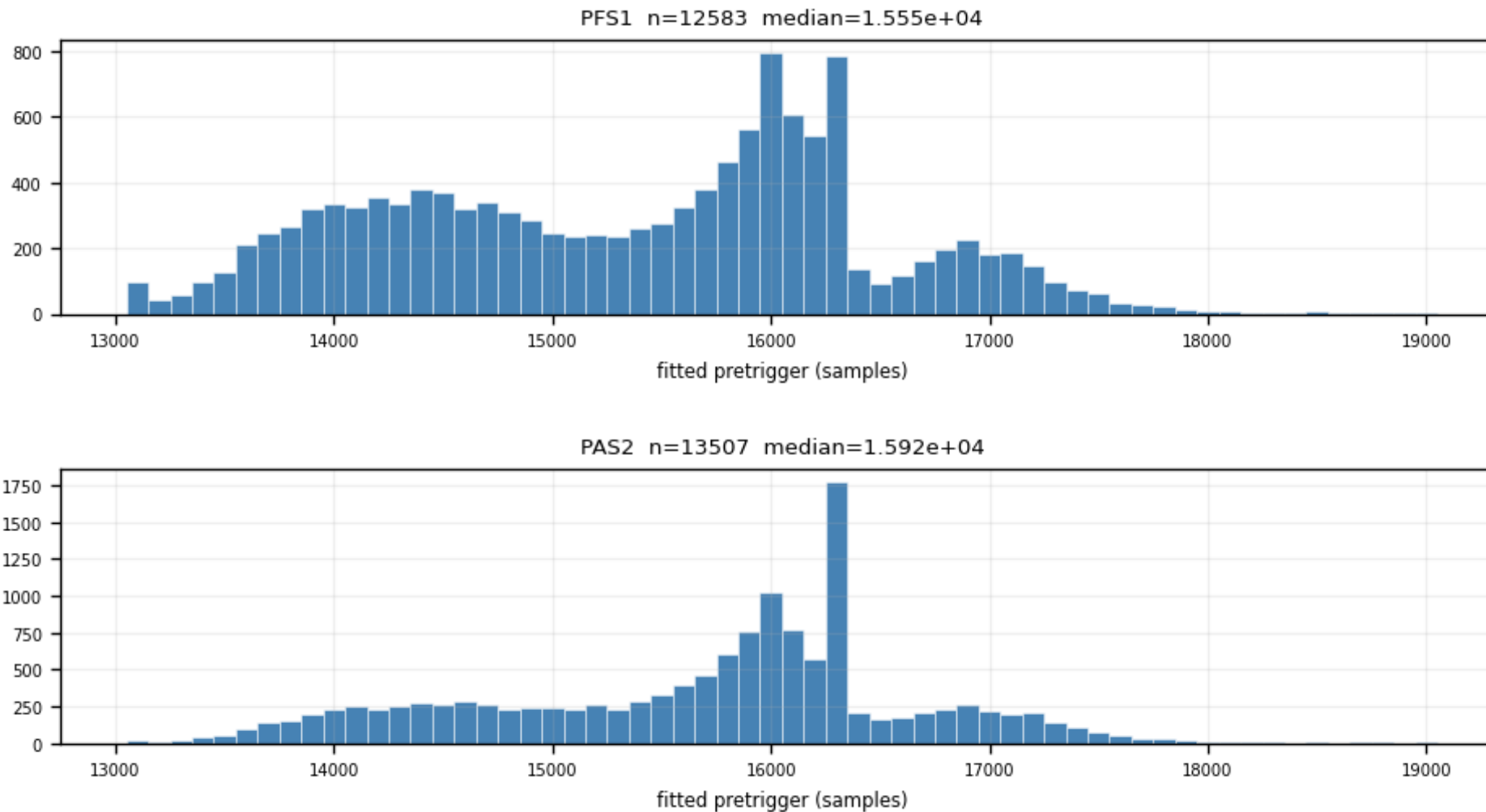
The onset is a FREE fit parameter; pinning it inside the fit would bias every other parameter.



Z4 · pretrigger: fitted onset per channel (3/6)

figure: zip4_pretrigger.png, channel panels 5–6 of 11, top to bottom

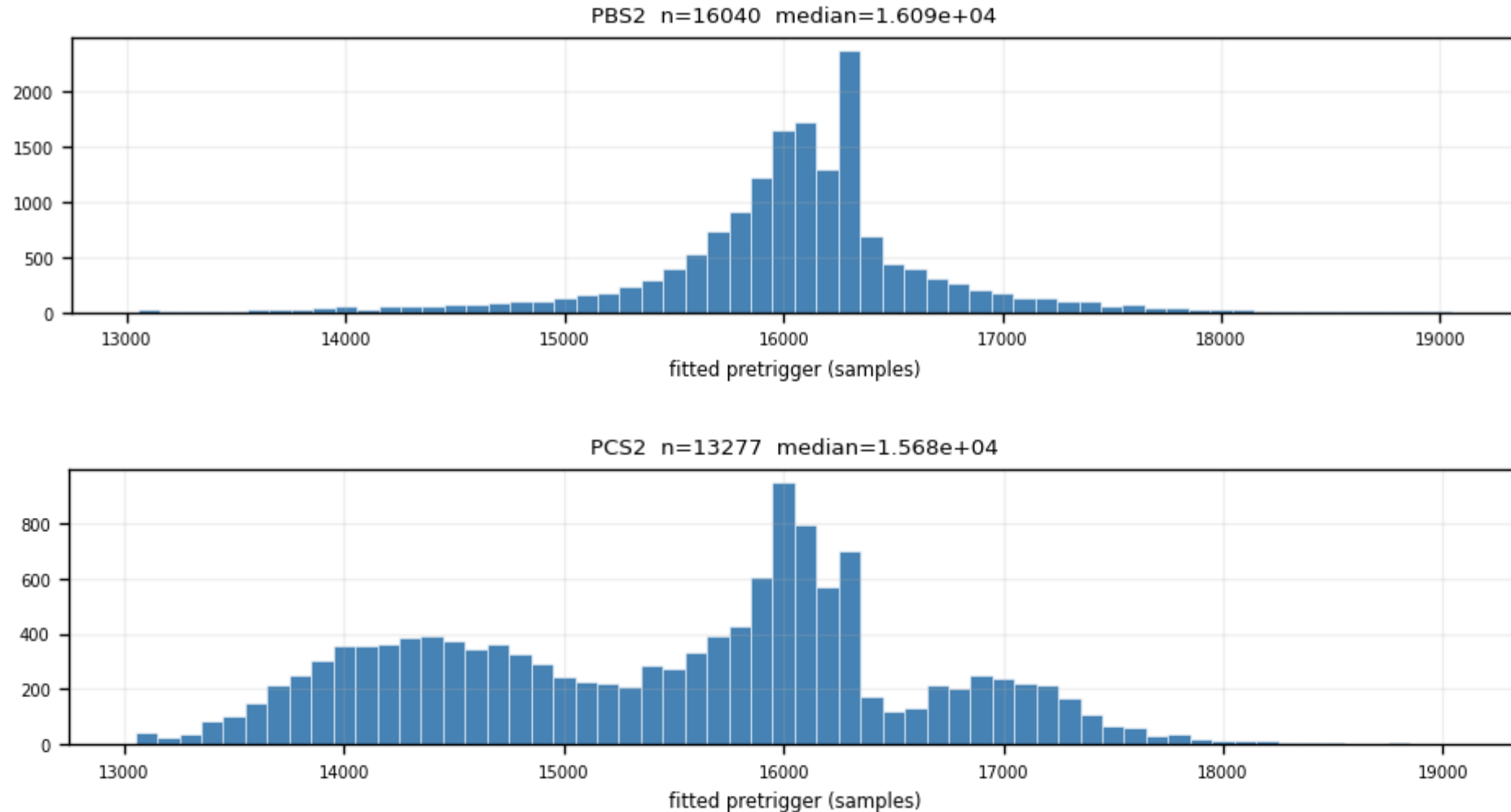
The onset is a FREE fit parameter; pinning it inside the fit would bias every other parameter.



Z4 · pretrigger: fitted onset per channel (4/6)

figure: zip4_pretrigger.png, channel panels 7–8 of 11, top to bottom

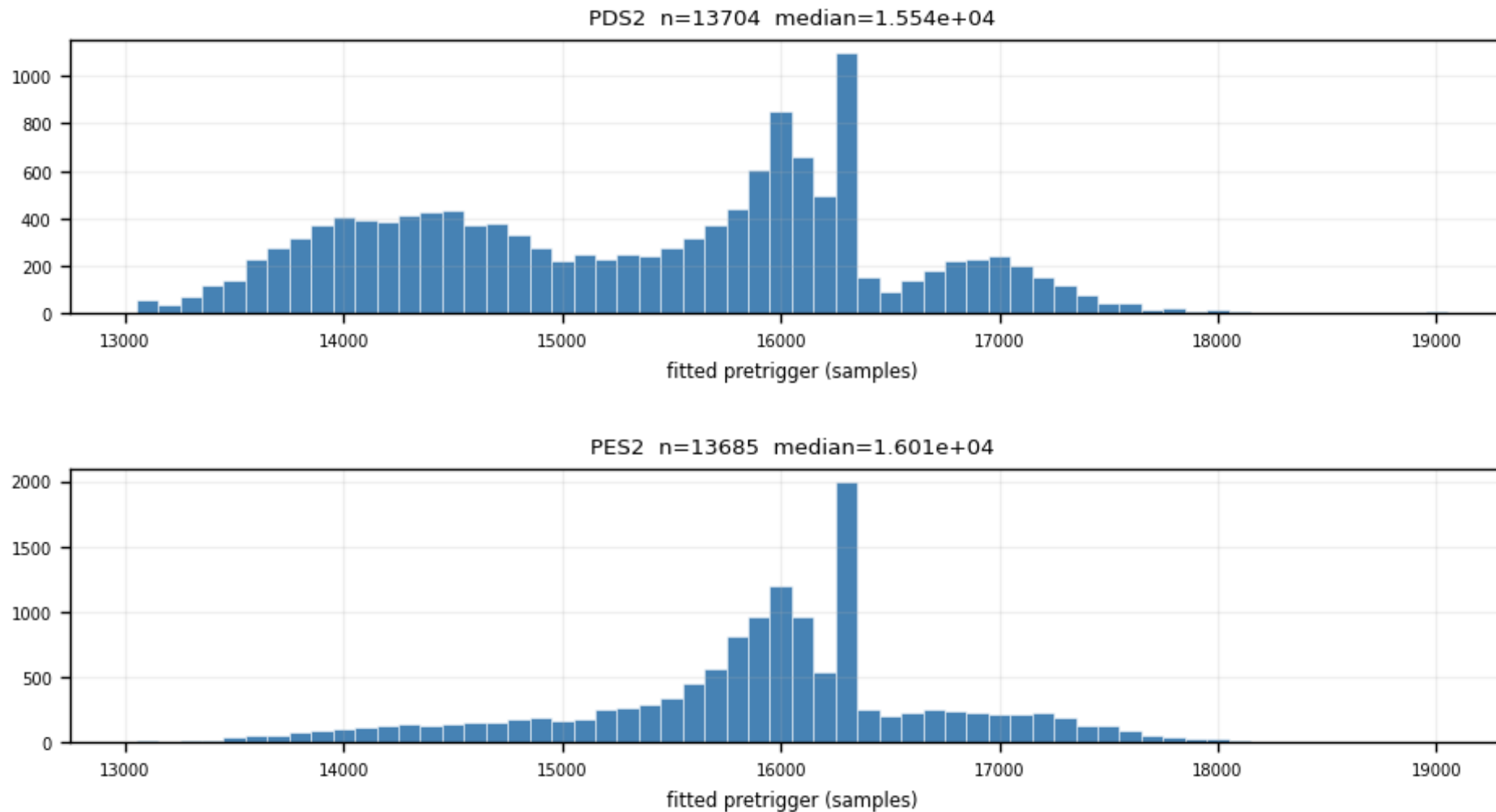
The onset is a FREE fit parameter; pinning it inside the fit would bias every other parameter.



Z4 · pretrigger: fitted onset per channel (5/6)

figure: zip4_pretrigger.png, channel panels 9–10 of 11, top to bottom

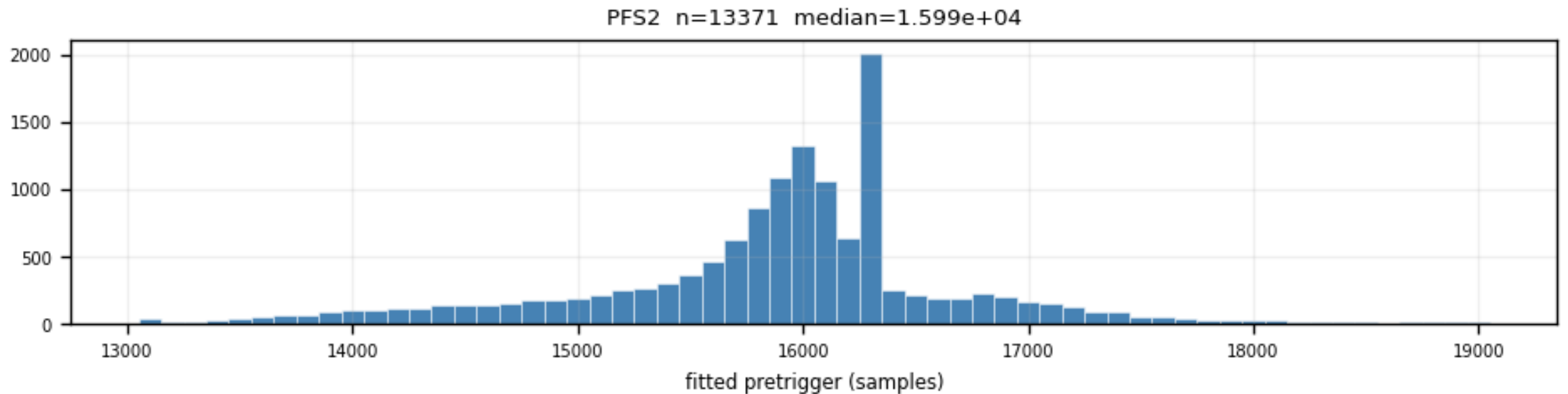
The onset is a FREE fit parameter; pinning it inside the fit would bias every other parameter.



Z4 · pretrigger: fitted onset per channel (6/6)

figure: zip4_pretrigger.png, channel panels 11–11 of 11, top to bottom

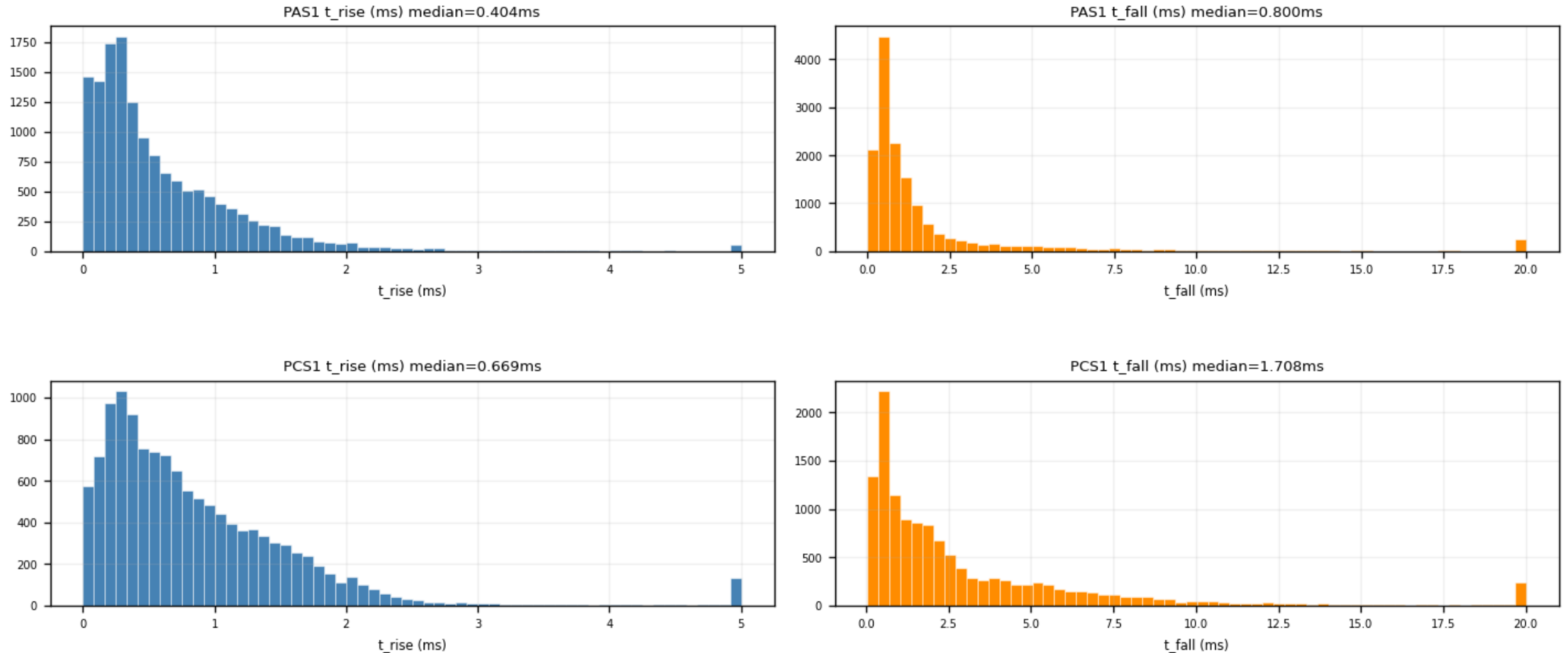
The onset is a FREE fit parameter; pinning it inside the fit would bias every other parameter.



Z4 · time_constants: τ_{rise} / τ_{fall} per channel (1/6)

figure: zip4_time_constants.png, channel panels 1–2 of 11, top to bottom

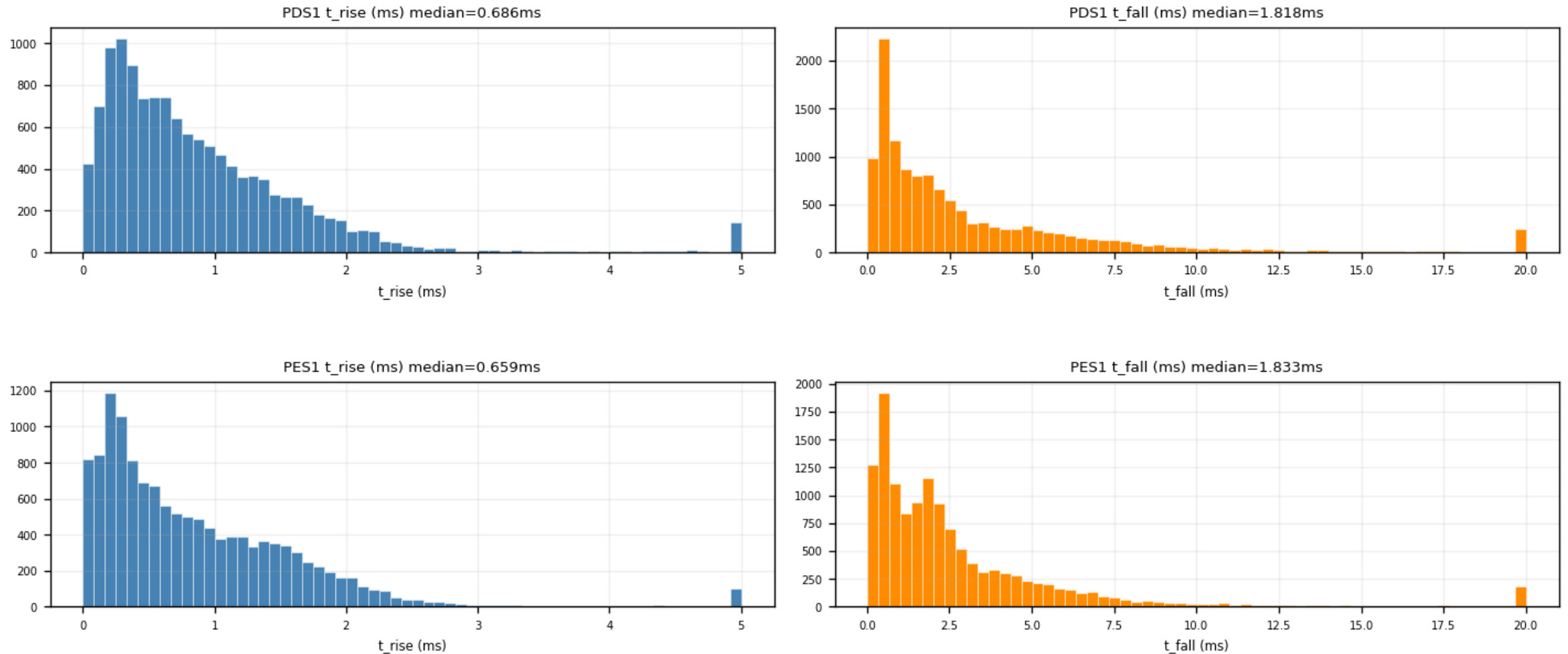
Fast-pulse population at $\tau_{\text{rise}} \approx 0.1$ ms; isolated spikes at the parameter bounds are noise fits pinned at the fit limits, removed by the NRMSE cut.



Z4 · time_constants: τ_{rise} / τ_{fall} per channel (2/6)

figure: zip4_time_constants.png, channel panels 3–4 of 11, top to bottom

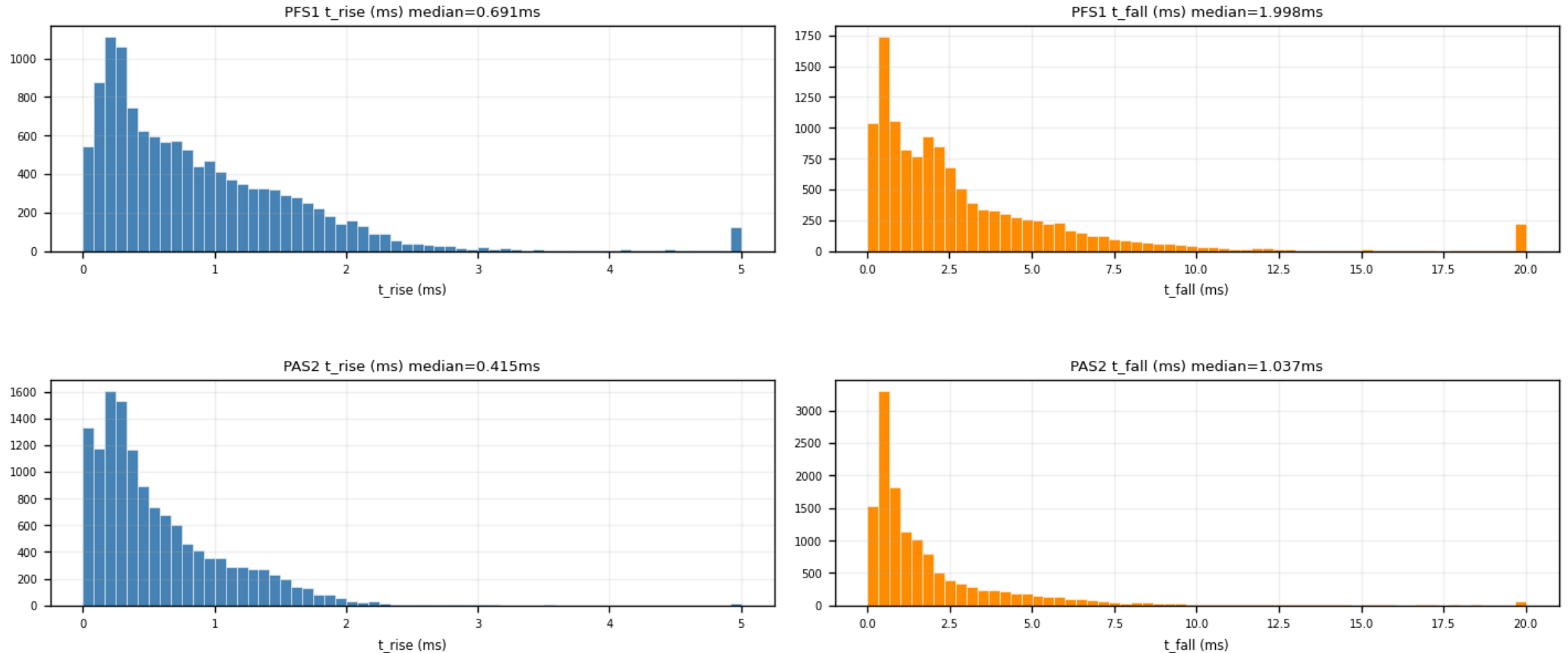
Fast-pulse population at $\tau_{\text{rise}} \approx 0.1$ ms; isolated spikes at the parameter bounds are noise fits pinned at the fit limits, removed by the NRMSE cut.



Z4 · time_constants: τ_{rise} / τ_{fall} per channel (3/6)

figure: zip4_time_constants.png, channel panels 5–6 of 11, top to bottom

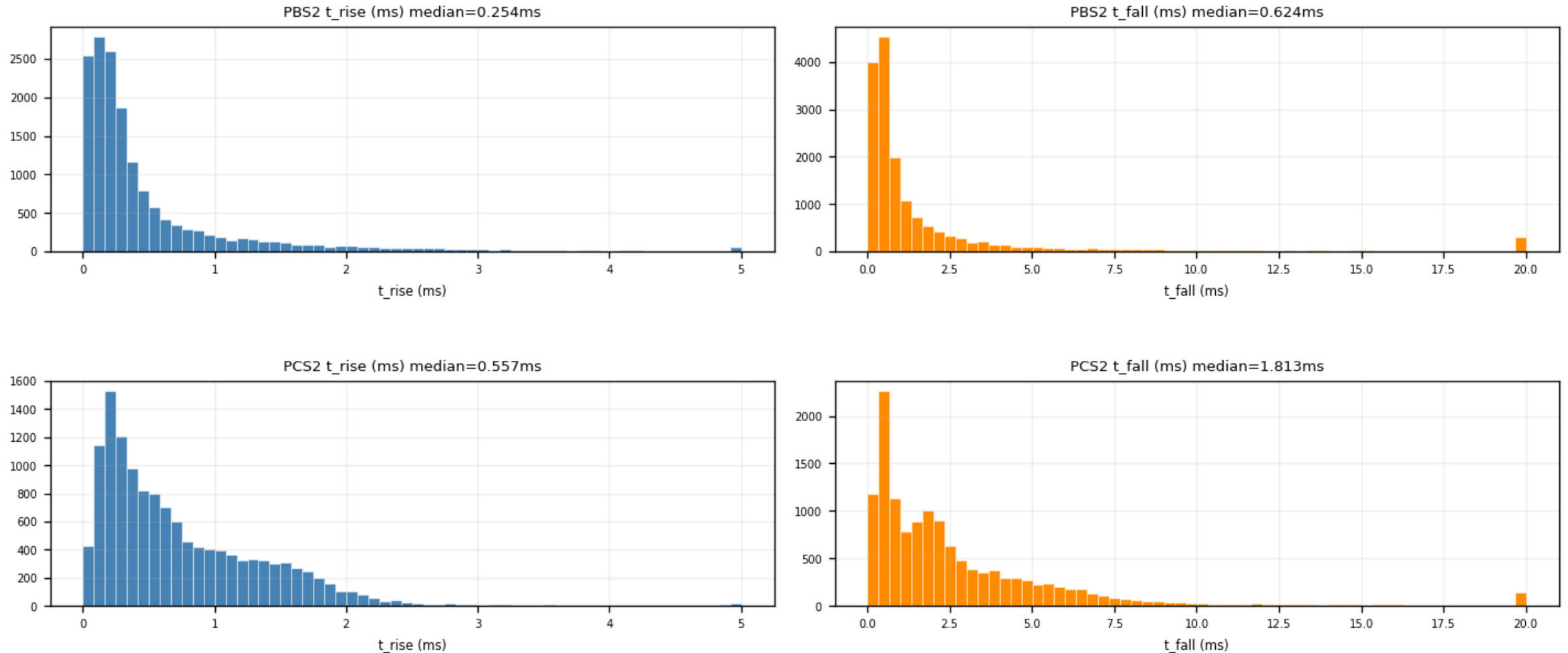
Fast-pulse population at $\tau_{\text{rise}} \approx 0.1$ ms; isolated spikes at the parameter bounds are noise fits pinned at the fit limits, removed by the NRMSE cut.



Z4 · time_constants: τ_{rise} / τ_{fall} per channel (4/6)

figure: zip4_time_constants.png, channel panels 7–8 of 11, top to bottom

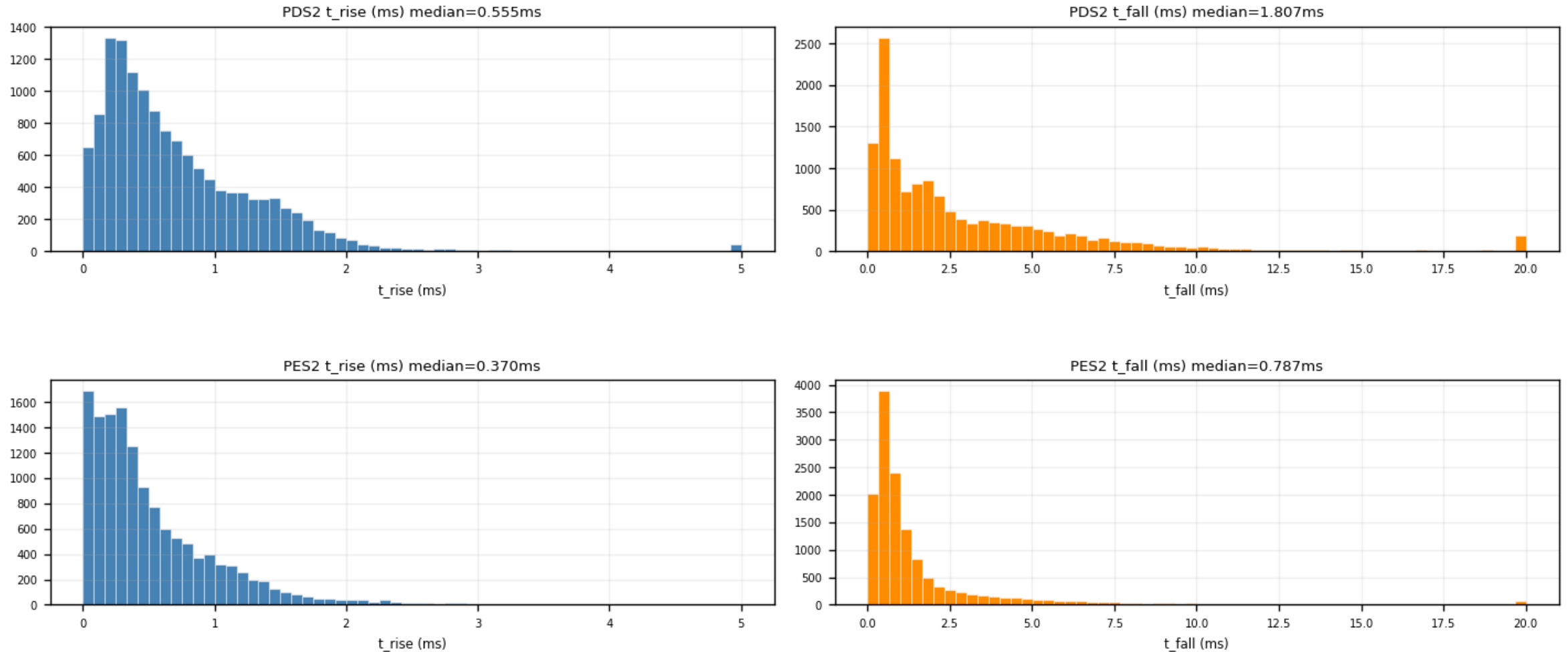
Fast-pulse population at $\tau_{\text{rise}} \approx 0.1$ ms; isolated spikes at the parameter bounds are noise fits pinned at the fit limits, removed by the NRMSE cut.



Z4 · time_constants: τ_{rise} / τ_{fall} per channel (5/6)

figure: zip4_time_constants.png, channel panels 9–10 of 11, top to bottom

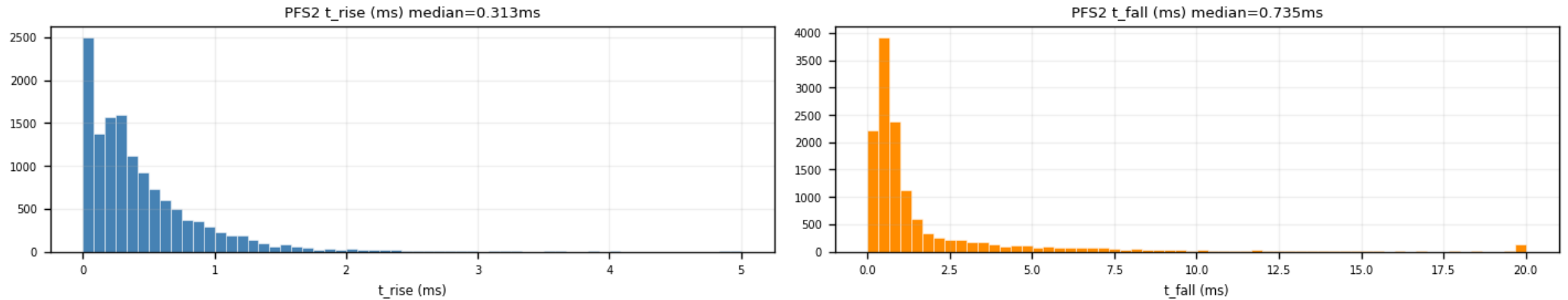
Fast-pulse population at $\tau_{\text{rise}} \approx 0.1$ ms; isolated spikes at the parameter bounds are noise fits pinned at the fit limits, removed by the NRMSE cut.



Z4 · time_constants: τ_{rise} / τ_{fall} per channel (6/6)

figure: zip4_time_constants.png, channel panels 11–11 of 11, top to bottom

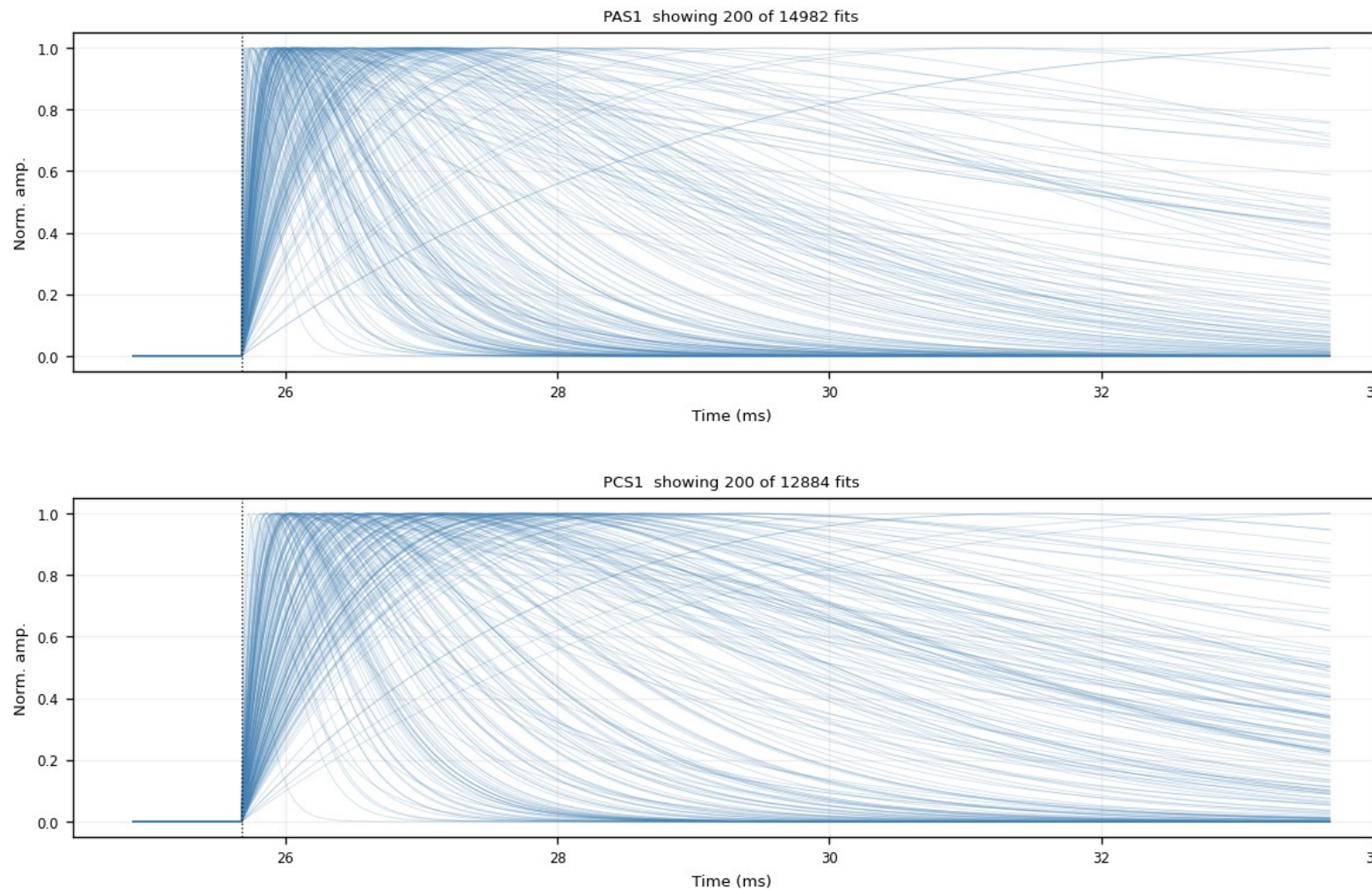
Fast-pulse population at $\tau_{\text{rise}} \approx 0.1$ ms; isolated spikes at the parameter bounds are noise fits pinned at the fit limits, removed by the NRMSE cut.



Z4 · fitted_curves_overlay: fan, no cut (1/6)

figure: zip4_fitted_curves_overlay.png, channel panels 1–2 of 11, top to bottom

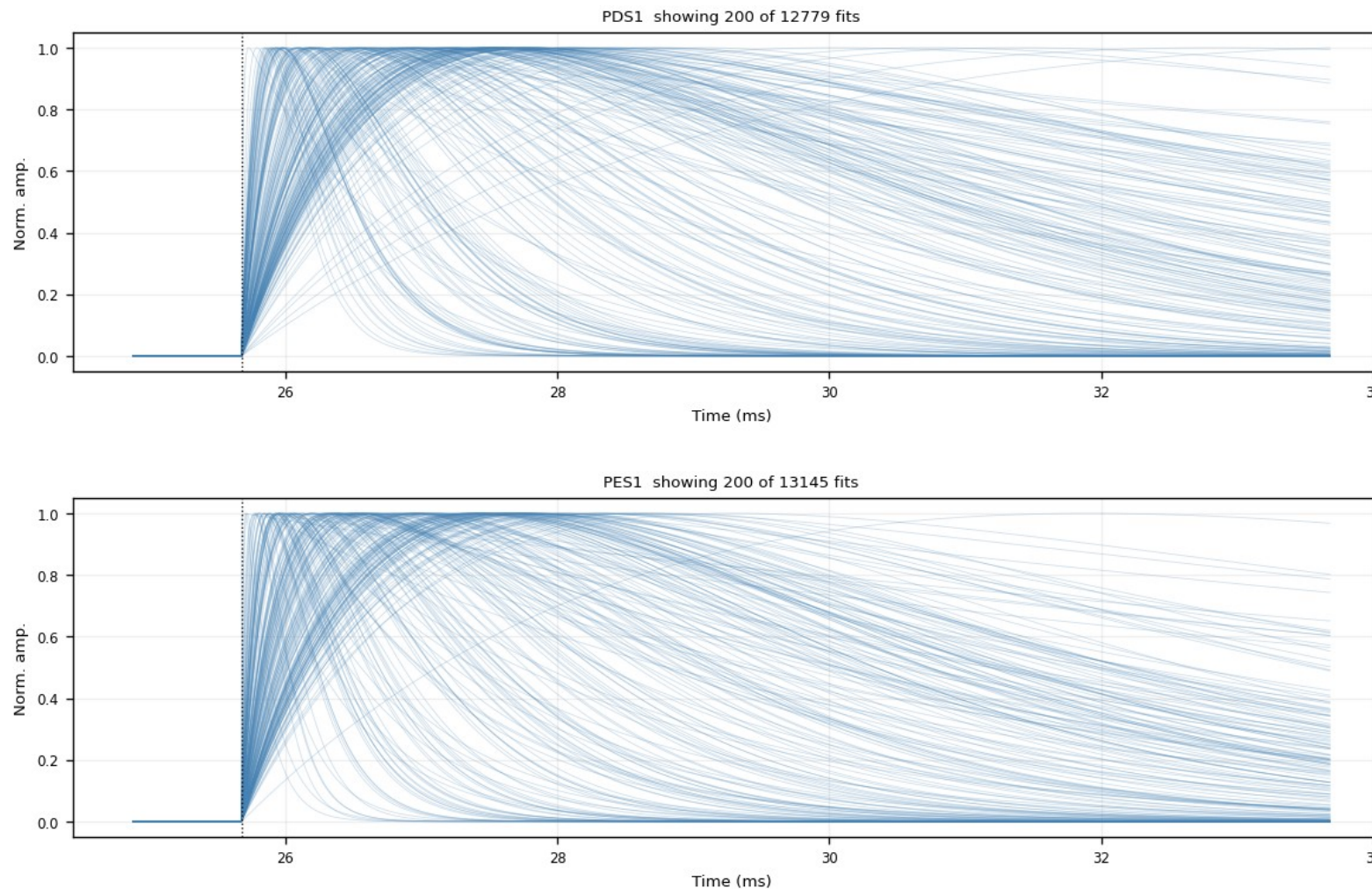
Every fit_ok fitted curve at the common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.



Z4 · fitted_curves_overlay: fan, no cut (2/6)

figure: zip4_fitted_curves_overlay.png, channel panels 3–4 of 11, top to bottom

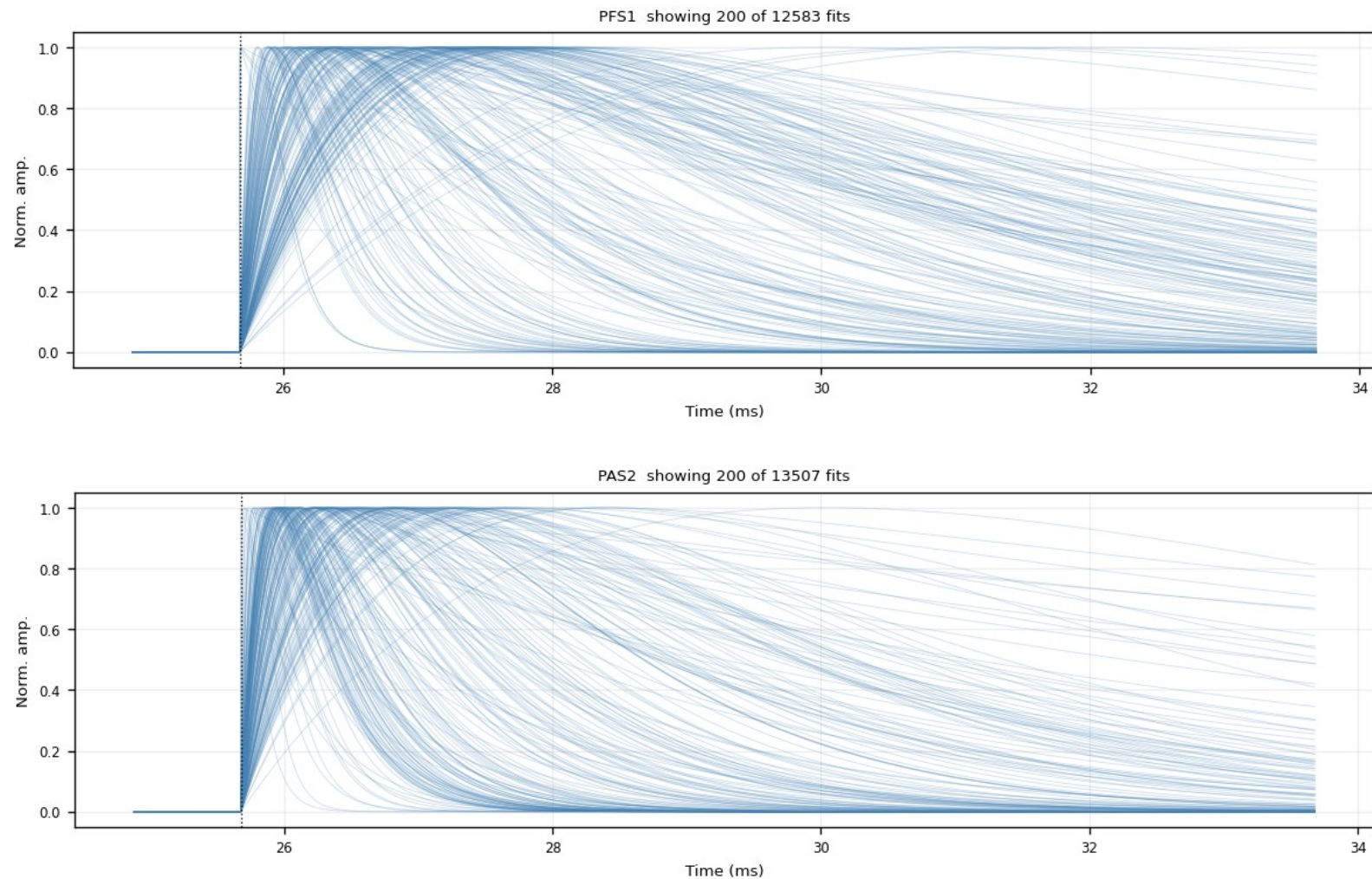
Every fit_ok fitted curve at the common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.



Z4 · fitted_curves_overlay: fan, no cut (3/6)

figure: zip4_fitted_curves_overlay.png, channel panels 5–6 of 11, top to bottom

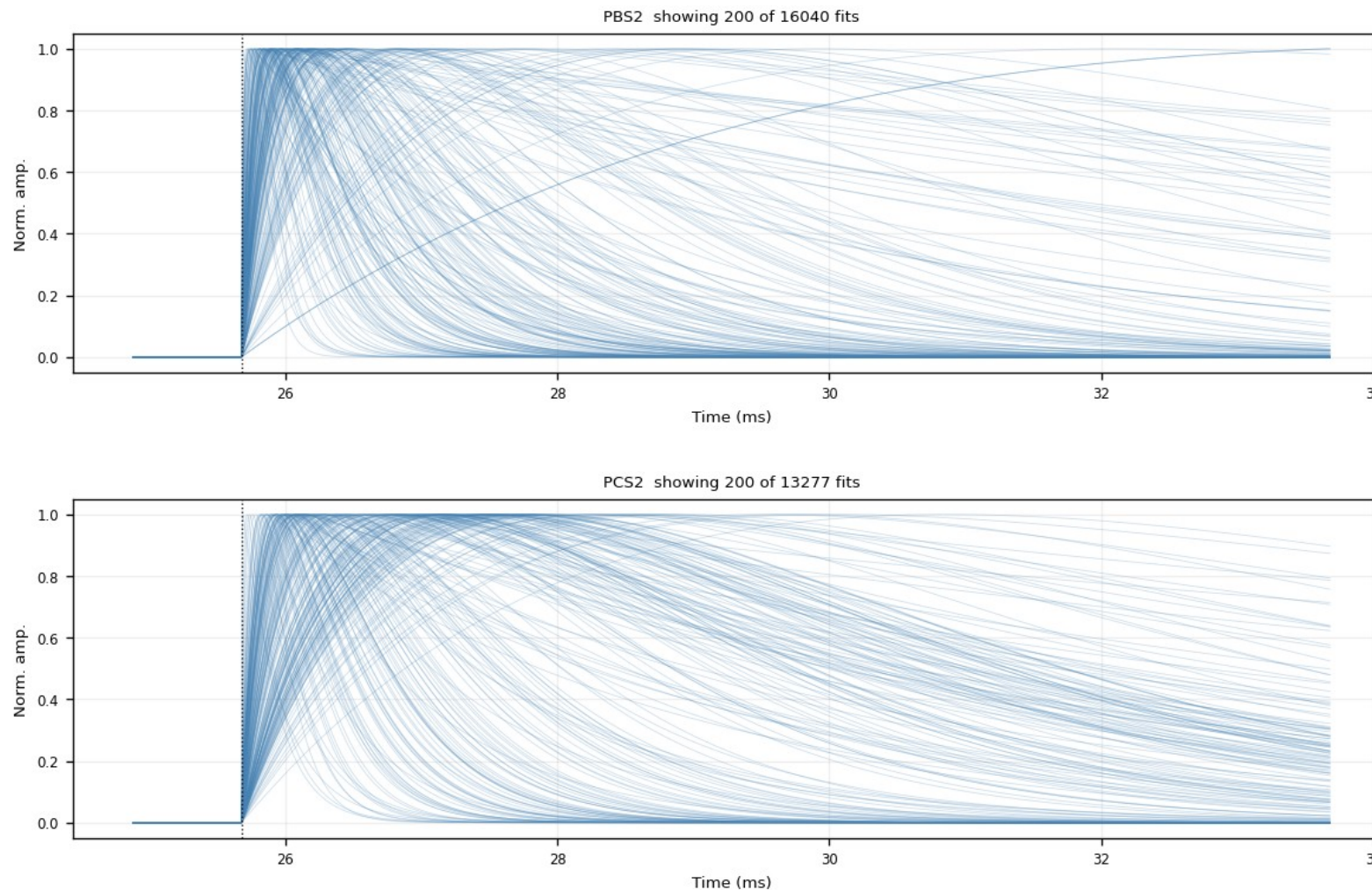
Every fit_ok fitted curve at the common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.



Z4 · fitted_curves_overlay: fan, no cut (4/6)

figure: zip4_fitted_curves_overlay.png, channel panels 7–8 of 11, top to bottom

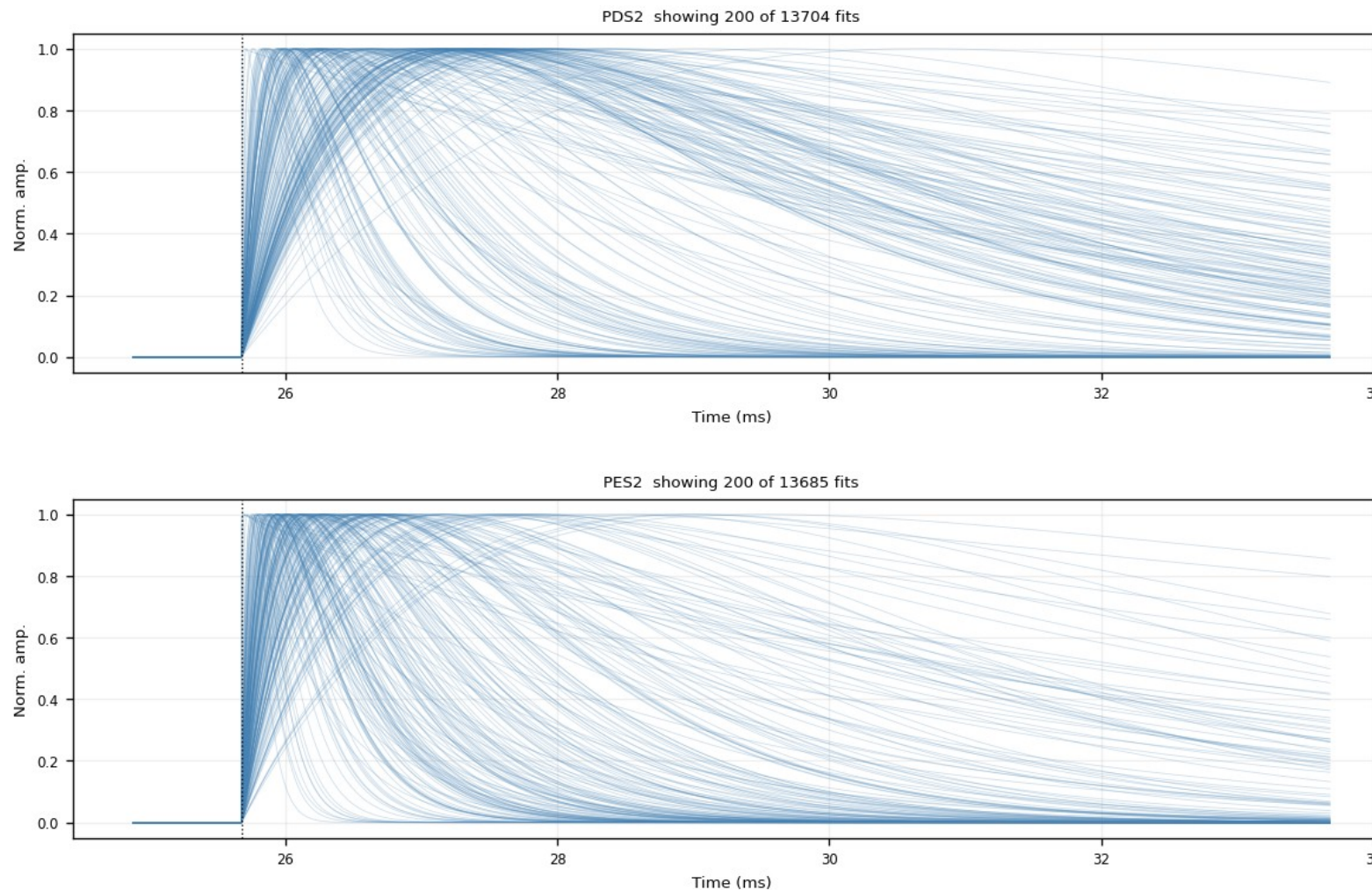
Every fit_ok fitted curve at the common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.



Z4 · fitted_curves_overlay: fan, no cut (5/6)

figure: zip4_fitted_curves_overlay.png, channel panels 9–10 of 11, top to bottom

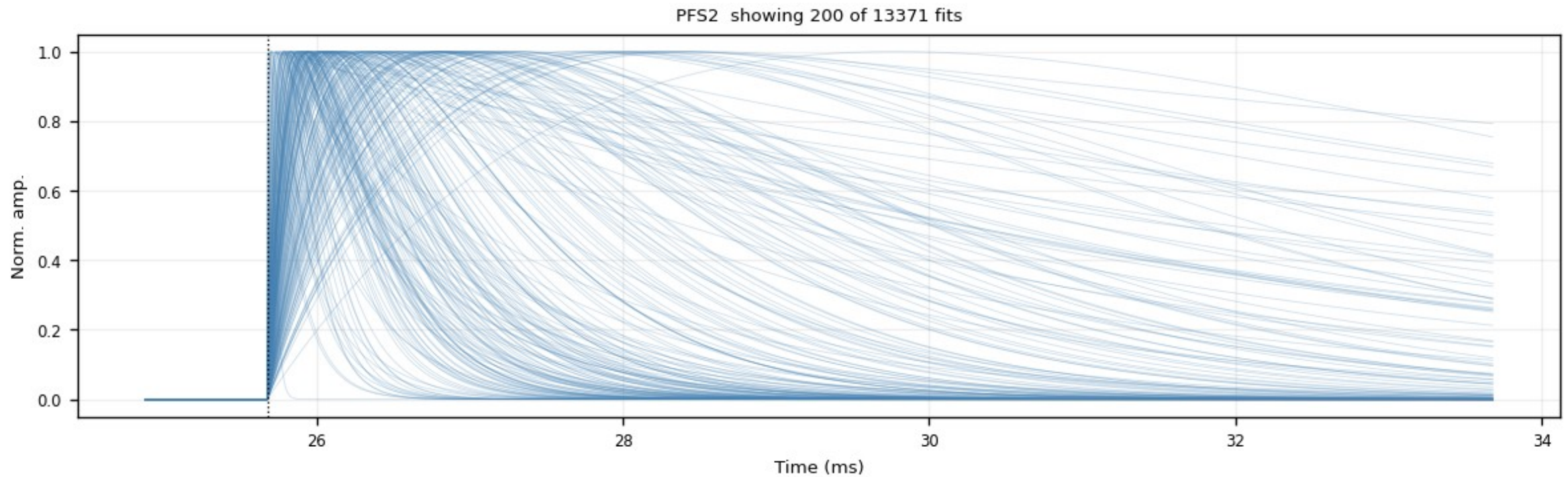
Every fit_ok fitted curve at the common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.



Z4 · fitted_curves_overlay: fan, no cut (6/6)

figure: zip4_fitted_curves_overlay.png, channel panels 11–11 of 11, top to bottom

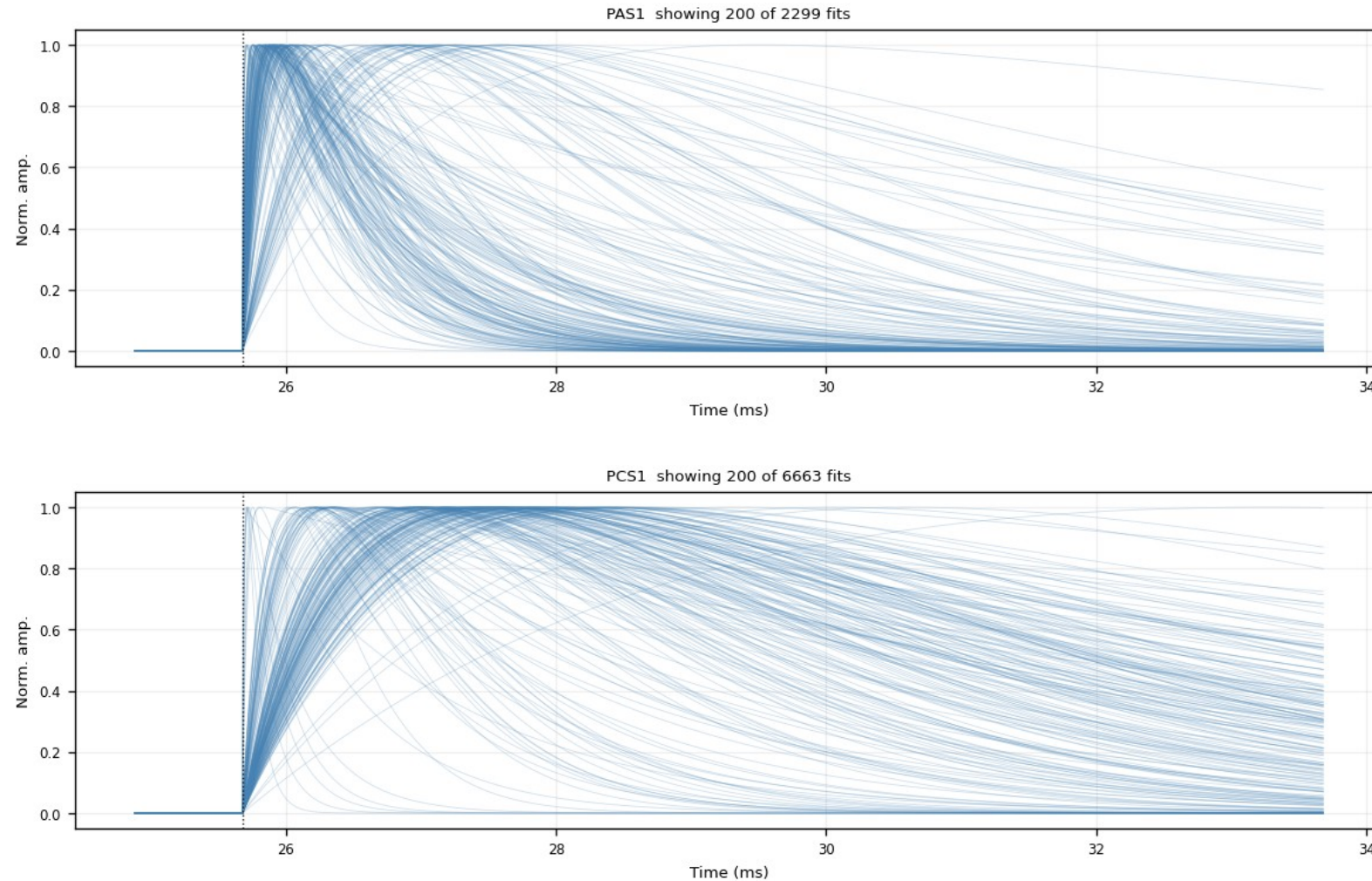
Every fit_ok fitted curve at the common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.



Z4 · fitted_curves_overlay: after $\text{NRMSE} \leq 0.4$ (1/6)

figure: zip4_fitted_curves_overlay_nrmse0.4.png, channel panels 1–2 of 11, top to bottom

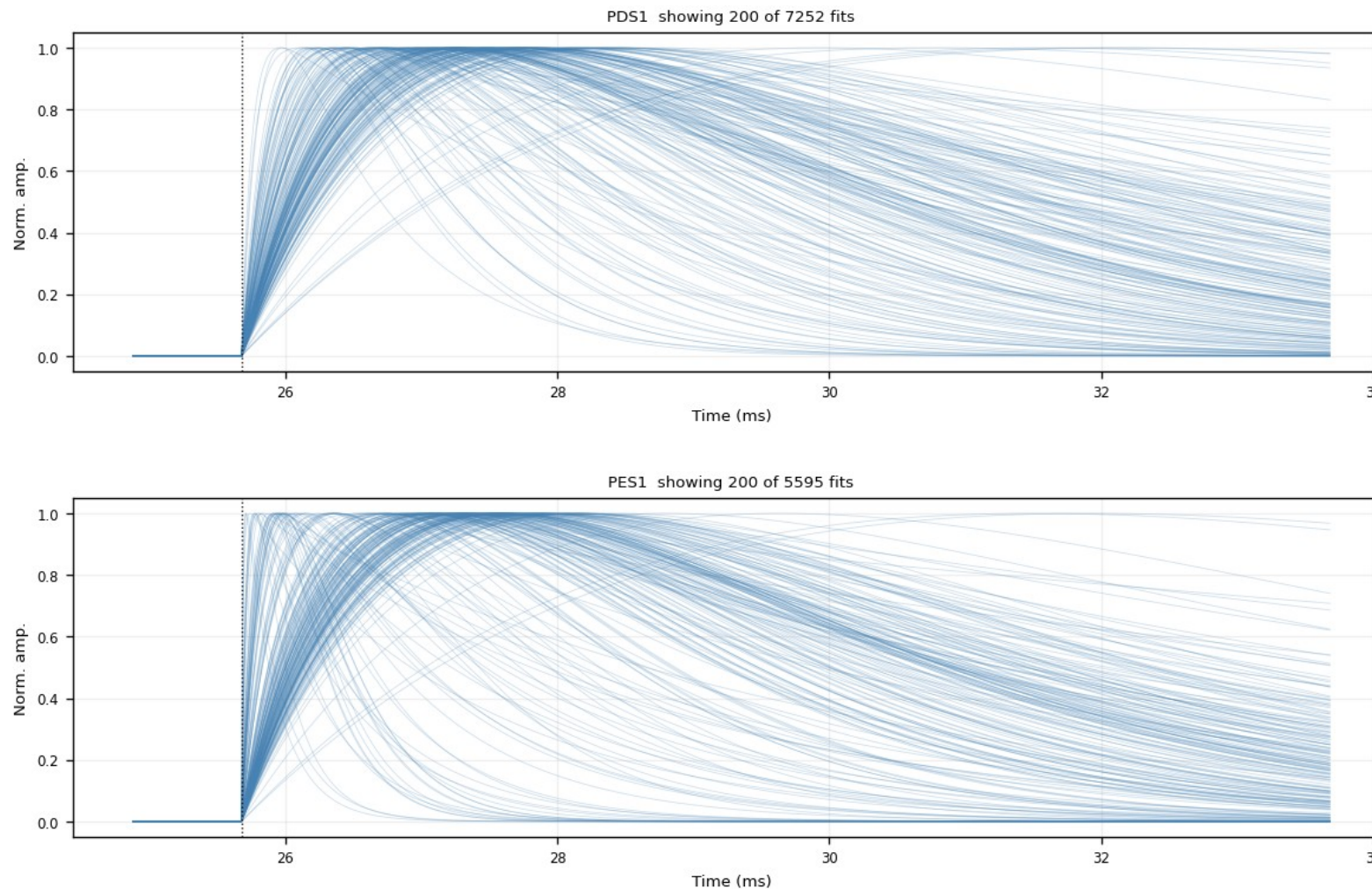
Same fan after quality cut 1: the noise fan collapses, the fast-pulse bundle remains.



Z4 · fitted_curves_overlay: after $\text{NRMSE} \leq 0.4$ (2/6)

figure: zip4_fitted_curves_overlay_nrmse0.4.png, channel panels 3–4 of 11, top to bottom

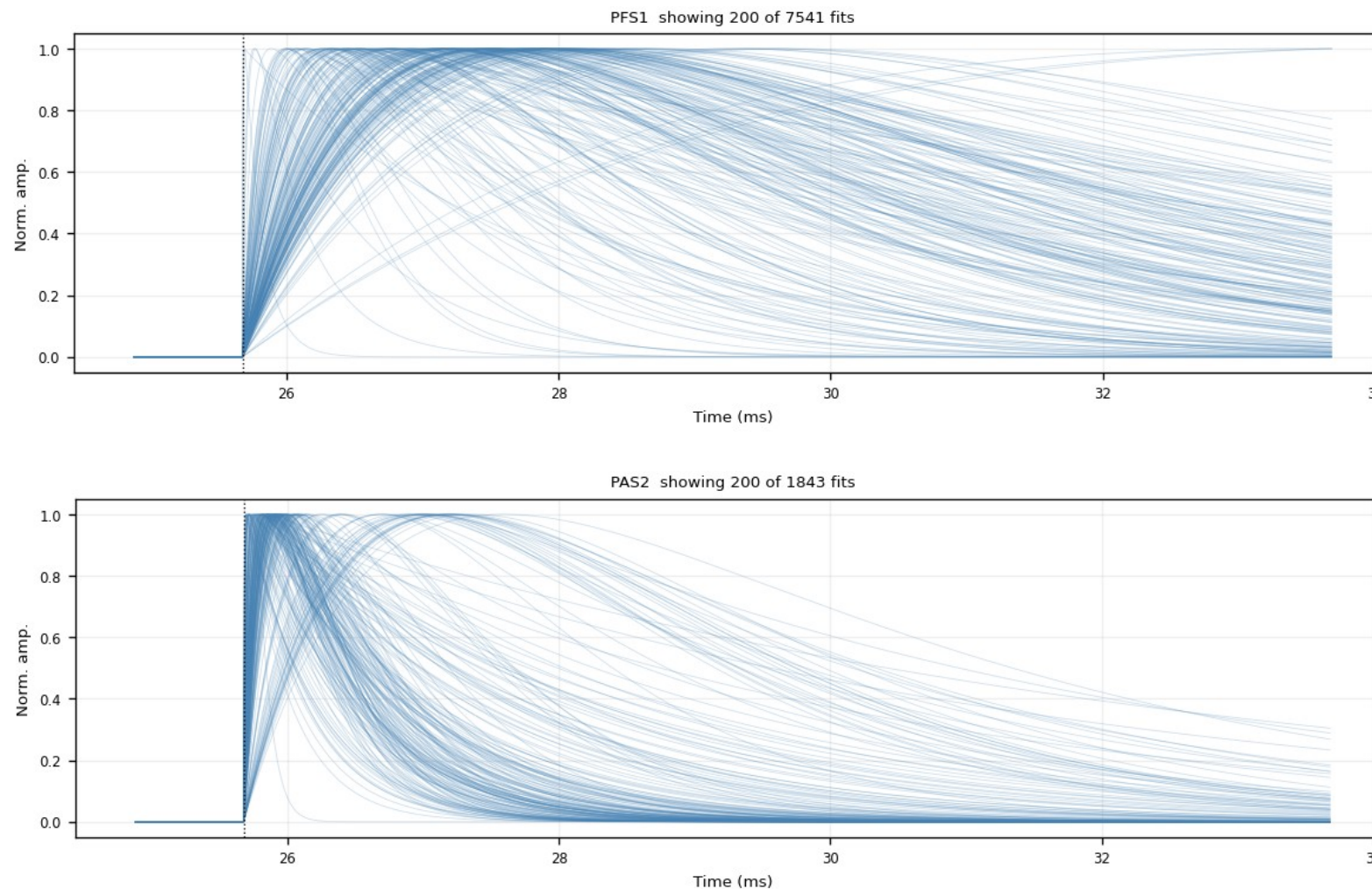
Same fan after quality cut 1: the noise fan collapses, the fast-pulse bundle remains.



Z4 · fitted_curves_overlay: after $\text{NRMSE} \leq 0.4$ (3/6)

figure: zip4_fitted_curves_overlay_nrmse0.4.png, channel panels 5–6 of 11, top to bottom

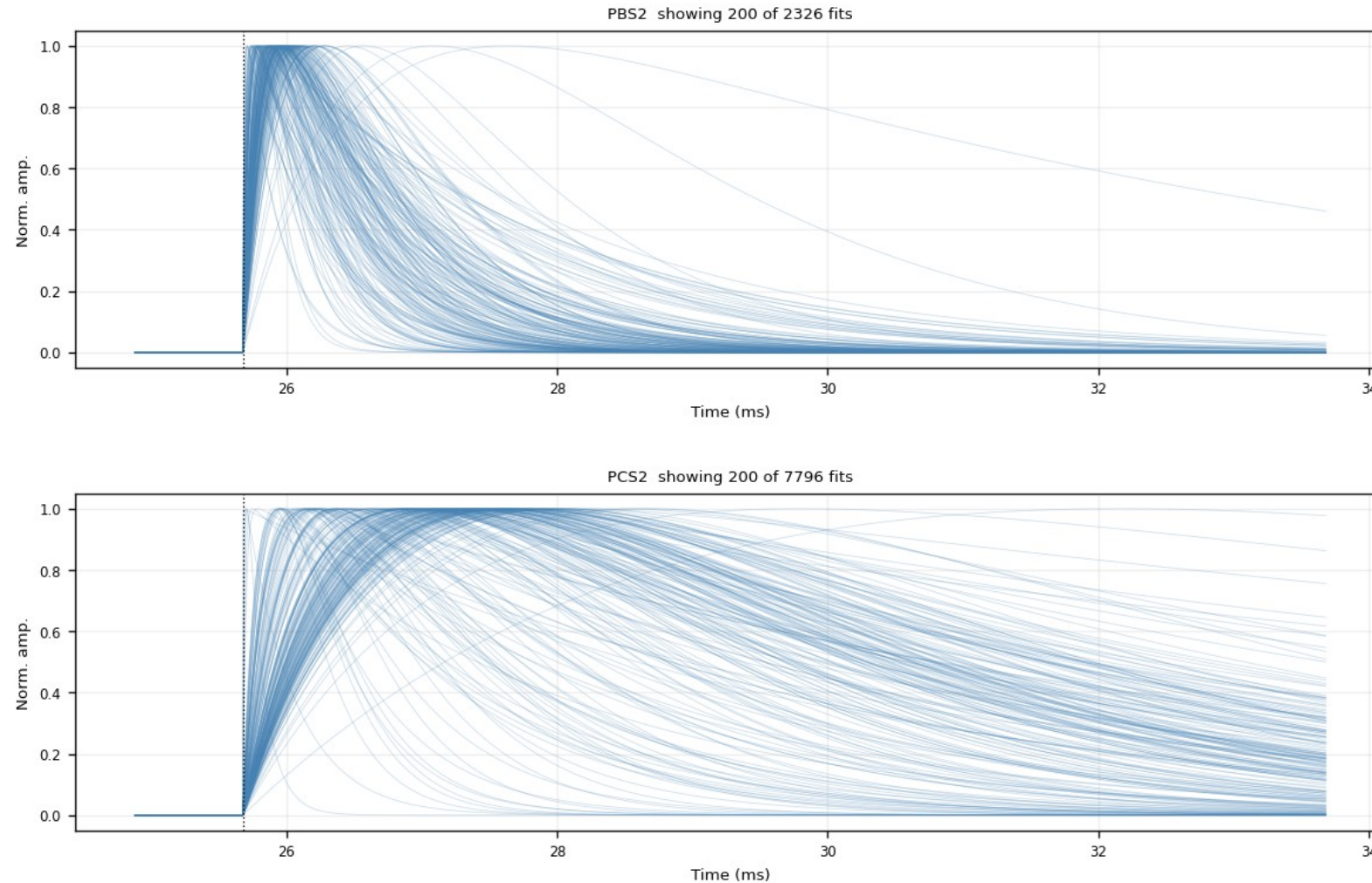
Same fan after quality cut 1: the noise fan collapses, the fast-pulse bundle remains.



Z4 · fitted_curves_overlay: after $\text{NRMSE} \leq 0.4$ (4/6)

figure: zip4_fitted_curves_overlay_nrmse0.4.png, channel panels 7–8 of 11, top to bottom

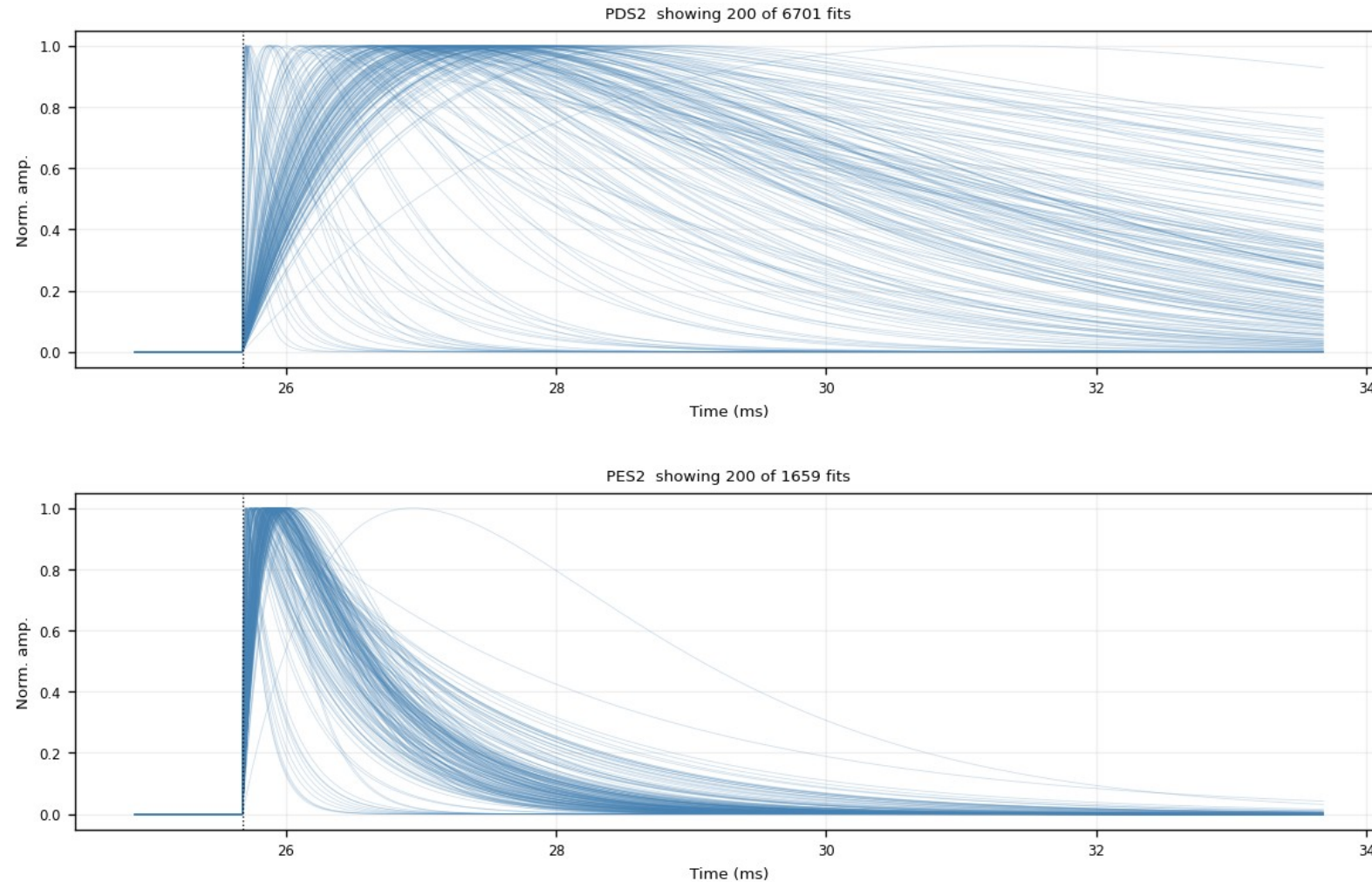
Same fan after quality cut 1: the noise fan collapses, the fast-pulse bundle remains.



Z4 · fitted_curves_overlay: after $\text{NRMSE} \leq 0.4$ (5/6)

figure: zip4_fitted_curves_overlay_nrmse0.4.png, channel panels 9–10 of 11, top to bottom

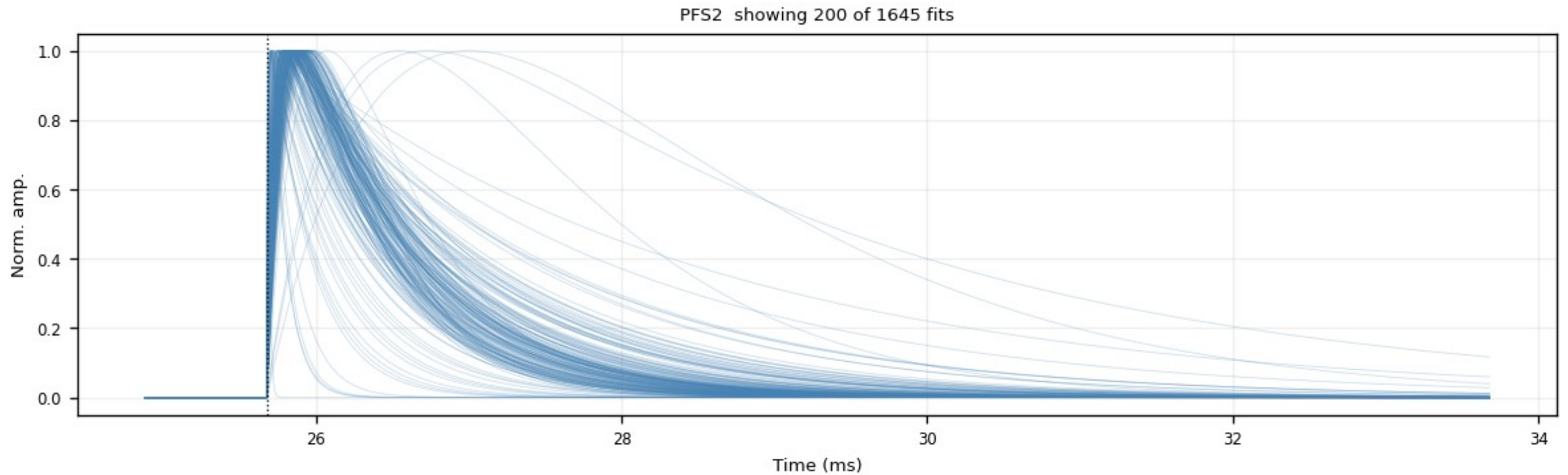
Same fan after quality cut 1: the noise fan collapses, the fast-pulse bundle remains.



Z4 · fitted_curves_overlay: after $\text{NRMSE} \leq 0.4$ (6/6)

figure: zip4_fitted_curves_overlay_nrmse0.4.png, channel panels 11–11 of 11, top to bottom

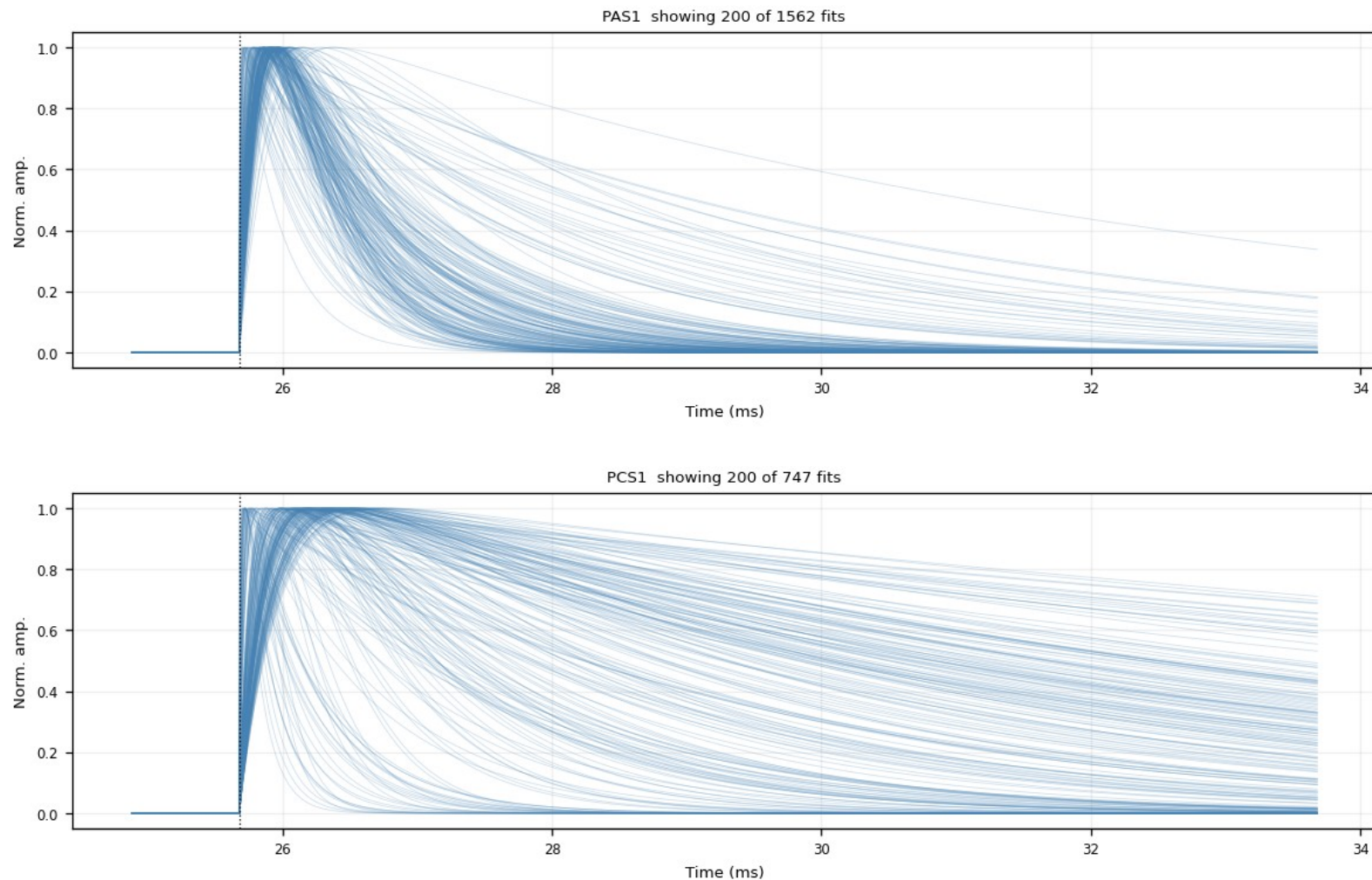
Same fan after quality cut 1: the noise fan collapses, the fast-pulse bundle remains.



Z4 · fitted_curves_overlay: after both cuts (1/6)

figure: zip4_fitted_curves_overlay_nrmse0.4_trise0.30ms.png, channel panels 1–2 of 11, top to bottom

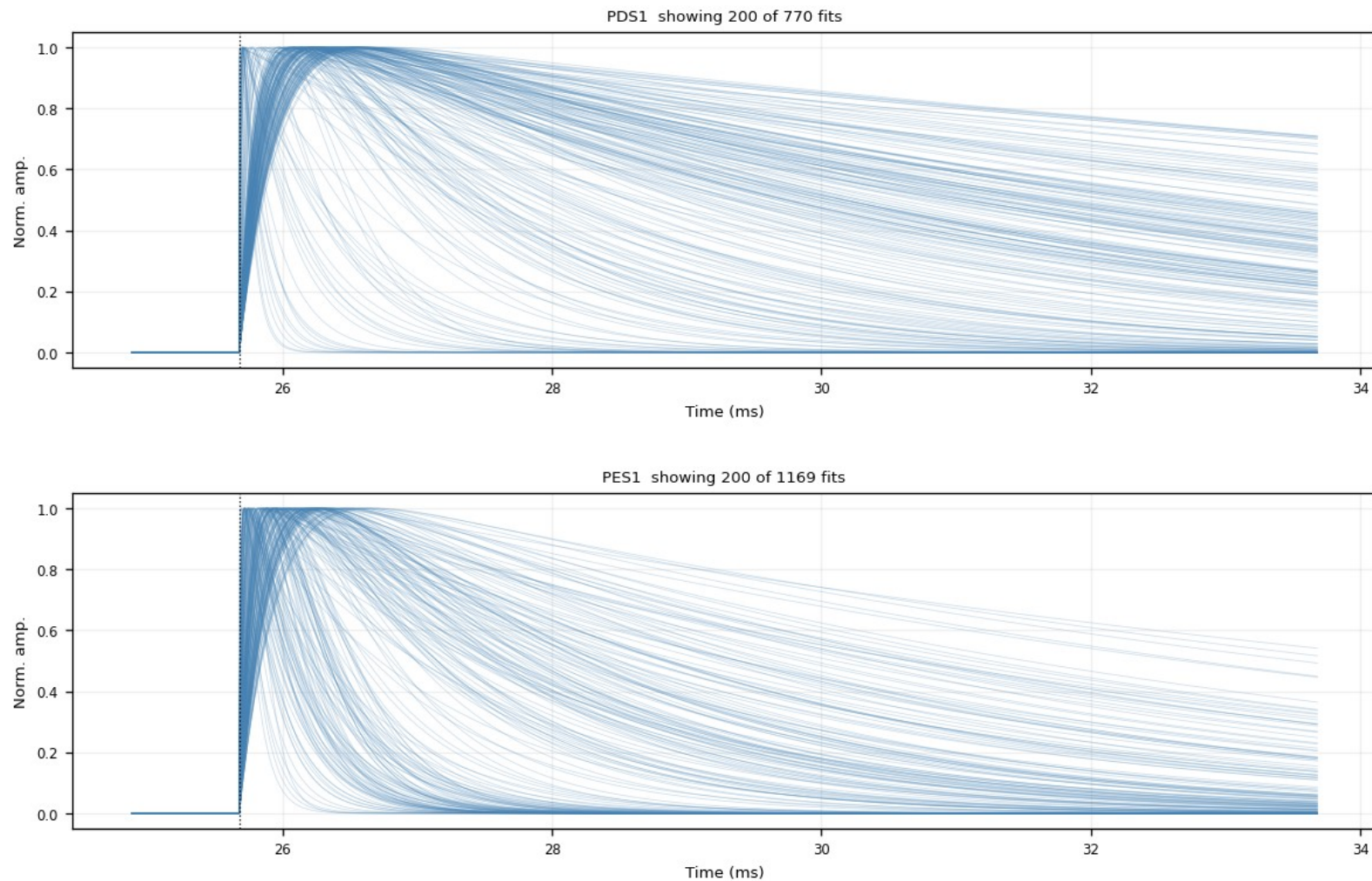
After $\text{NRMSE} \leq 0.4$ AND $\tau_{\text{rise}} \leq 0.3$ ms, the exact PCA input population: the clean fast-pulse bundle only.



Z4 · fitted_curves_overlay: after both cuts (2/6)

figure: zip4_fitted_curves_overlay_nrmse0.4_trise0.30ms.png, channel panels 3–4 of 11, top to bottom

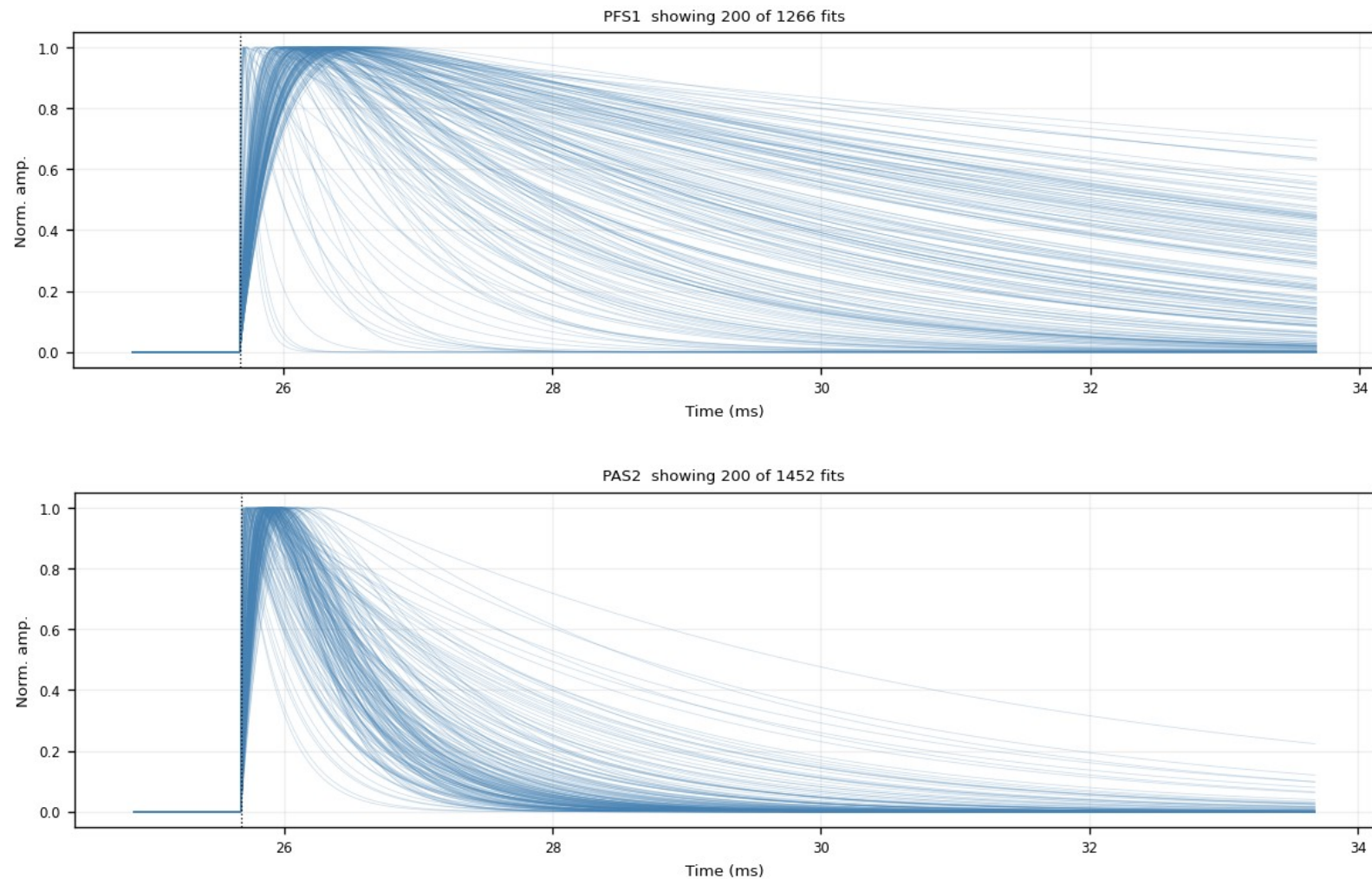
After $\text{NRMSE} \leq 0.4$ AND $\tau_{\text{rise}} \leq 0.3$ ms, the exact PCA input population: the clean fast-pulse bundle only.



Z4 · fitted_curves_overlay: after both cuts (3/6)

figure: zip4_fitted_curves_overlay_nrmse0.4_trise0.30ms.png, channel panels 5–6 of 11, top to bottom

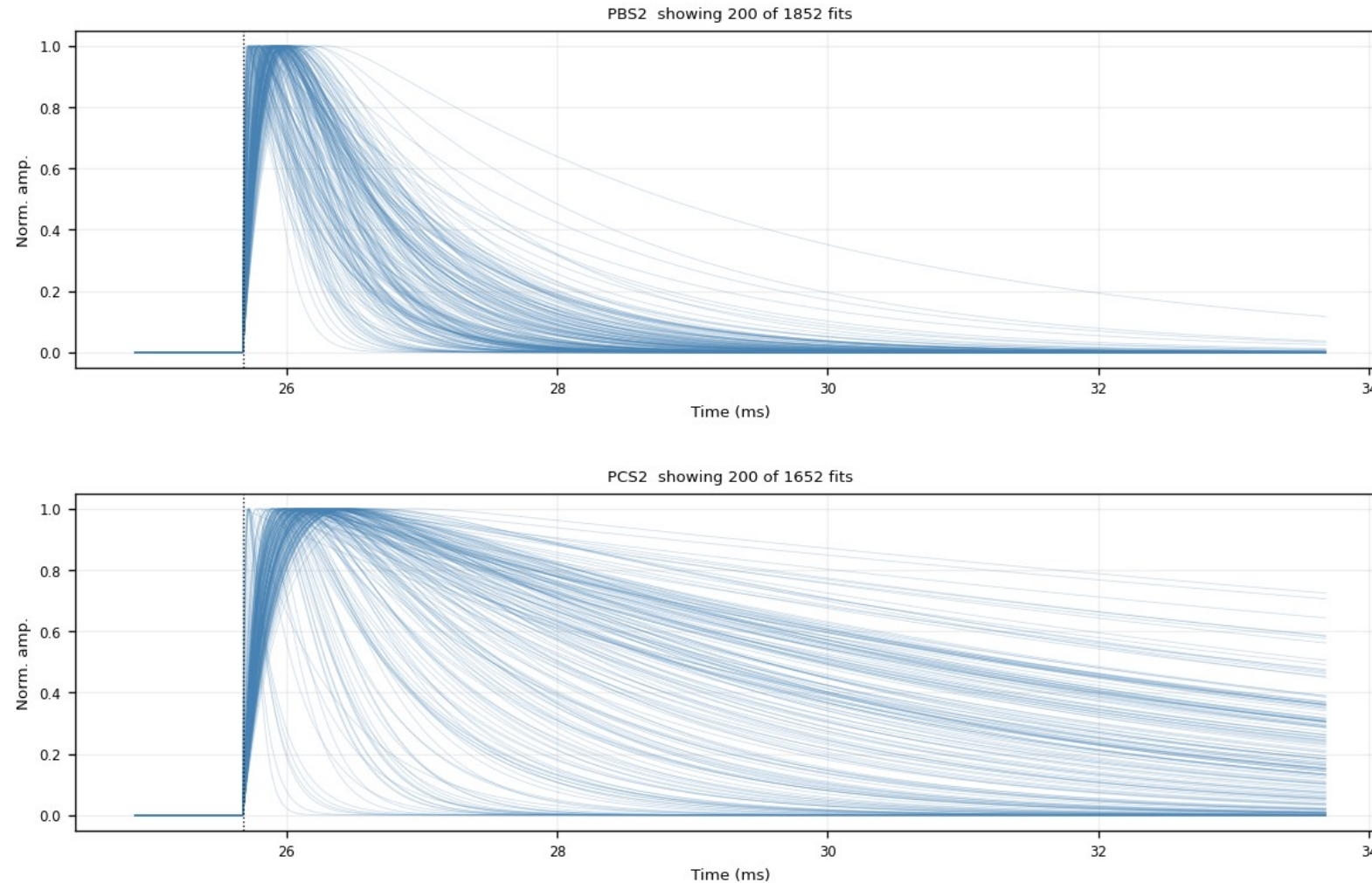
After $\text{NRMSE} \leq 0.4$ AND $\tau_{\text{rise}} \leq 0.3$ ms, the exact PCA input population: the clean fast-pulse bundle only.



Z4 · fitted_curves_overlay: after both cuts (4/6)

figure: zip4_fitted_curves_overlay_nrmse0.4_trise0.30ms.png, channel panels 7–8 of 11, top to bottom

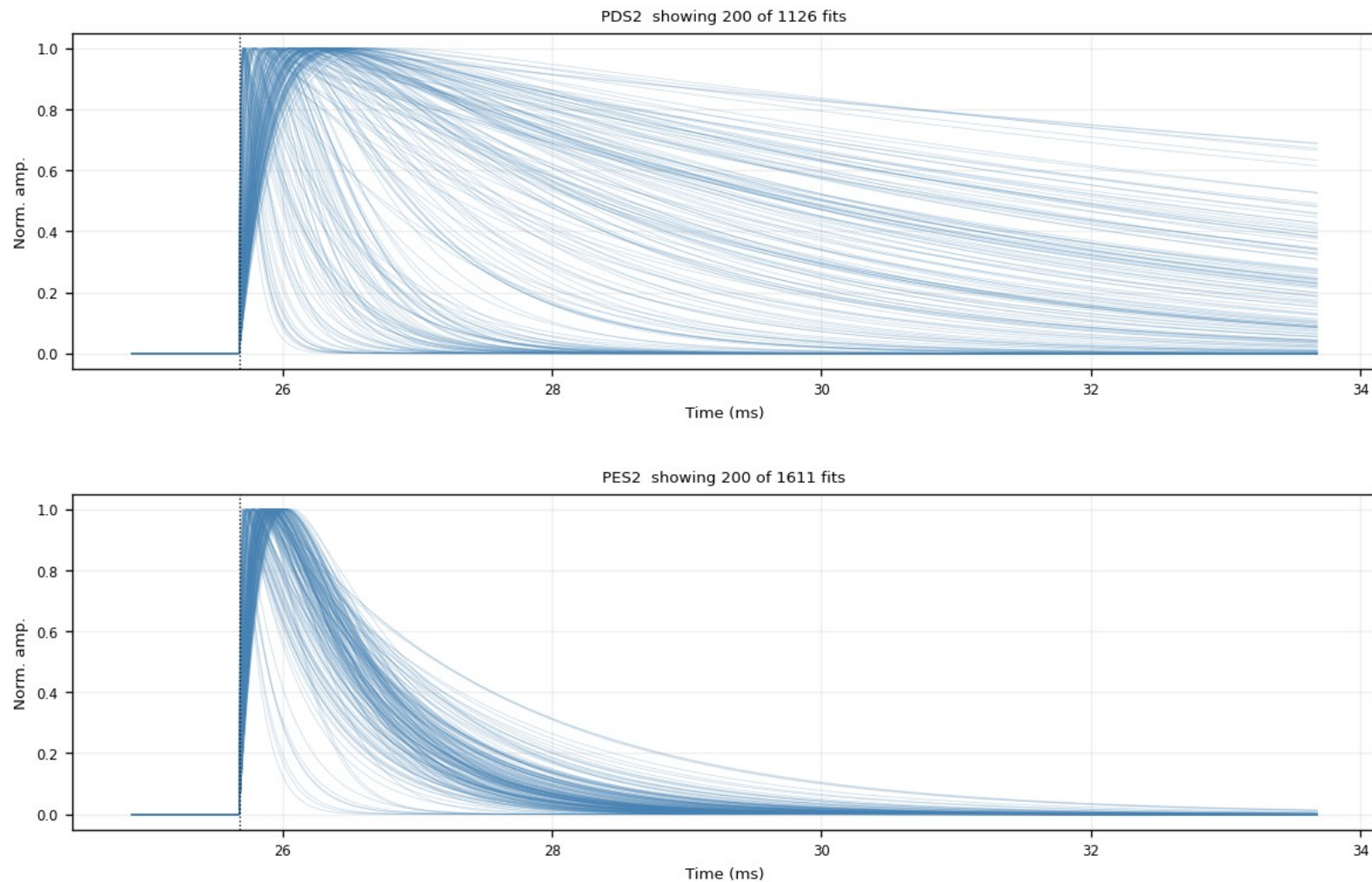
After $\text{NRMSE} \leq 0.4$ AND $\tau_{\text{rise}} \leq 0.3$ ms, the exact PCA input population: the clean fast-pulse bundle only.



Z4 · fitted_curves_overlay: after both cuts (5/6)

figure: zip4_fitted_curves_overlay_nrmse0.4_trise0.30ms.png, channel panels 9–10 of 11, top to bottom

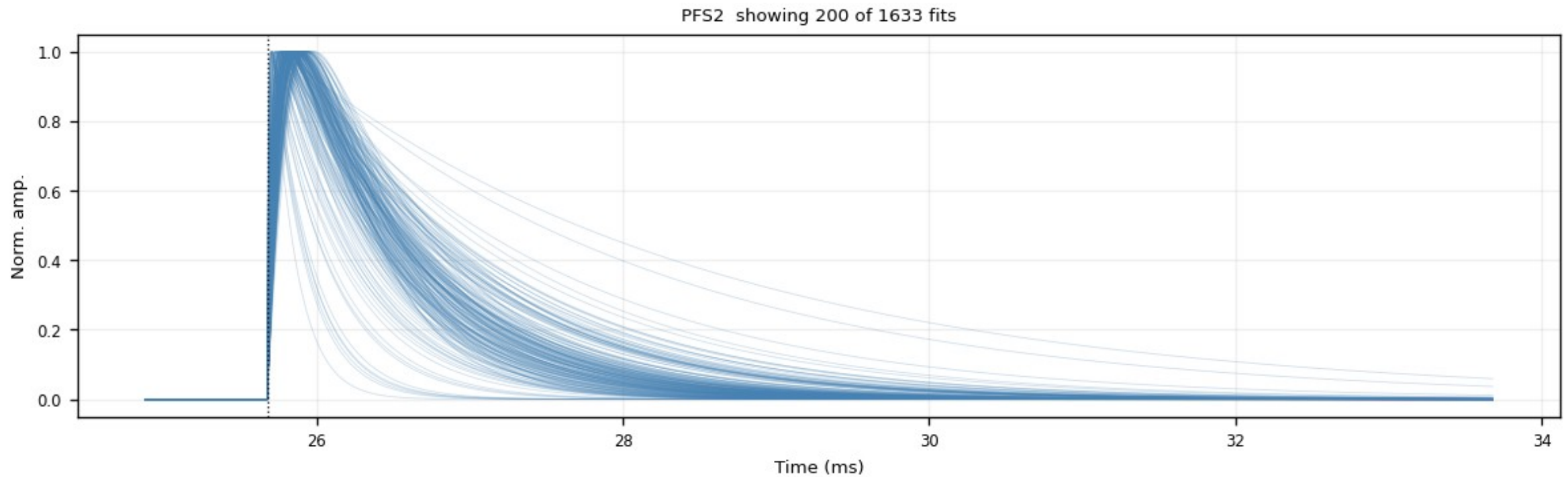
After $\text{NRMSE} \leq 0.4$ AND $\tau_{\text{rise}} \leq 0.3$ ms, the exact PCA input population: the clean fast-pulse bundle only.



Z4 · fitted_curves_overlay: after both cuts (6/6)

figure: zip4_fitted_curves_overlay_nrmse0.4_trise0.30ms.png, channel panels 11–11 of 11, top to bottom

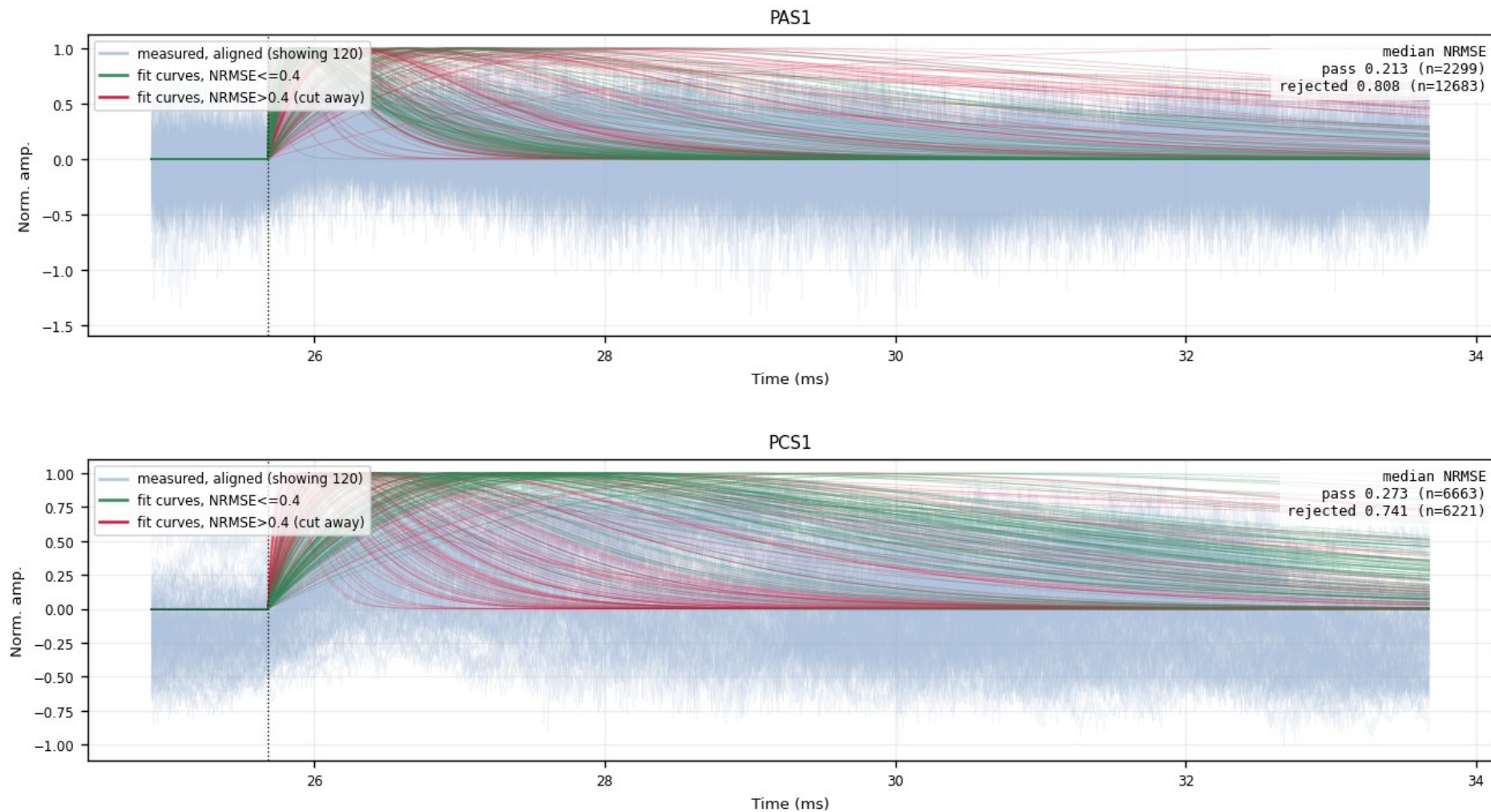
After $\text{NRMSE} \leq 0.4$ AND $\tau_{\text{rise}} \leq 0.3$ ms, the exact PCA input population: the clean fast-pulse bundle only.



Z4 · overlay_fan_cut: kept vs rejected fits (1/6)

figure: zip4_overlay_fan_cut_nrmse0.4.png, channel panels 1–2 of 11, top to bottom

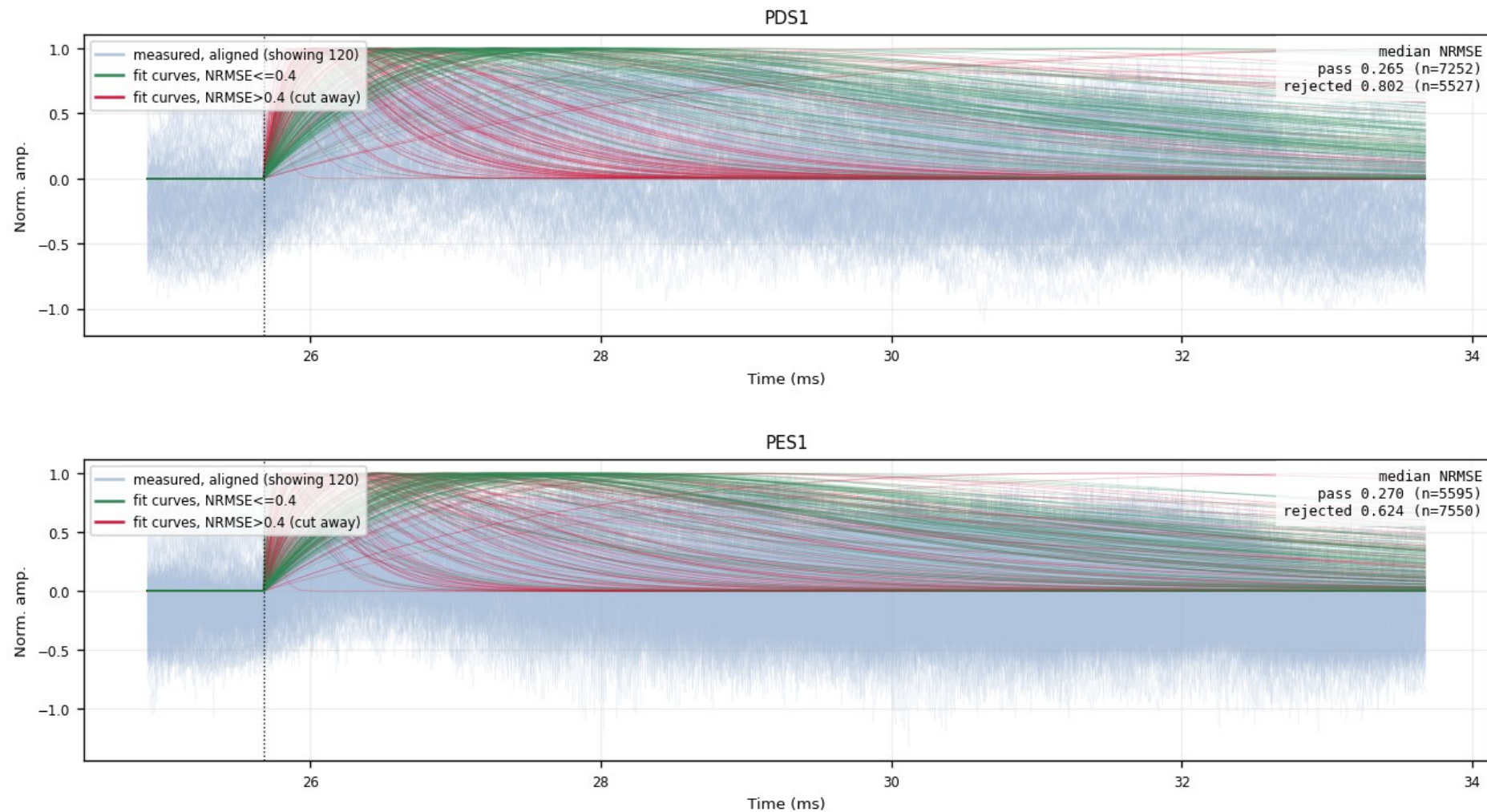
Gray-blue = aligned measured traces; green = fits with $\text{NRMSE} \leq 0.4$ (kept); red = $\text{NRMSE} > 0.4$ (cut away). Median NRMSE and counts of both populations stamped per channel.



Z4 · overlay_fan_cut: kept vs rejected fits (2/6)

figure: zip4_overlay_fan_cut_nrmse0.4.png, channel panels 3–4 of 11, top to bottom

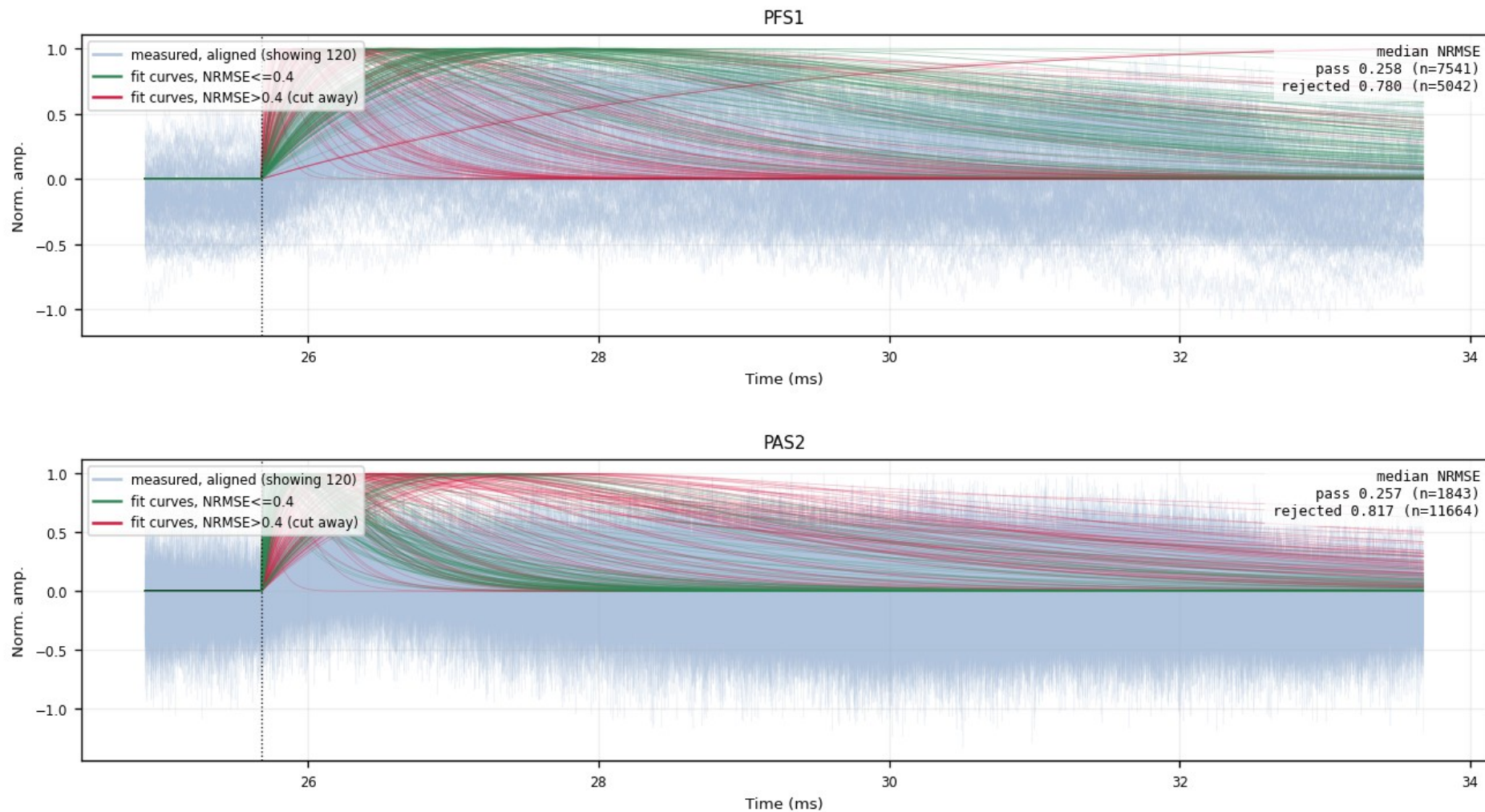
Gray-blue = aligned measured traces; green = fits with $\text{NRMSE} \leq 0.4$ (kept); red = $\text{NRMSE} > 0.4$ (cut away). Median NRMSE and counts of both populations stamped per channel.



Z4 · overlay_fan_cut: kept vs rejected fits (3/6)

figure: zip4_overlay_fan_cut_nrmse0.4.png, channel panels 5–6 of 11, top to bottom

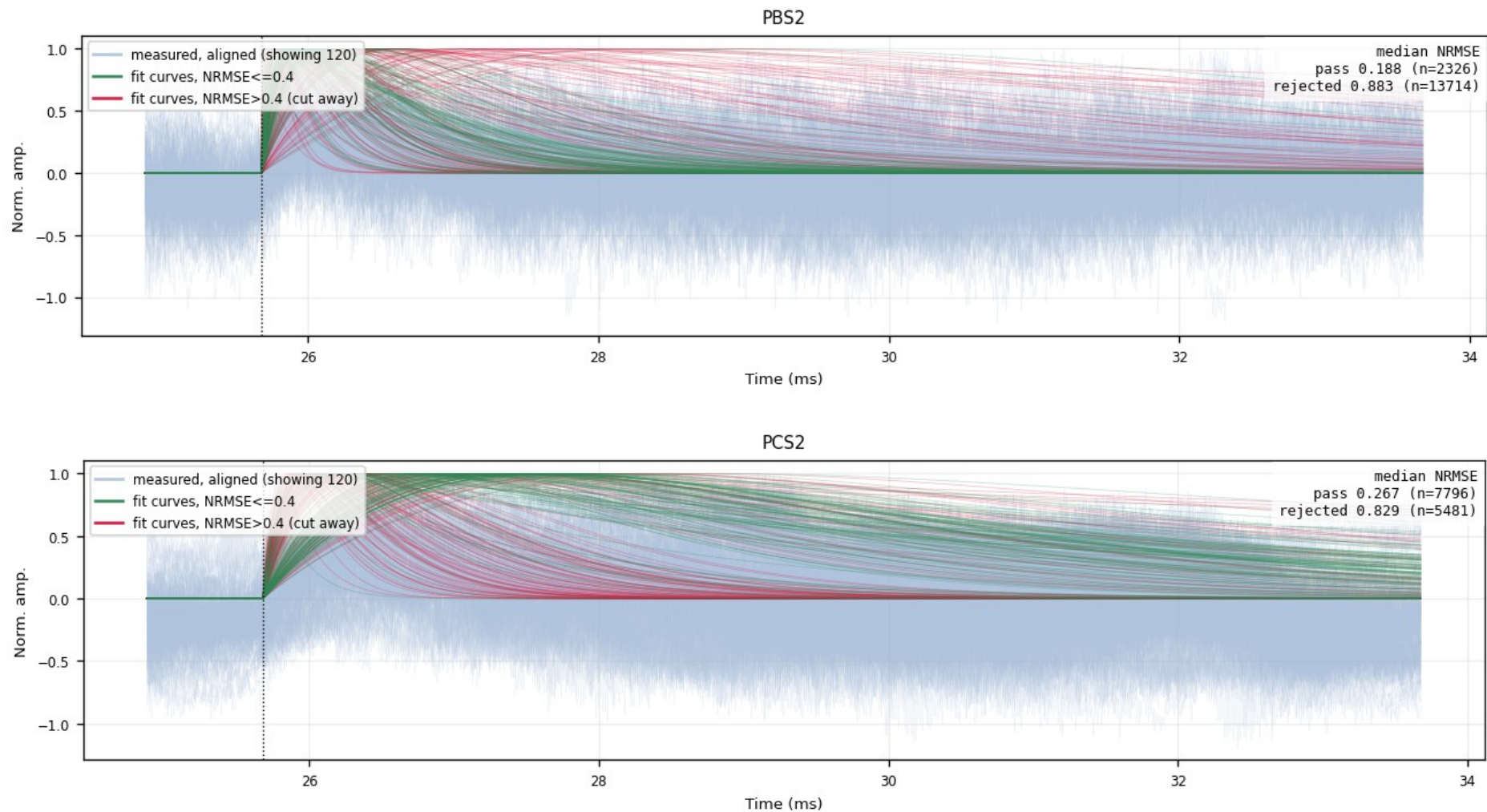
Gray-blue = aligned measured traces; green = fits with $\text{NRMSE} \leq 0.4$ (kept); red = $\text{NRMSE} > 0.4$ (cut away). Median NRMSE and counts of both populations stamped per channel.



Z4 · overlay_fan_cut: kept vs rejected fits (4/6)

figure: zip4_overlay_fan_cut_nrmse0.4.png, channel panels 7–8 of 11, top to bottom

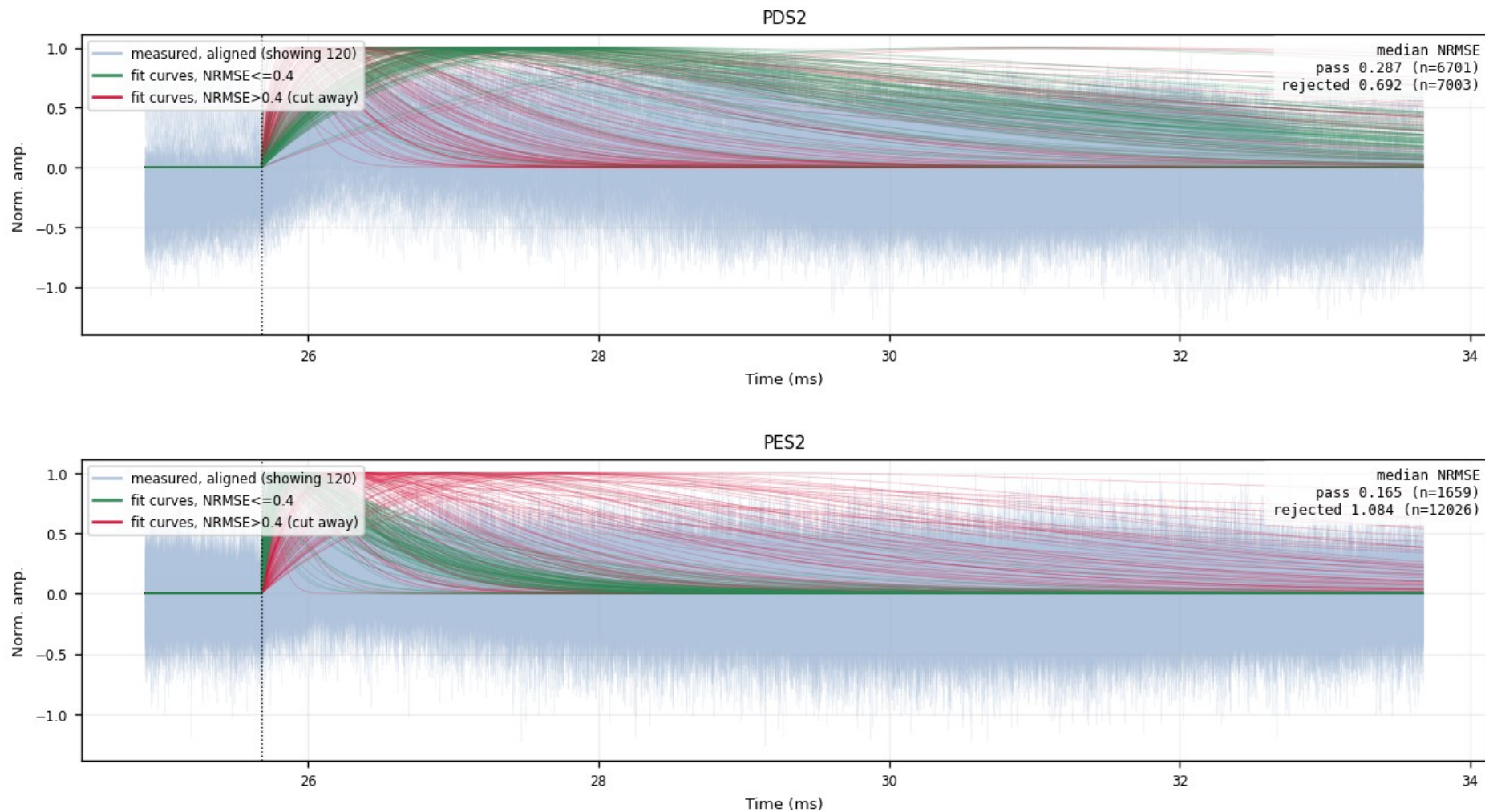
Gray-blue = aligned measured traces; green = fits with $\text{NRMSE} \leq 0.4$ (kept); red = $\text{NRMSE} > 0.4$ (cut away). Median NRMSE and counts of both populations stamped per channel.



Z4 · overlay_fan_cut: kept vs rejected fits (5/6)

figure: zip4_overlay_fan_cut_nrmse0.4.png, channel panels 9–10 of 11, top to bottom

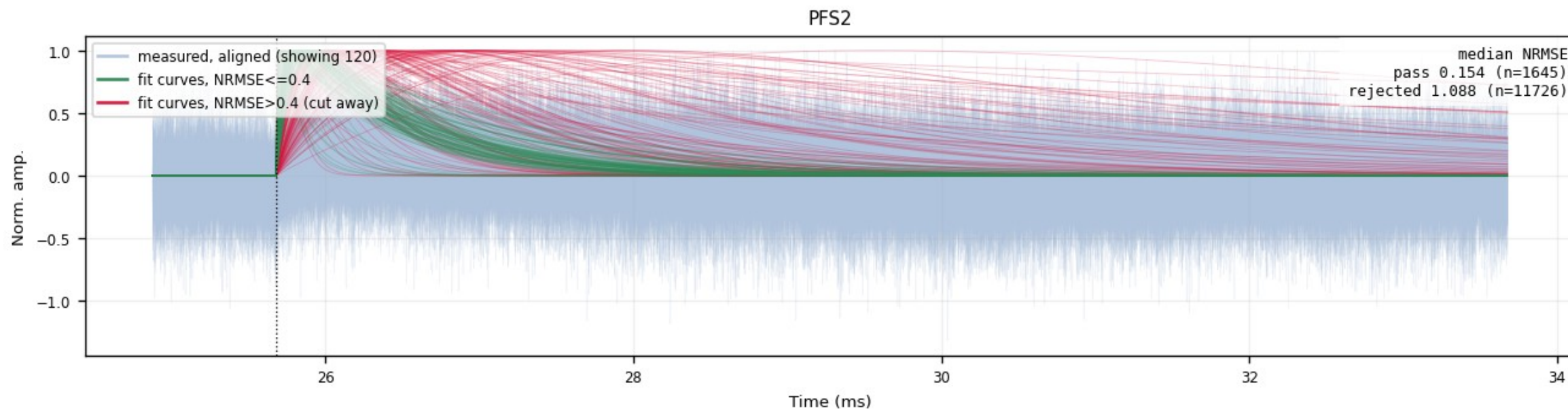
Gray-blue = aligned measured traces; green = fits with $\text{NRMSE} \leq 0.4$ (kept); red = $\text{NRMSE} > 0.4$ (cut away). Median NRMSE and counts of both populations stamped per channel.



Z4 · overlay_fan_cut: kept vs rejected fits (6/6)

figure: zip4_overlay_fan_cut_nrmse0.4.png, channel panels 11–11 of 11, top to bottom

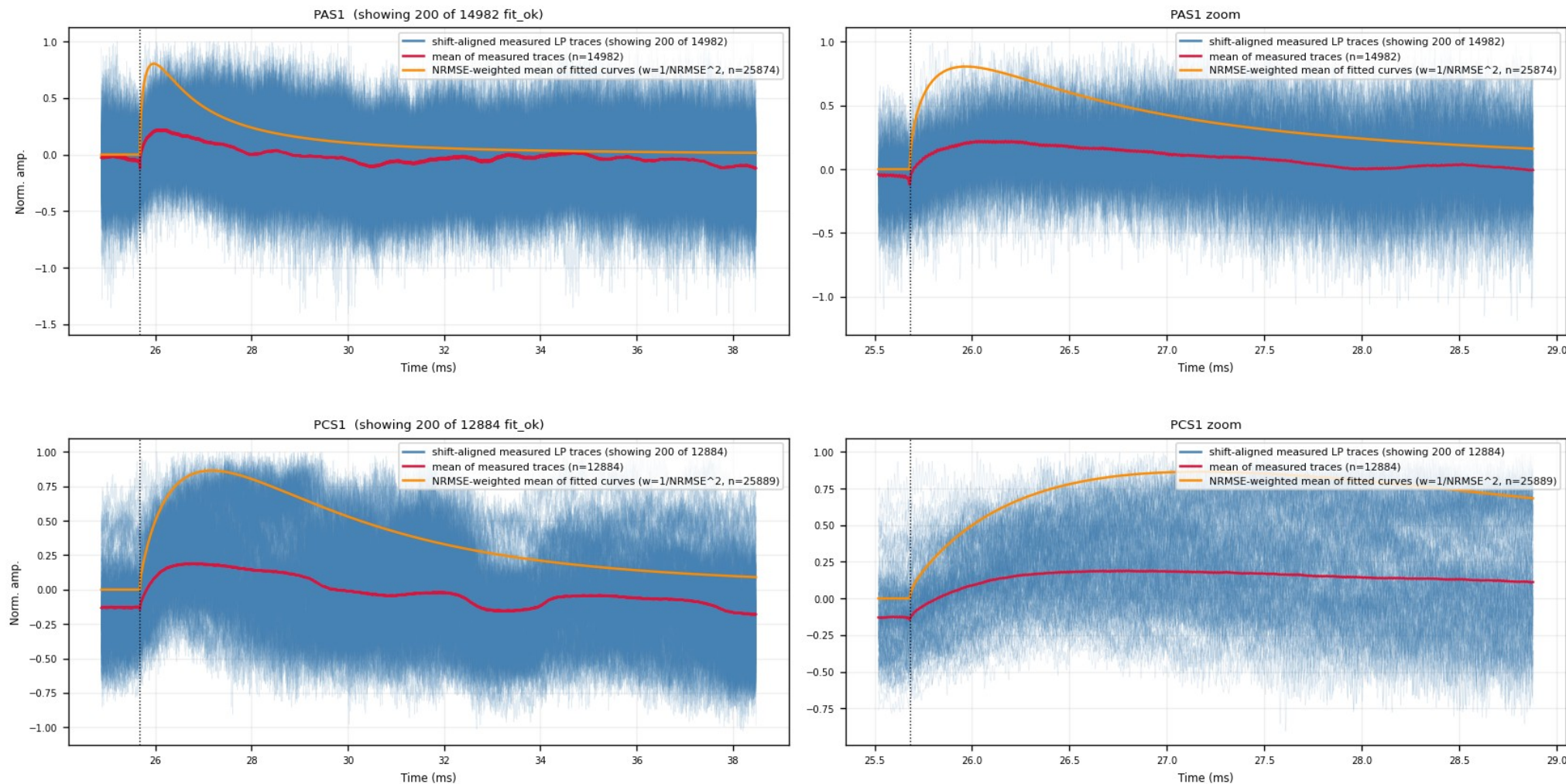
Gray-blue = aligned measured traces; green = fits with $\text{NRMSE} \leq 0.4$ (kept); red = $\text{NRMSE} > 0.4$ (cut away). Median NRMSE and counts of both populations stamped per channel.



Z4 · aligned_overlay: aligned measured traces (1/6)

figure: zip4_lp_aligned_overlay.png, channel panels 1–2 of 11, top to bottom

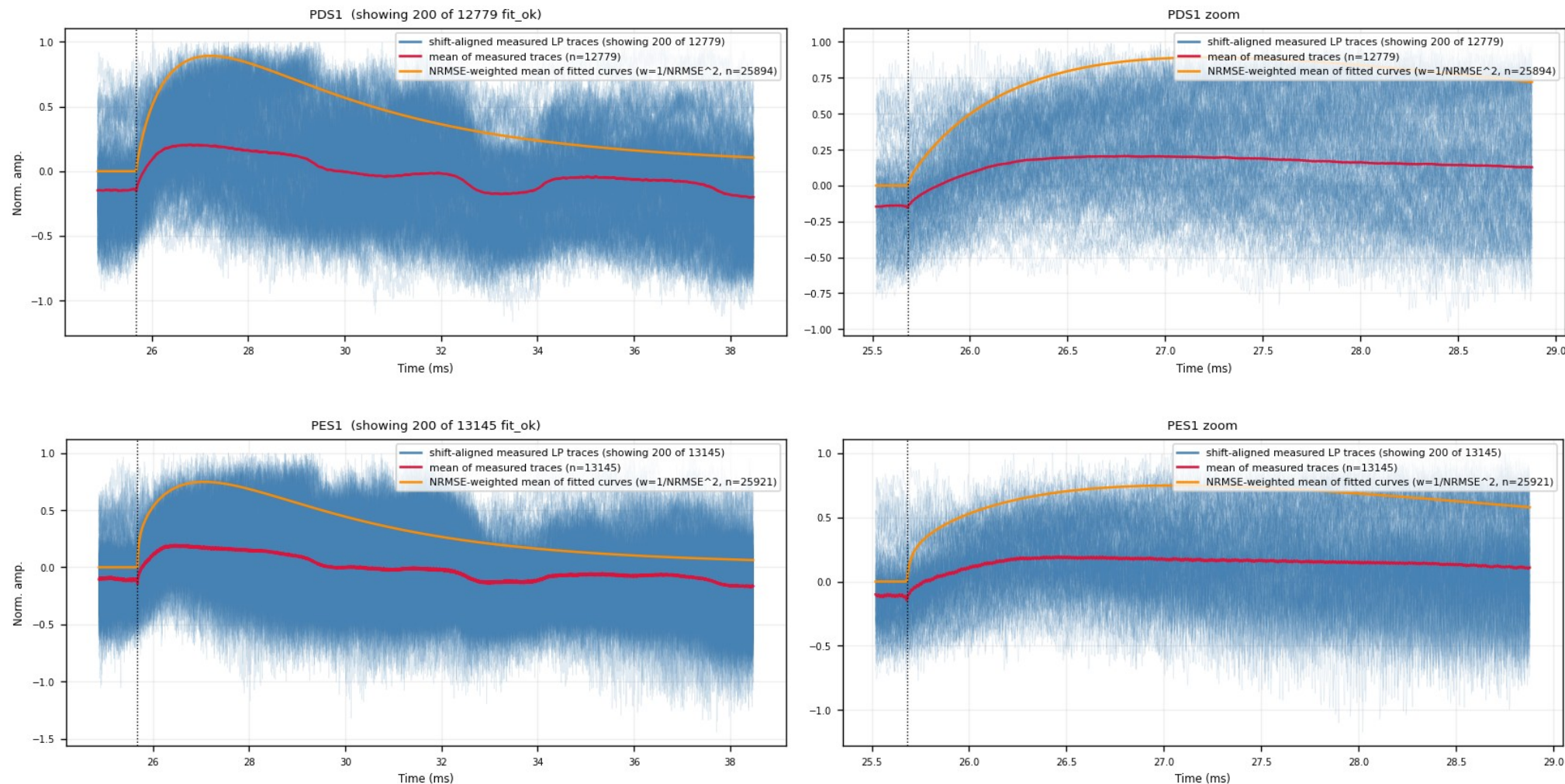
Blue = measured LP traces shifted by (fitted pretrigger – 16050); red = mean of all fit_ok events; orange = NRMSE-weighted mean of the fitted curves. Rise edges line up at 16050.



Z4 · aligned_overlay: aligned measured traces (2/6)

figure: zip4_lp_aligned_overlay.png, channel panels 3–4 of 11, top to bottom

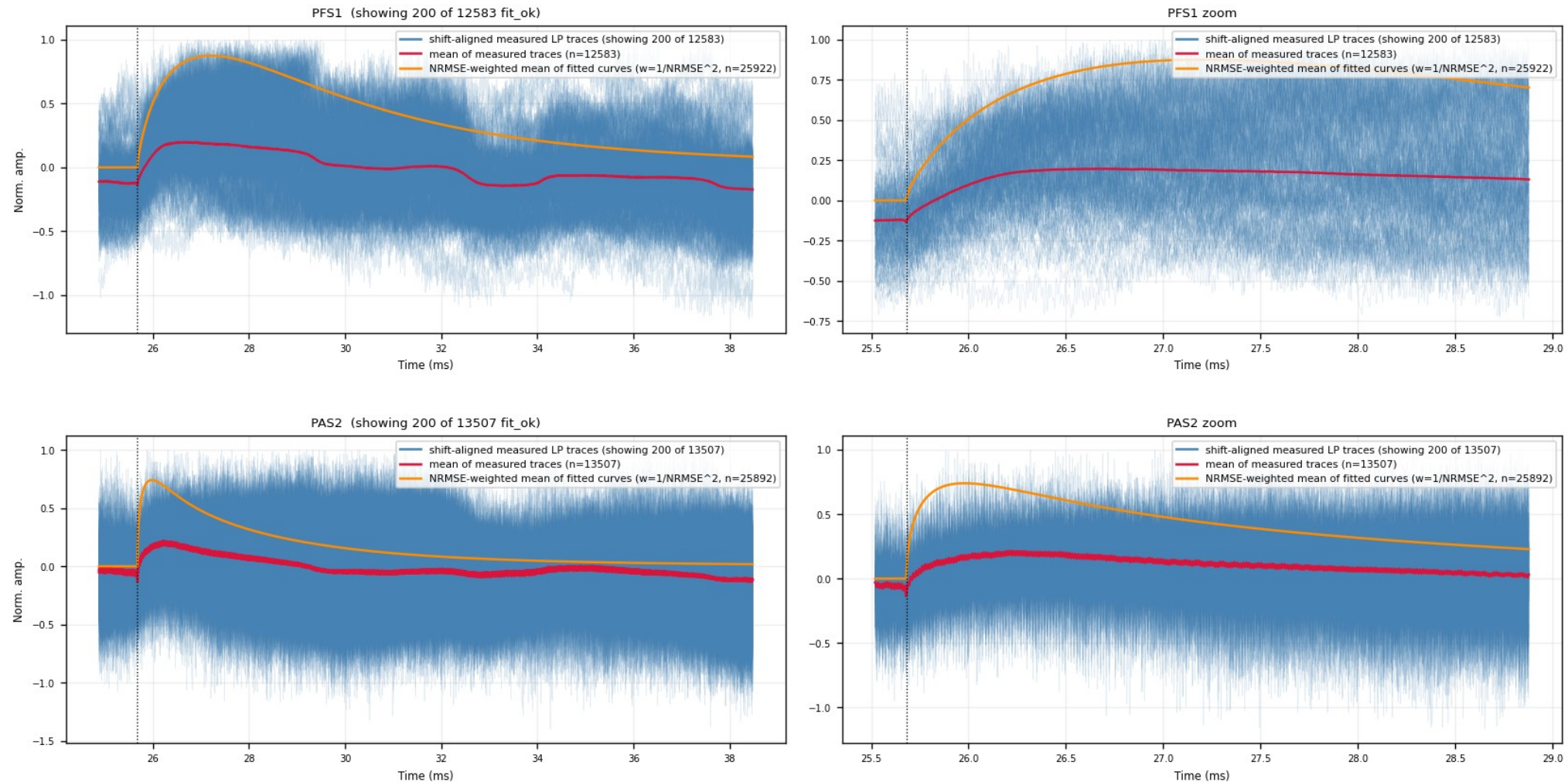
Blue = measured LP traces shifted by (fitted pretrigger – 16050); red = mean of all fit_ok events; orange = NRMSE-weighted mean of the fitted curves. Rise edges line up at 16050.



Z4 · aligned_overlay: aligned measured traces (3/6)

figure: zip4_lp_aligned_overlay.png, channel panels 5–6 of 11, top to bottom

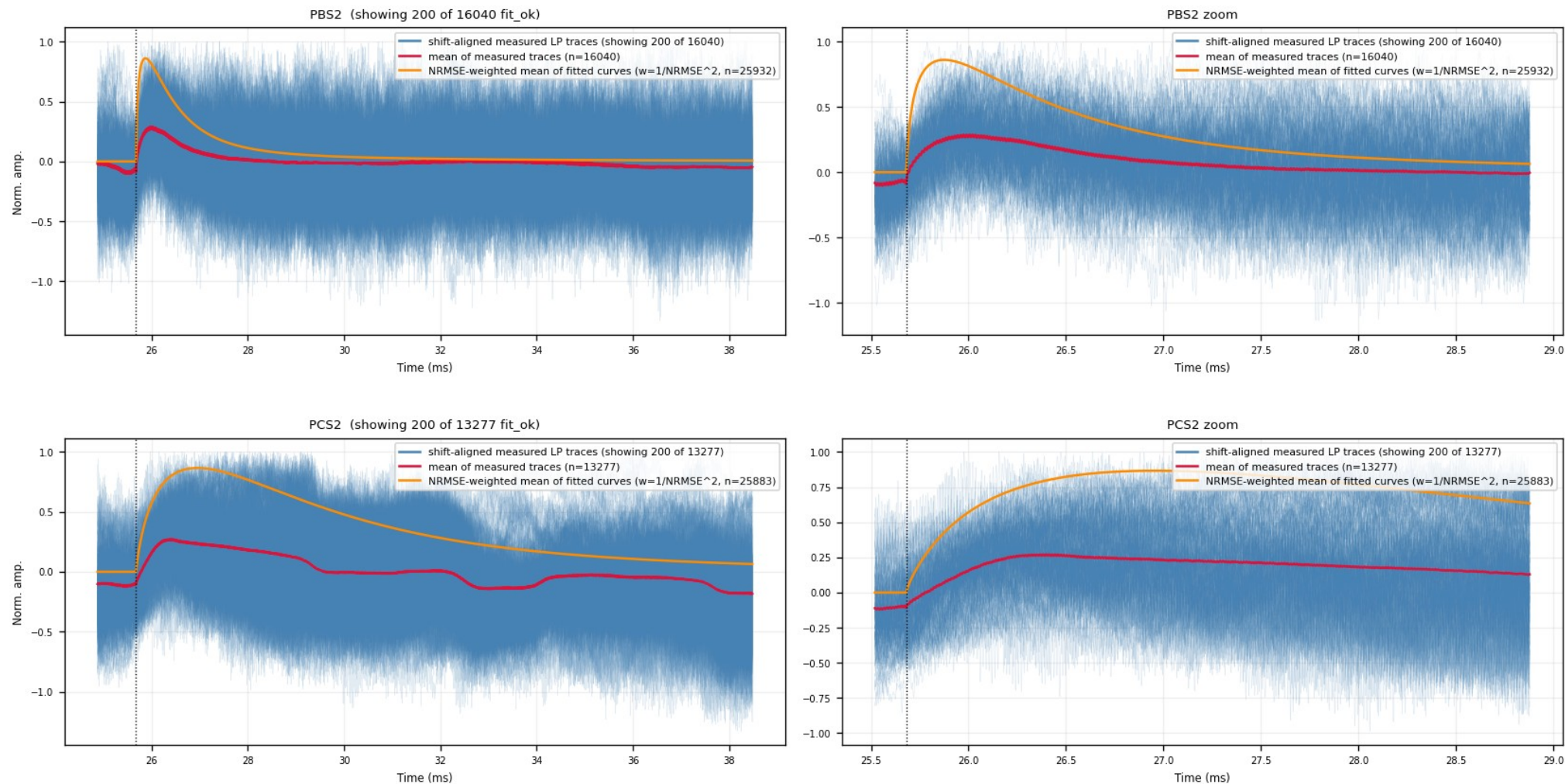
Blue = measured LP traces shifted by (fitted pretrigger – 16050); red = mean of all fit_ok events; orange = NRMSE-weighted mean of the fitted curves. Rise edges line up at 16050.



Z4 · aligned_overlay: aligned measured traces (4/6)

figure: zip4_lp_aligned_overlay.png, channel panels 7–8 of 11, top to bottom

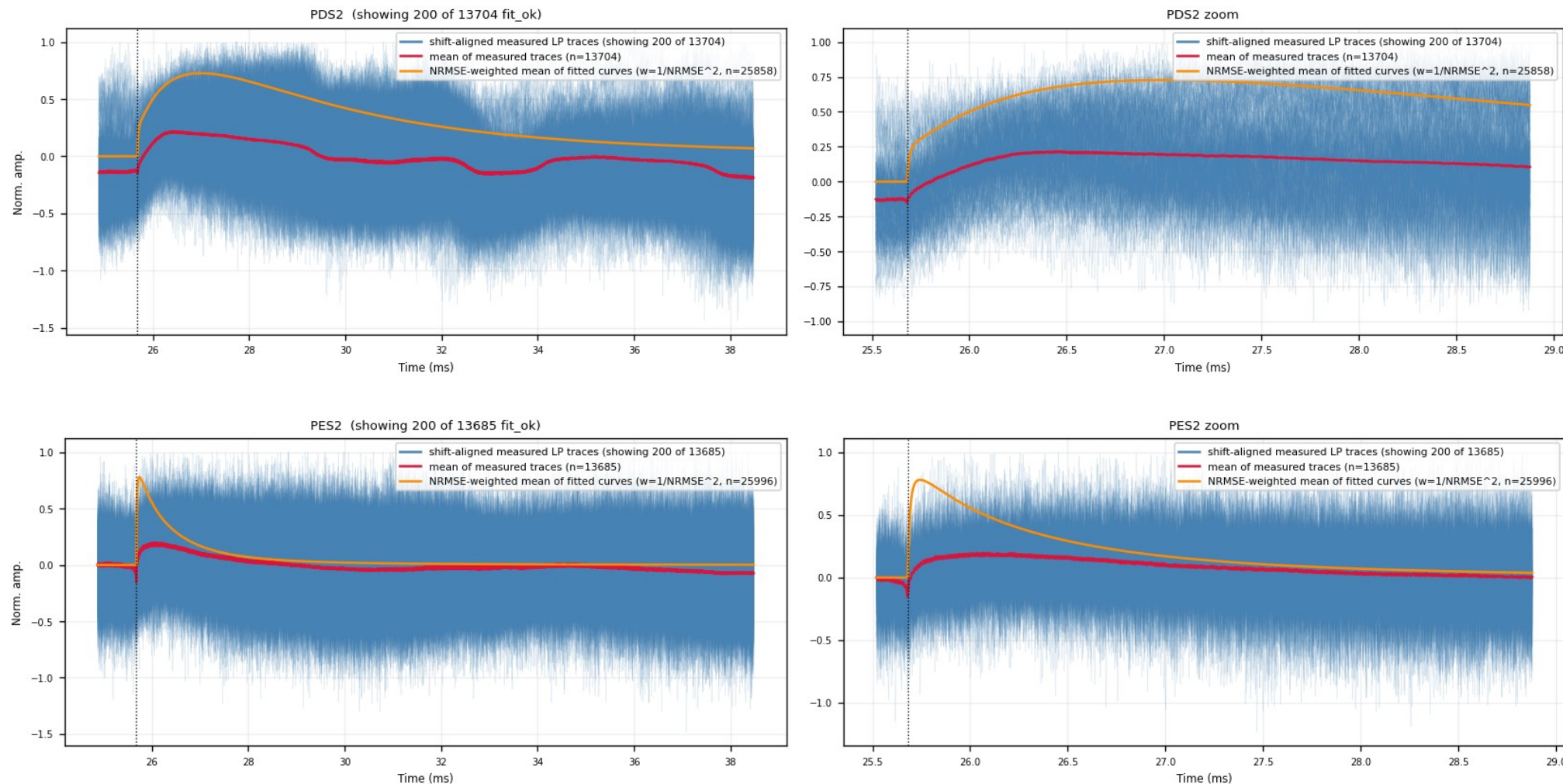
Blue = measured LP traces shifted by (fitted pretrigger – 16050); red = mean of all fit_ok events; orange = NRMSE-weighted mean of the fitted curves. Rise edges line up at 16050.



Z4 · aligned_overlay: aligned measured traces (5/6)

figure: zip4_lp_aligned_overlay.png, channel panels 9–10 of 11, top to bottom

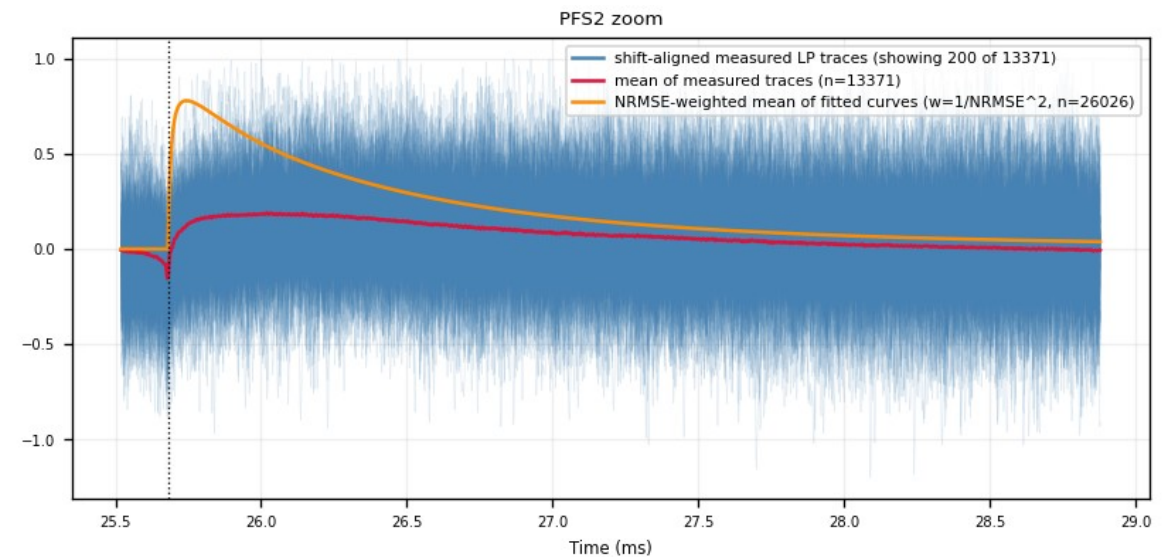
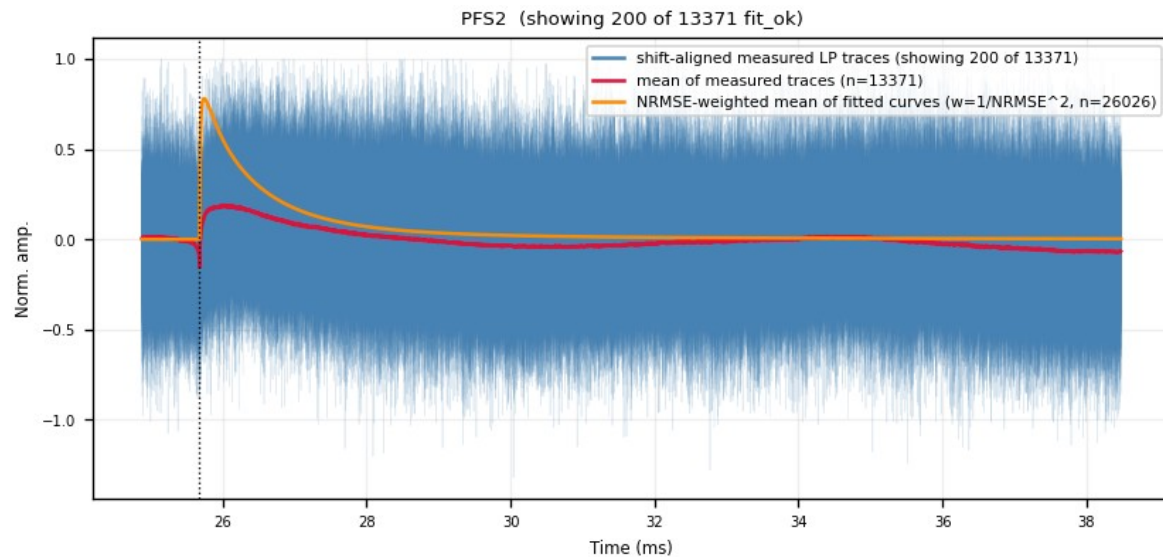
Blue = measured LP traces shifted by (fitted pretrigger – 16050); red = mean of all fit_ok events; orange = NRMSE-weighted mean of the fitted curves. Rise edges line up at 16050.



Z4 · aligned_overlay: aligned measured traces (6/6)

figure: zip4_lp_aligned_overlay.png, channel panels 11–11 of 11, top to bottom

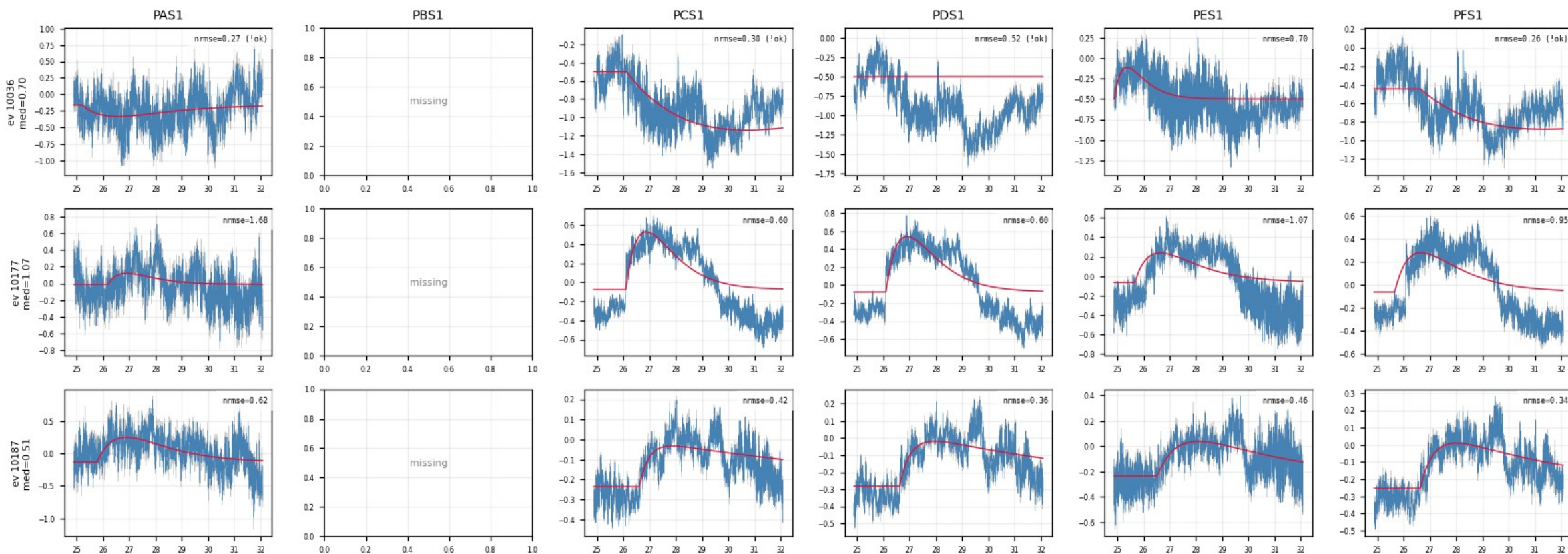
Blue = measured LP traces shifted by (fitted pretrigger – 16050); red = mean of all fit_ok events; orange = NRMSE-weighted mean of the fitted curves. Rise edges line up at 16050.



Z4 · slow_rise_events: NRMSE-rejected events (1/10)

figure: zip4_slow_rise_events.png, event rows 1–3 of 15, left half of the channels

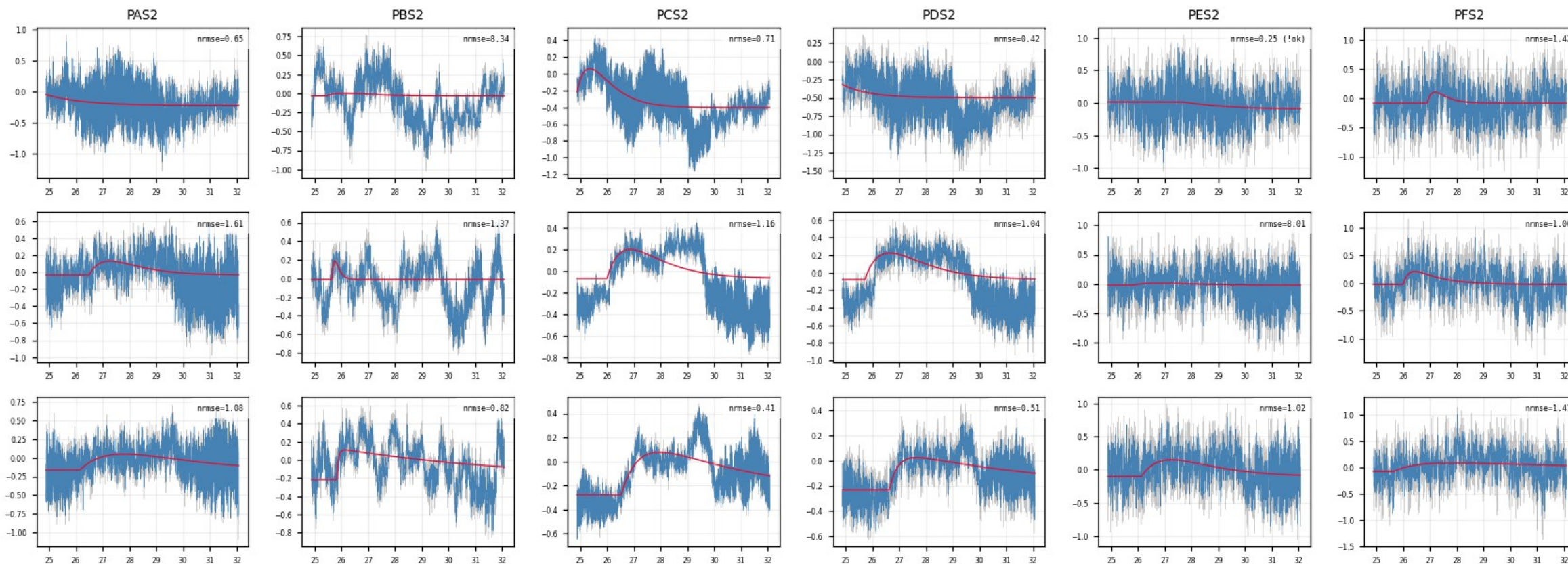
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.



Z4 · slow_rise_events: NRMSE-rejected events (2/10)

figure: zip4_slow_rise_events.png, event rows 1–3 of 15, right half of the channels

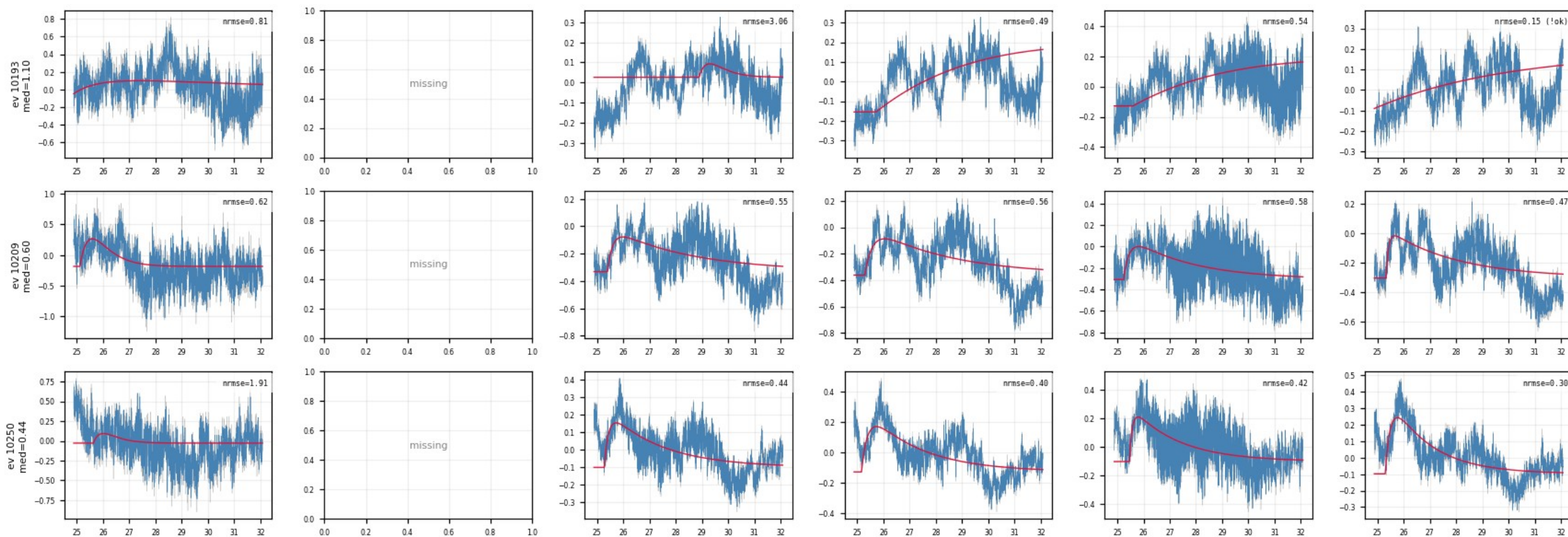
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.



Z4 · slow_rise_events: NRMSE-rejected events (3/10)

figure: zip4_slow_rise_events.png, event rows 4–6 of 15, left half of the channels

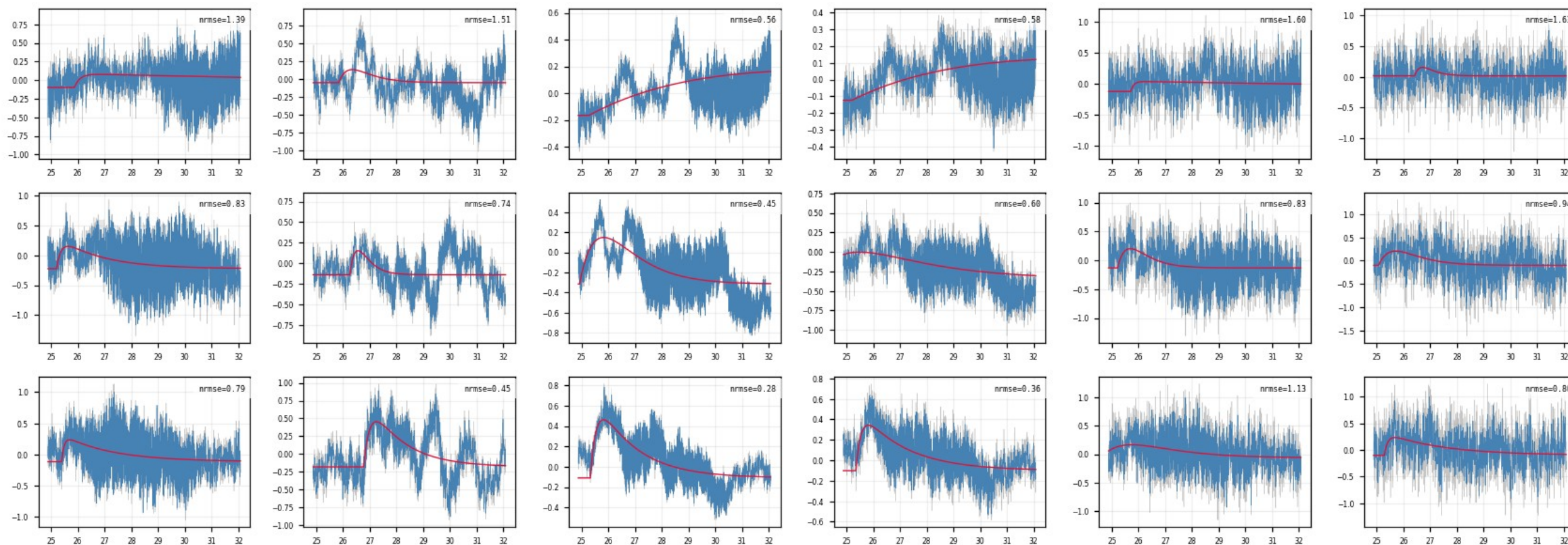
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.



Z4 · slow_rise_events: NRMSE-rejected events (4/10)

figure: zip4_slow_rise_events.png, event rows 4–6 of 15, right half of the channels

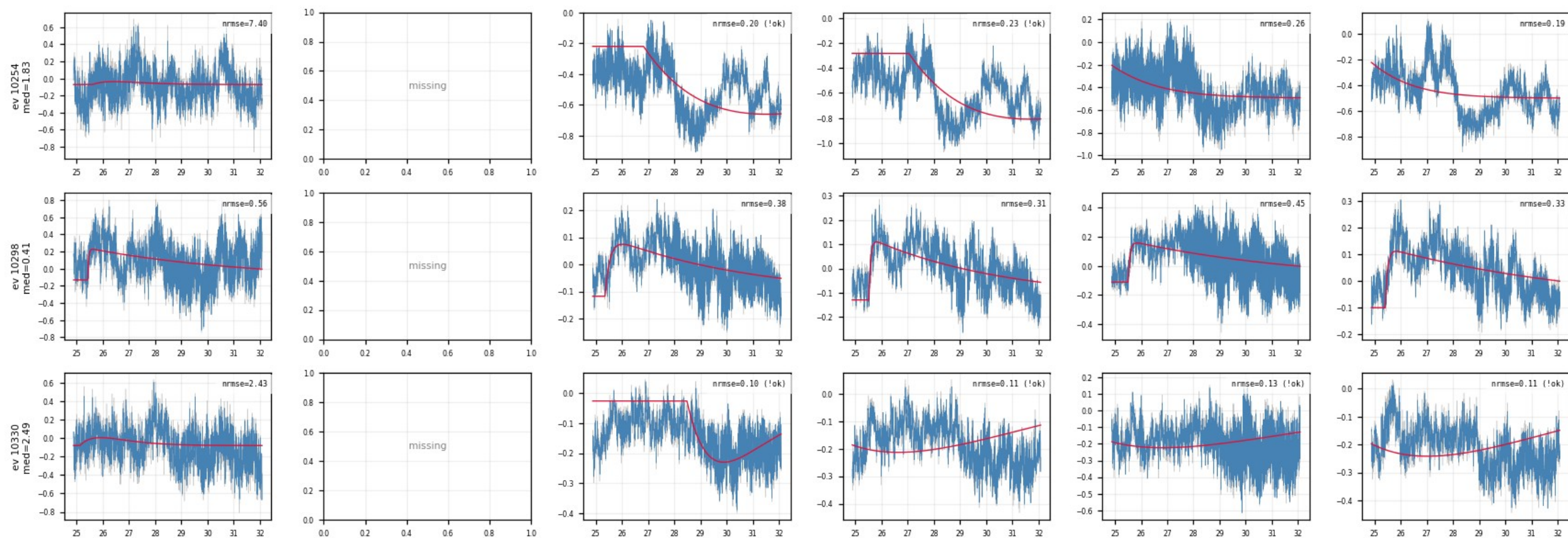
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.



Z4 · slow_rise_events: NRMSE-rejected events (5/10)

figure: zip4_slow_rise_events.png, event rows 7–9 of 15, left half of the channels

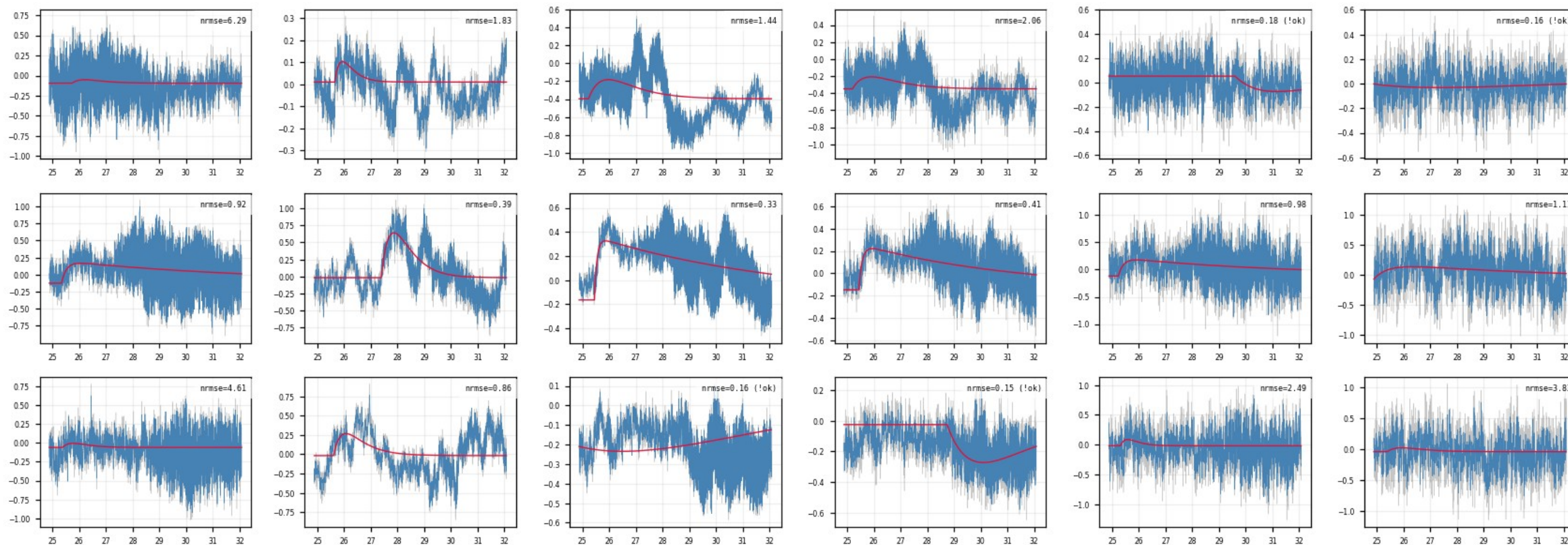
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.



Z4 · slow_rise_events: NRMSE-rejected events (6/10)

figure: zip4_slow_rise_events.png, event rows 7–9 of 15, right half of the channels

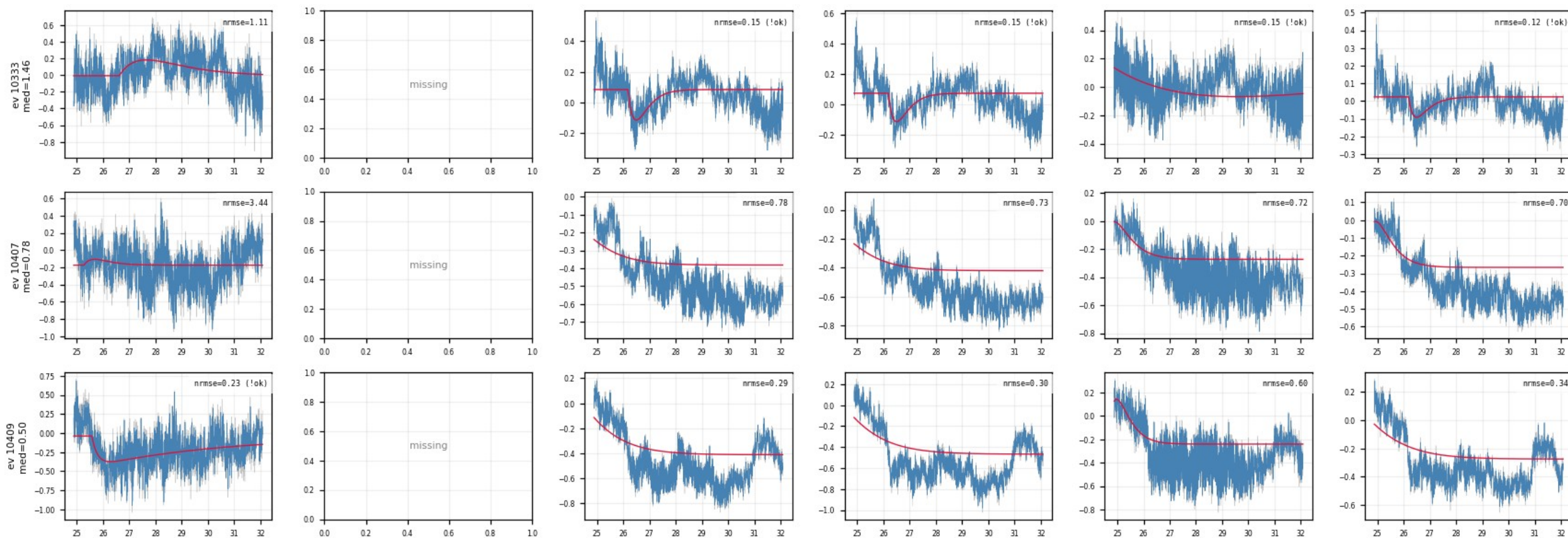
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.



Z4 · slow_rise_events: NRMSE-rejected events (7/10)

figure: zip4_slow_rise_events.png, event rows 10–12 of 15, left half of the channels

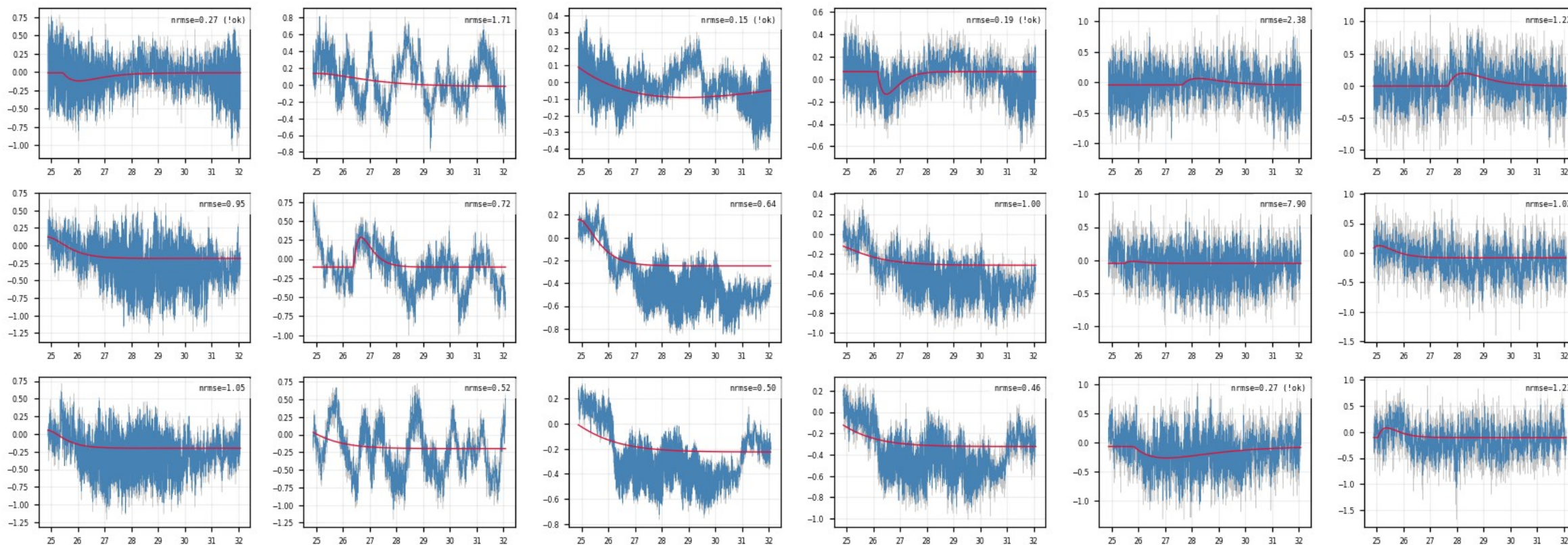
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.



Z4 · slow_rise_events: NRMSE-rejected events (8/10)

figure: zip4_slow_rise_events.png, event rows 10–12 of 15, right half of the channels

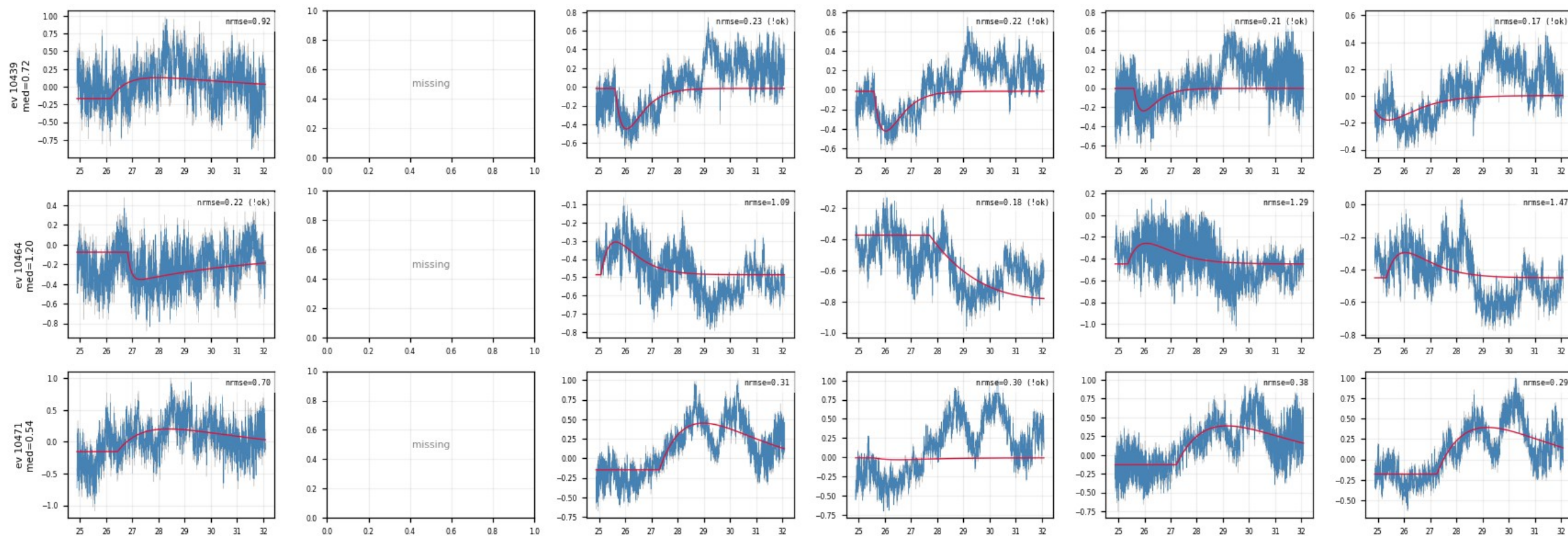
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.



Z4 · slow_rise_events: NRMSE-rejected events (9/10)

figure: zip4_slow_rise_events.png, event rows 13–15 of 15, left half of the channels

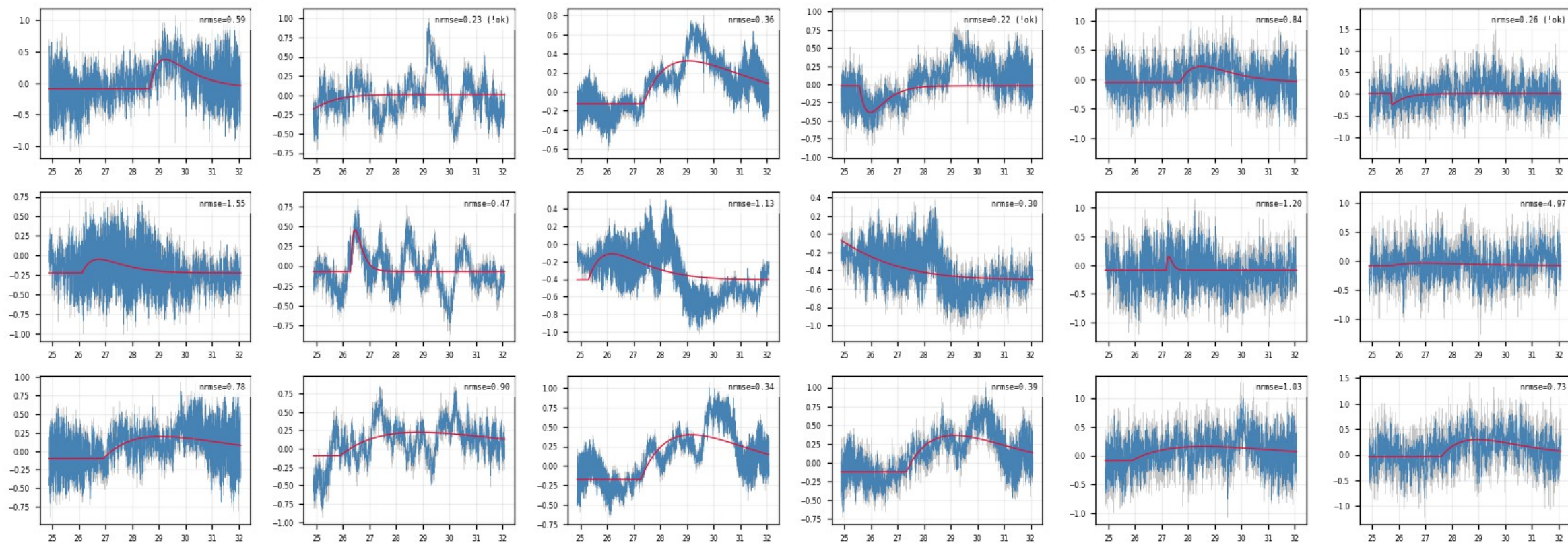
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.



Z4 · slow_rise_events: NRMSE-rejected events (10/10)

figure: zip4_slow_rise_events.png, event rows 13–15 of 15, right half of the channels

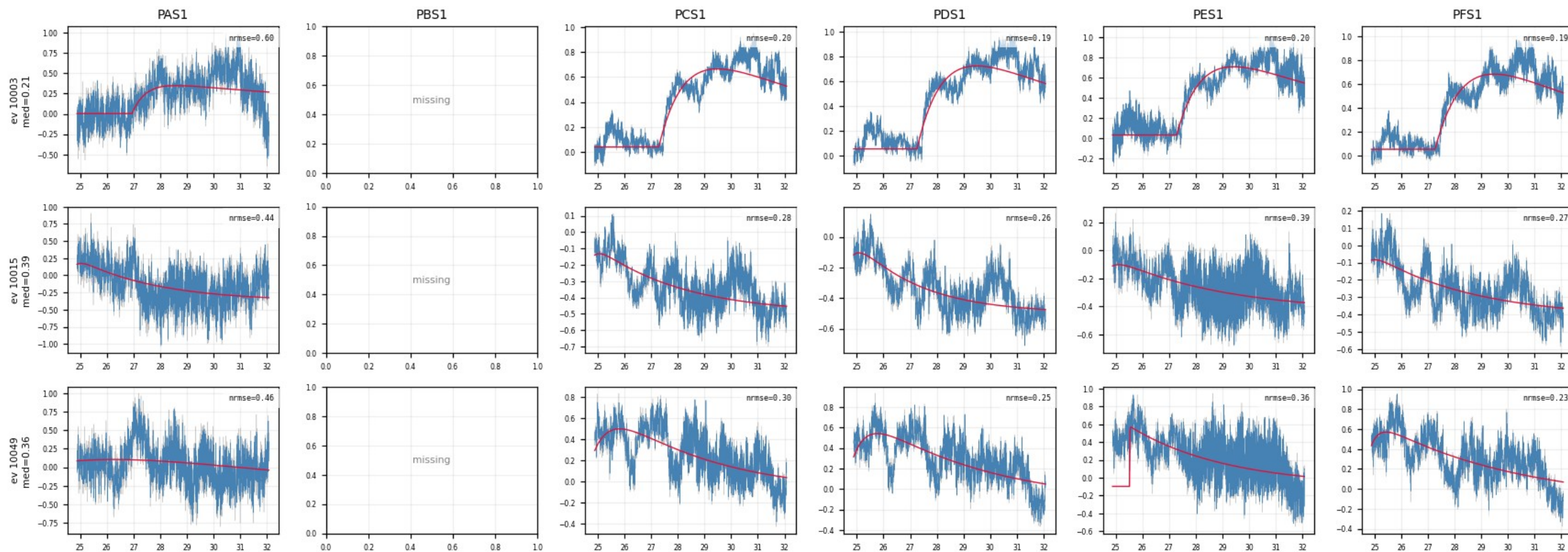
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.



Z4 · shadow_events: well-fit slow-rise events (1/10)

figure: zip4_shadow_events.png, event rows 1–3 of 15, left half of the channels

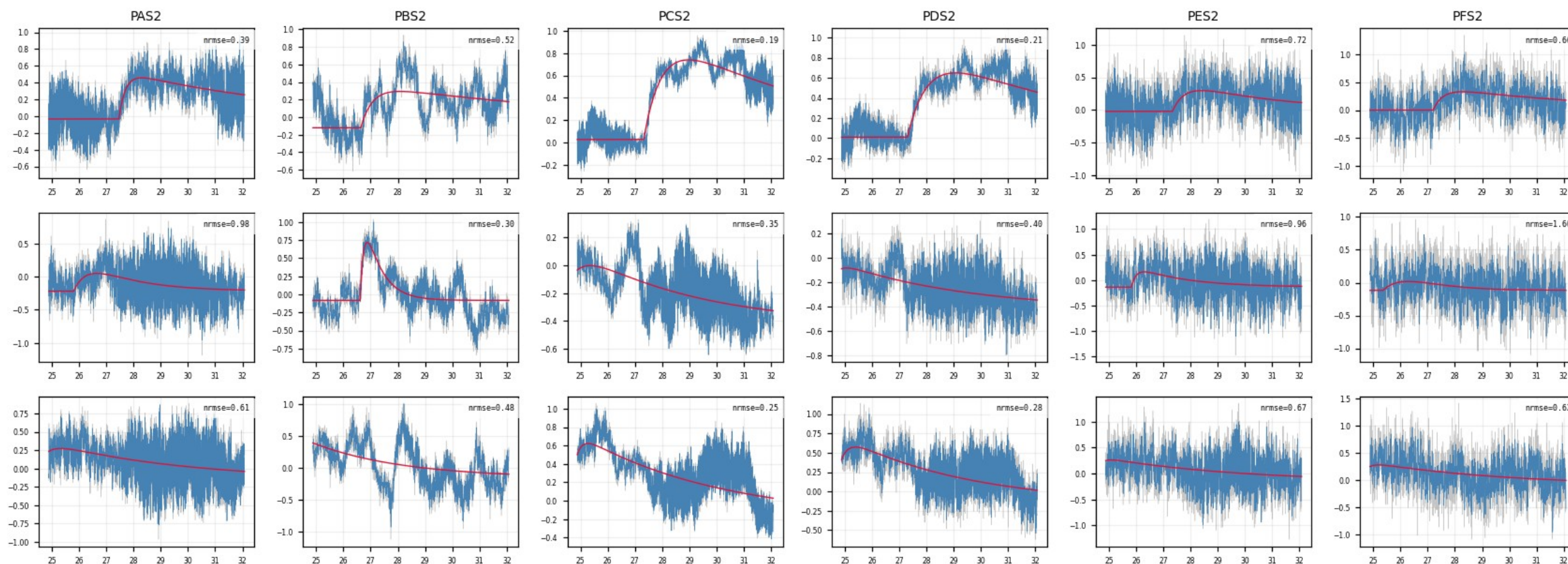
Median NRMSE ≤ 0.4 AND median $\tau_{\text{rise}} > 0.2$ ms: genuinely slow, well-fit pulses ; kept, and exactly what the NxM multi-templates are for.



Z4 · shadow_events: well-fit slow-rise events (2/10)

figure: zip4_shadow_events.png, event rows 1–3 of 15, right half of the channels

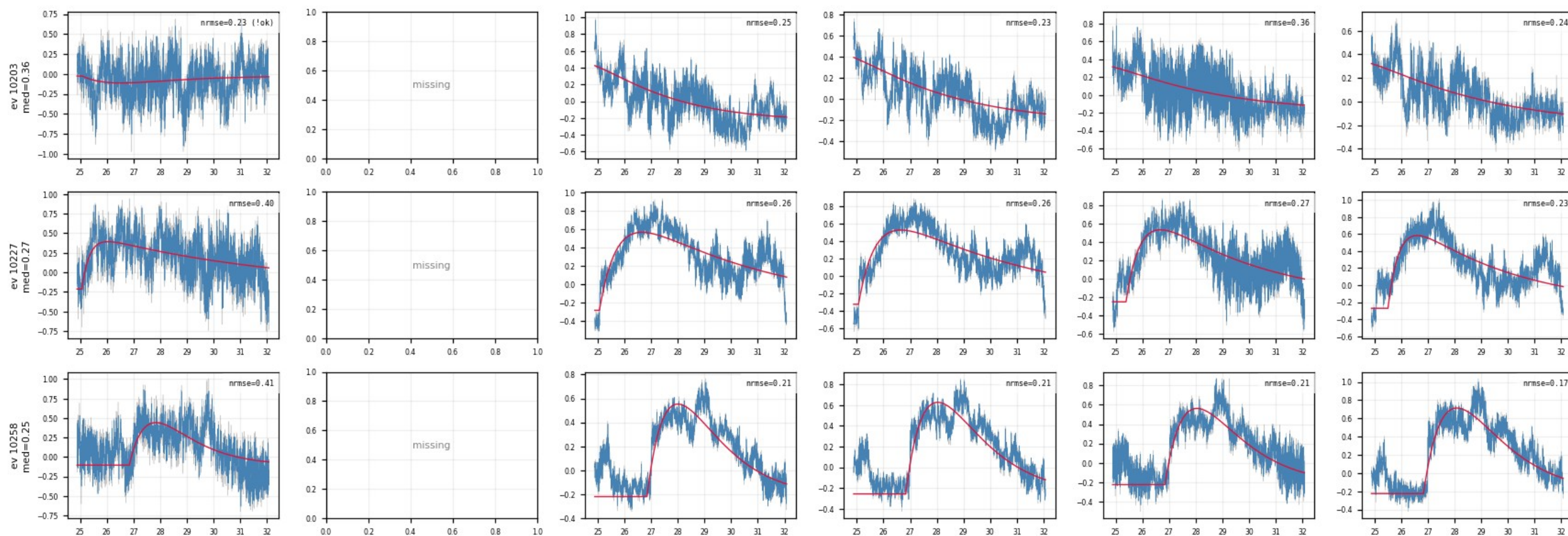
Median NRMSE ≤ 0.4 AND median $\tau_{\text{rise}} > 0.2$ ms: genuinely slow, well-fit pulses ; kept, and exactly what the NxM multi-templates are for.



Z4 · shadow_events: well-fit slow-rise events (3/10)

figure: zip4_shadow_events.png, event rows 4–6 of 15, left half of the channels

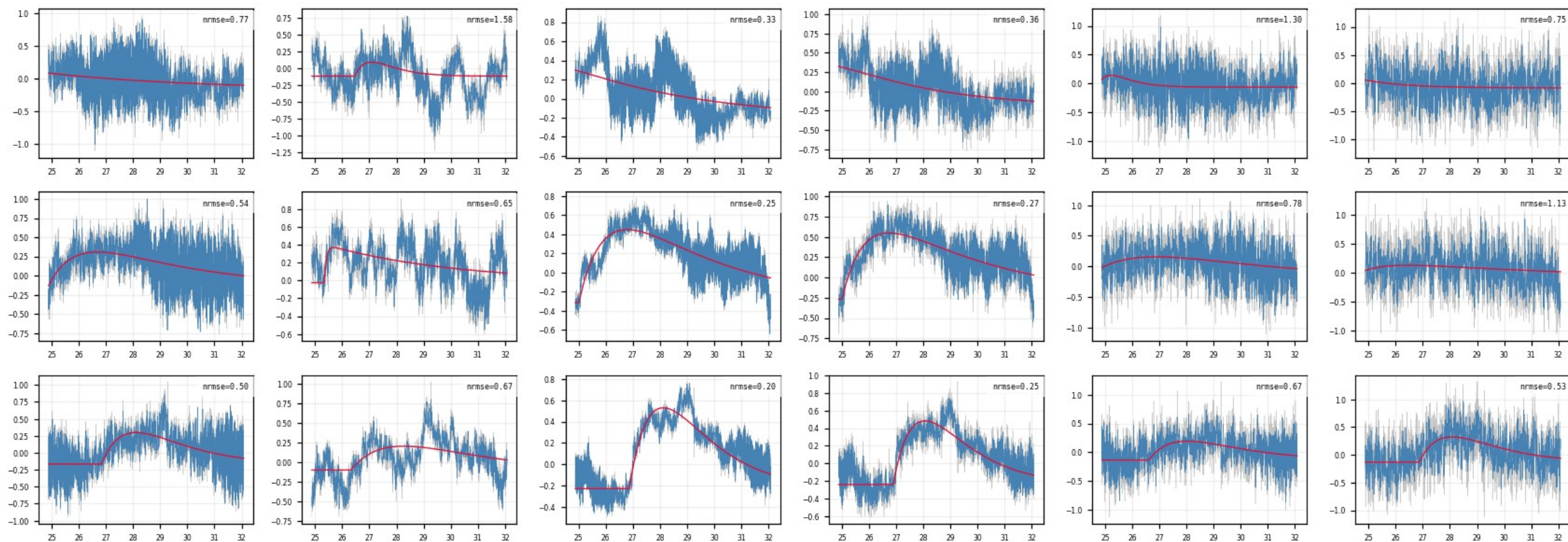
Median NRMSE ≤ 0.4 AND median $\tau_{\text{rise}} > 0.2$ ms: genuinely slow, well-fit pulses ; kept, and exactly what the NxM multi-templates are for.



Z4 · shadow_events: well-fit slow-rise events (4/10)

figure: zip4_shadow_events.png, event rows 4–6 of 15, right half of the channels

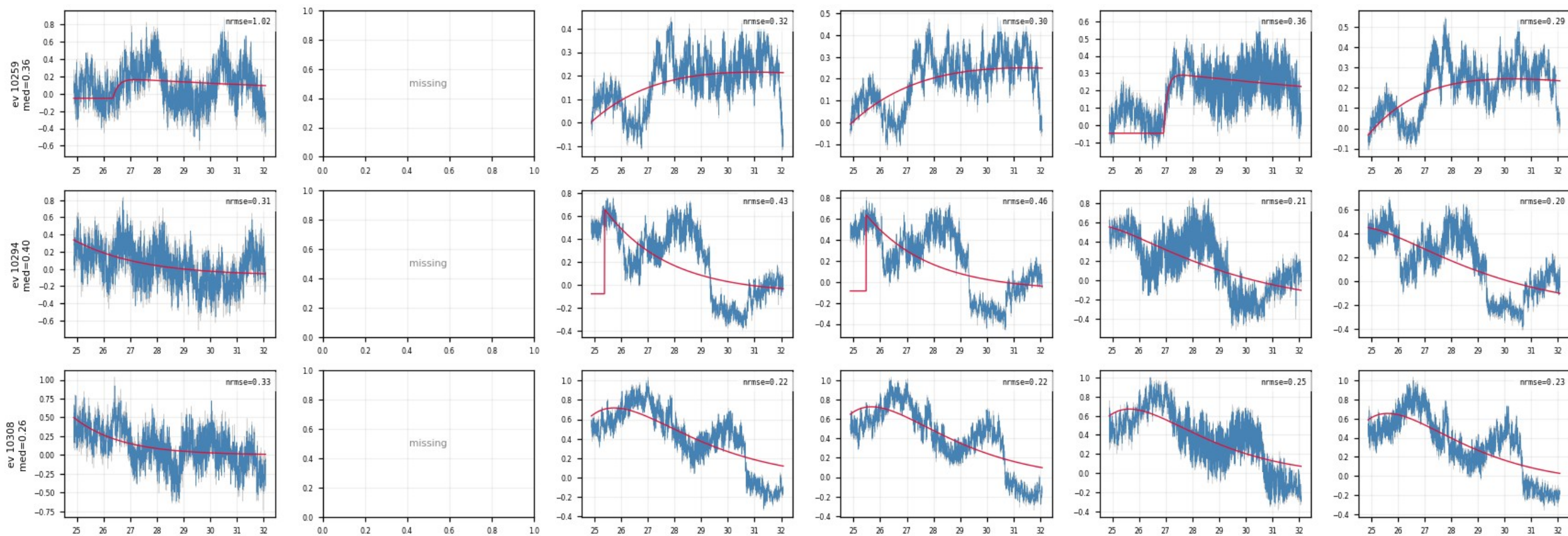
Median NRMSE ≤ 0.4 AND median $\tau_{\text{rise}} > 0.2$ ms: genuinely slow, well-fit pulses ; kept, and exactly what the NxM multi-templates are for.



Z4 · shadow_events: well-fit slow-rise events (5/10)

figure: zip4_shadow_events.png, event rows 7–9 of 15, left half of the channels

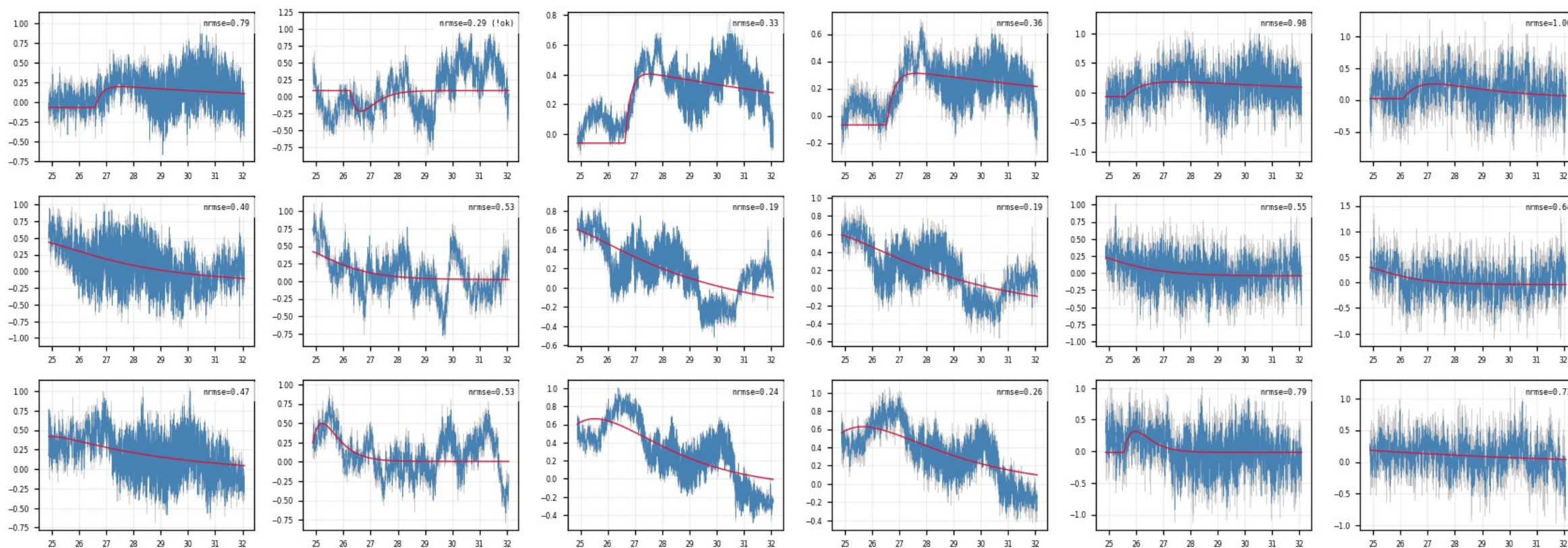
Median NRMSE ≤ 0.4 AND median $\tau_{\text{rise}} > 0.2$ ms: genuinely slow, well-fit pulses ; kept, and exactly what the NxM multi-templates are for.



Z4 · shadow_events: well-fit slow-rise events (6/10)

figure: zip4_shadow_events.png, event rows 7–9 of 15, right half of the channels

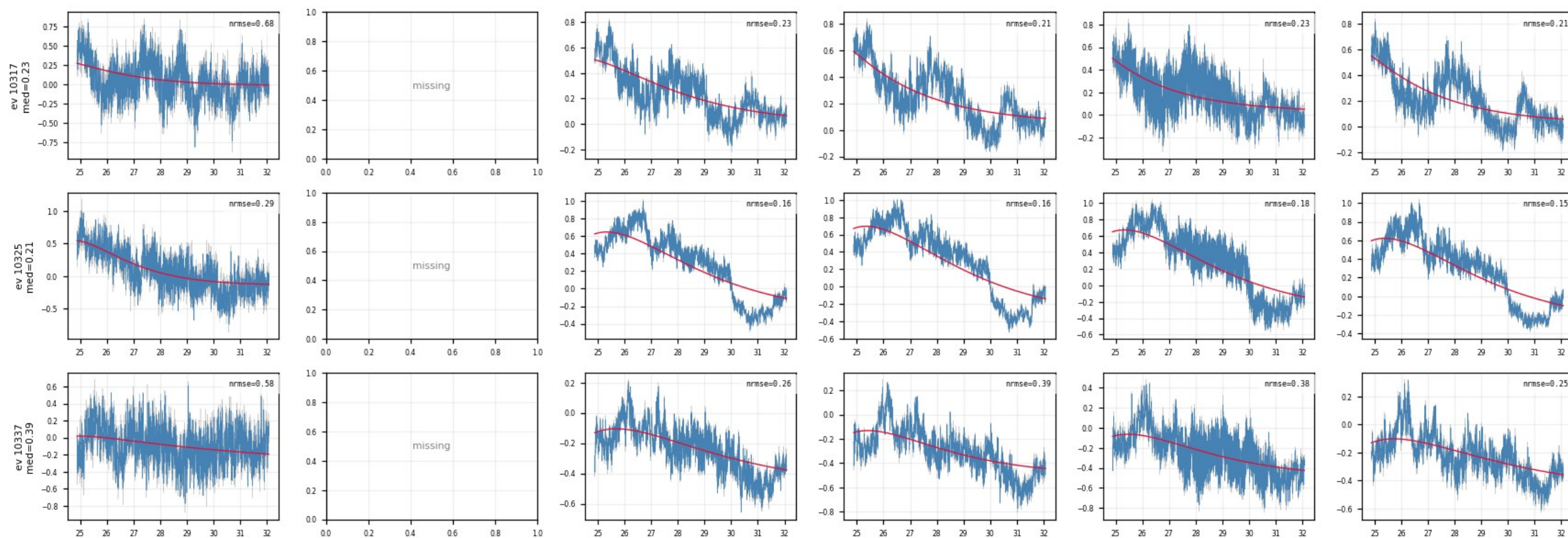
Median NRMSE ≤ 0.4 AND median $\tau_{\text{rise}} > 0.2$ ms: genuinely slow, well-fit pulses ; kept, and exactly what the NxM multi-templates are for.



Z4 · shadow_events: well-fit slow-rise events (7/10)

figure: zip4_shadow_events.png, event rows 10–12 of 15, left half of the channels

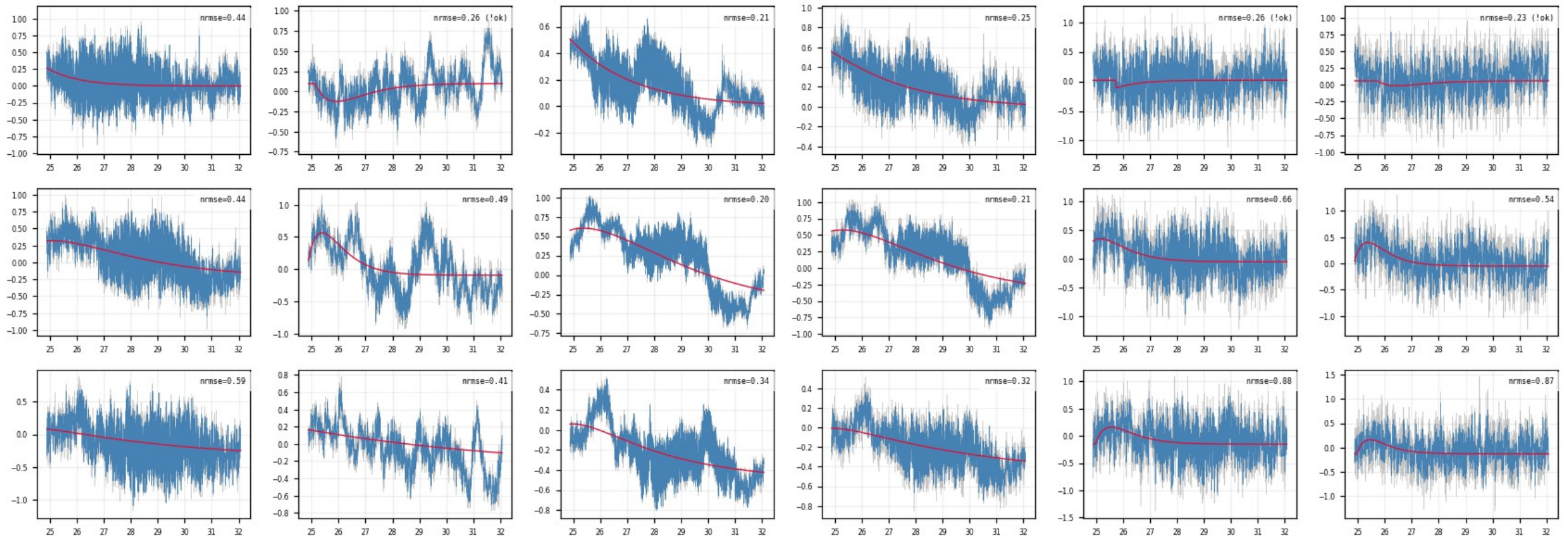
Median NRMSE ≤ 0.4 AND median $\tau_{\text{rise}} > 0.2$ ms: genuinely slow, well-fit pulses ; kept, and exactly what the NxM multi-templates are for.



Z4 · shadow_events: well-fit slow-rise events (8/10)

figure: zip4_shadow_events.png, event rows 10–12 of 15, right half of the channels

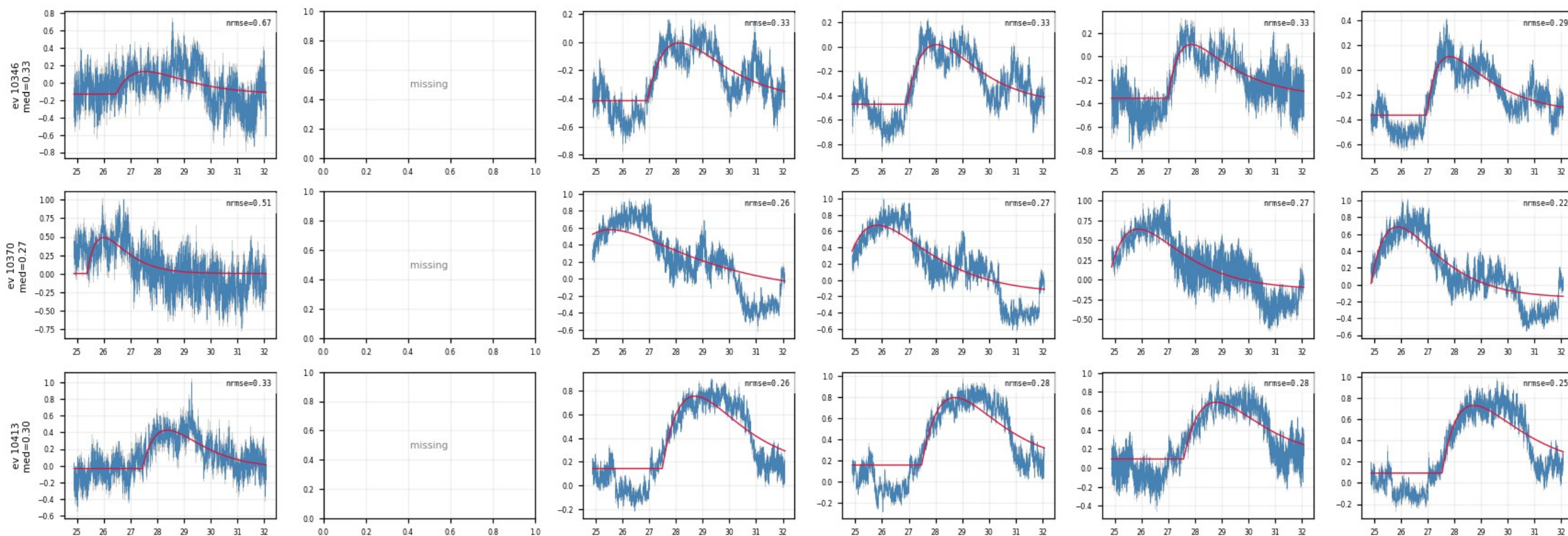
Median NRMSE ≤ 0.4 AND median $\tau_{\text{rise}} > 0.2$ ms: genuinely slow, well-fit pulses ; kept, and exactly what the NxM multi-templates are for.



Z4 · shadow_events: well-fit slow-rise events (9/10)

figure: zip4_shadow_events.png, event rows 13–15 of 15, left half of the channels

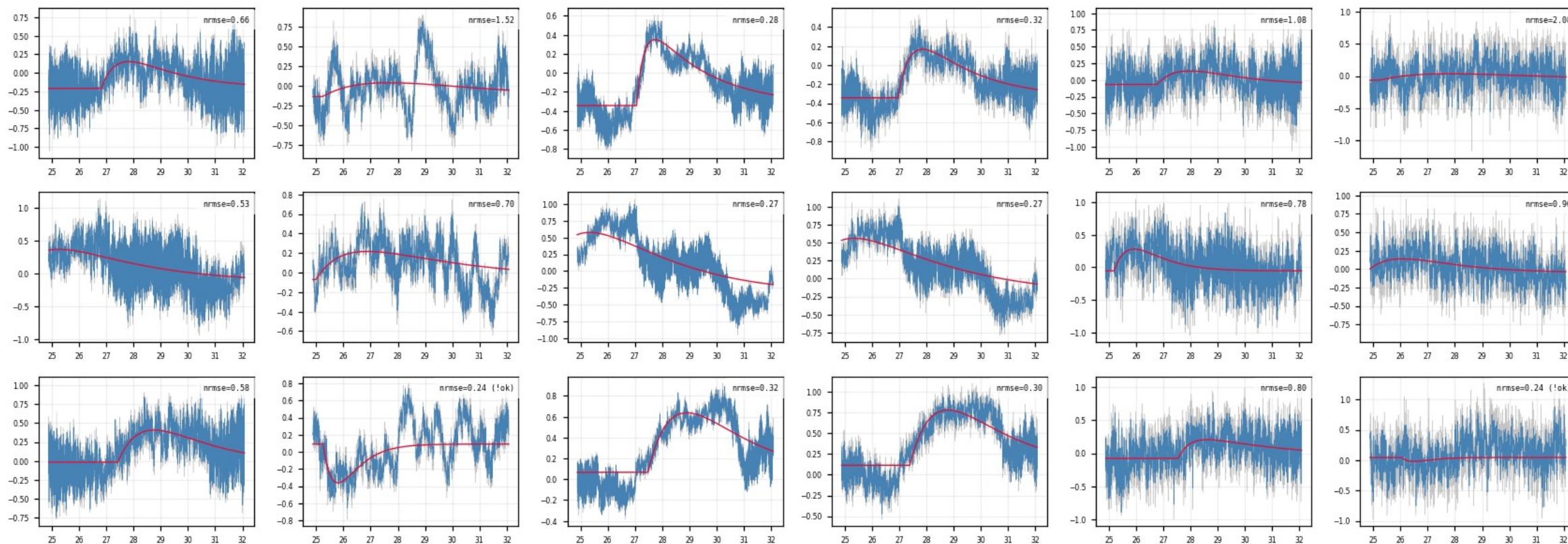
Median NRMSE ≤ 0.4 AND median $\tau_{\text{rise}} > 0.2$ ms: genuinely slow, well-fit pulses ; kept, and exactly what the NxM multi-templates are for.



Z4 · shadow_events: well-fit slow-rise events (10/10)

figure: zip4_shadow_events.png, event rows 13–15 of 15, right half of the channels

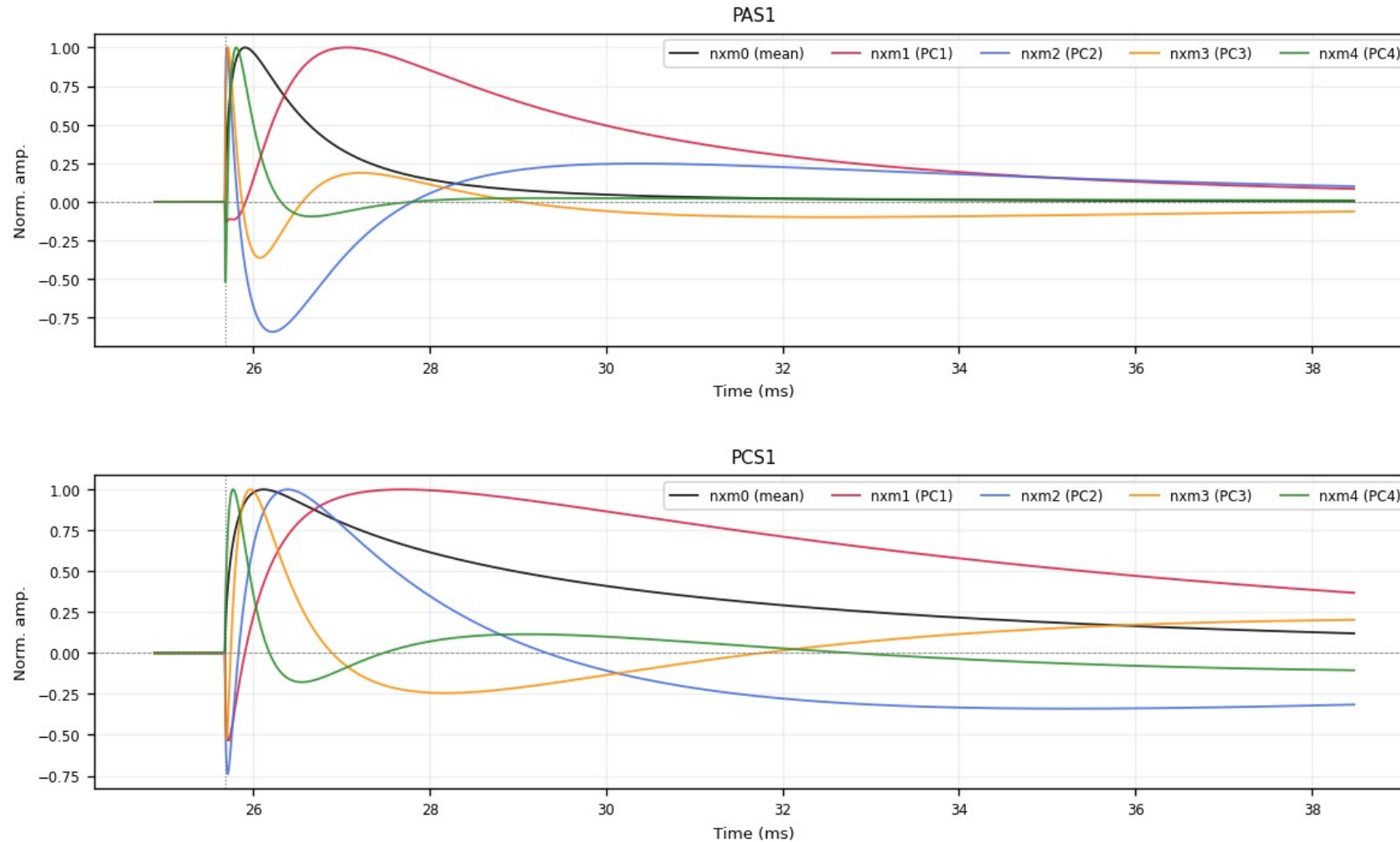
Median NRMSE ≤ 0.4 AND median $\tau_{\text{rise}} > 0.2$ ms: genuinely slow, well-fit pulses ; kept, and exactly what the NxM multi-templates are for.



Z4 · deliverables/nxm: PCA templates nxm0–4 (1/6)

figure: zip4_pca_templates.png, channel panels 1–2 of 11, top to bottom

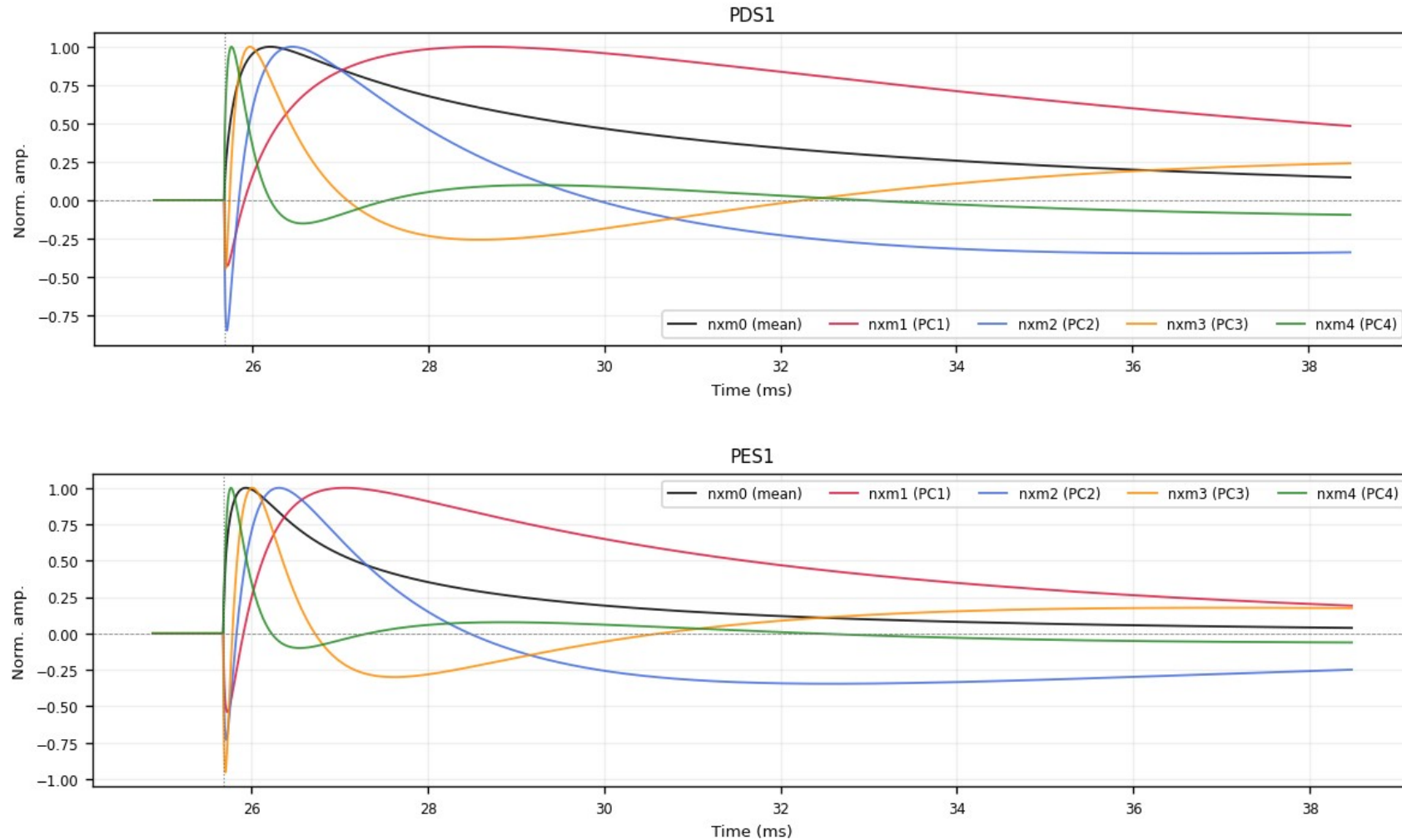
nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized; population = $\text{fit_ok} + \text{NRMSE} \leq 0.4 + \tau_{\text{rise}} \leq 0.3 \text{ ms}$.



Z4 · deliverables/nxm: PCA templates nxm0–4 (2/6)

figure: zip4_pca_templates.png, channel panels 3–4 of 11, top to bottom

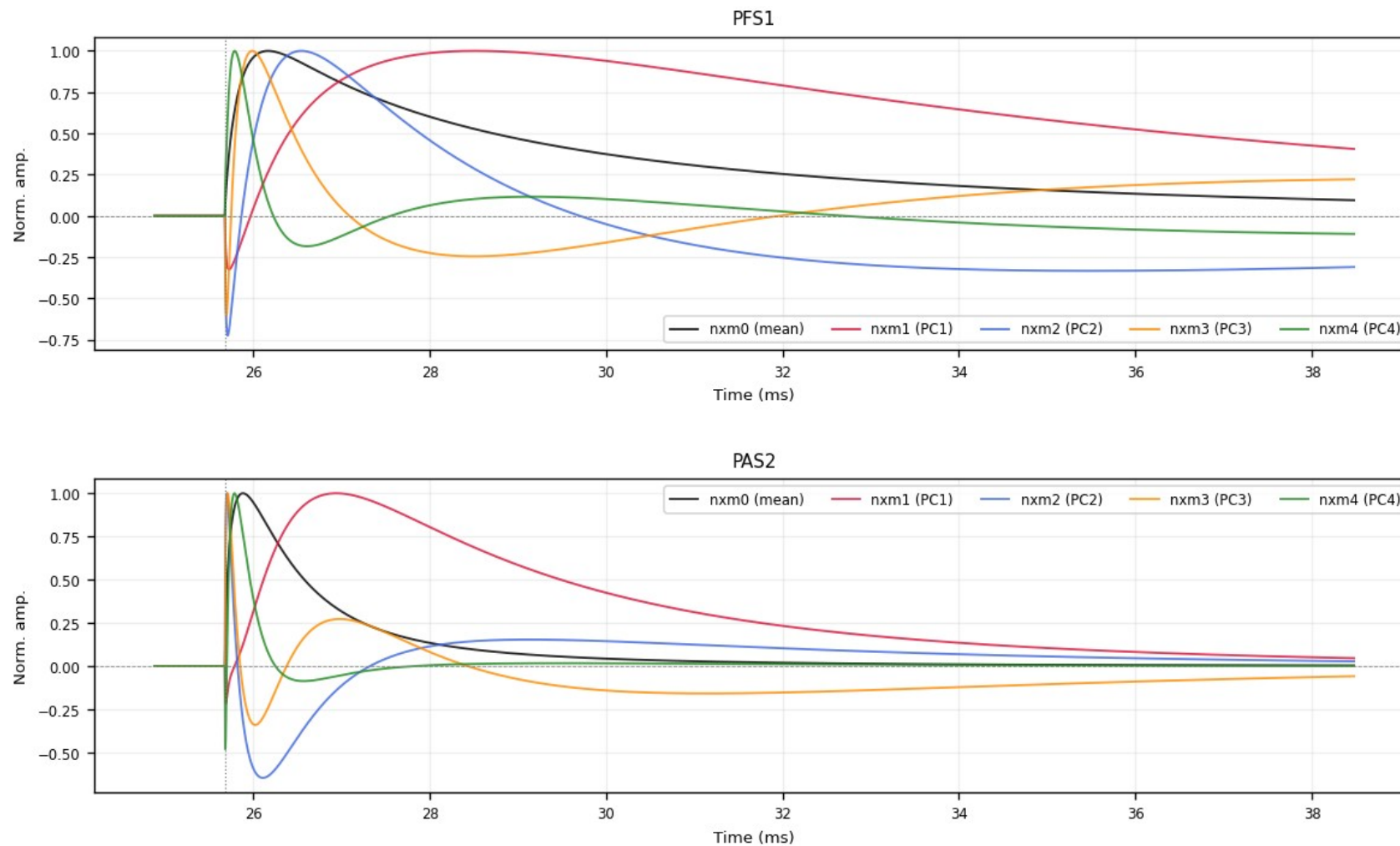
nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized; population = $\text{fit_ok} + \text{NRMSE} \leq 0.4 + \tau_{\text{rise}} \leq 0.3 \text{ ms}$.



Z4 · deliverables/nxm: PCA templates nxm0–4 (3/6)

figure: zip4_pca_templates.png, channel panels 5–6 of 11, top to bottom

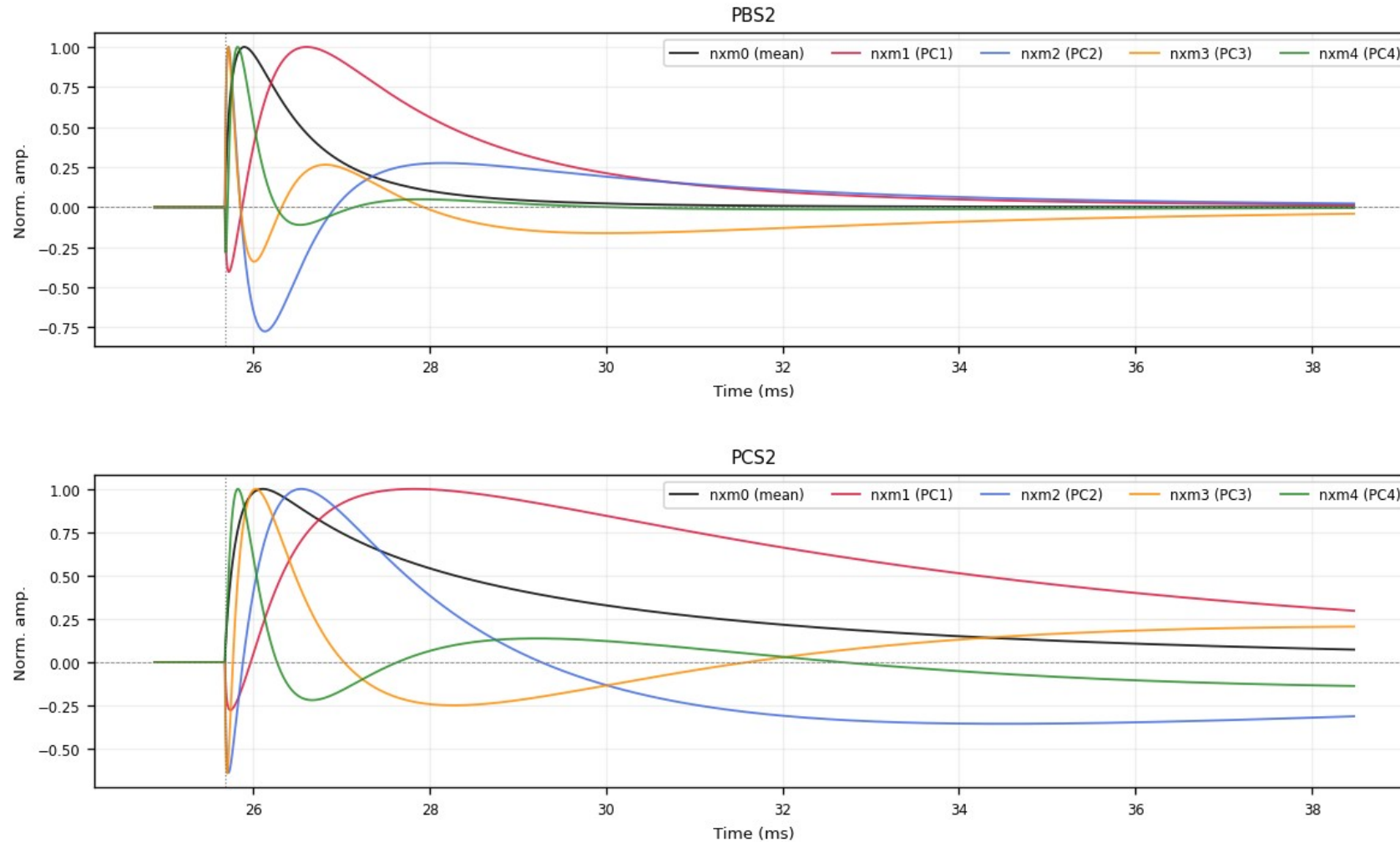
nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized; population = $\text{fit_ok} + \text{NRMSE} \leq 0.4 + \tau_{\text{rise}} \leq 0.3 \text{ ms}$.



Z4 · deliverables/nxm: PCA templates nxm0–4 (4/6)

figure: zip4_pca_templates.png, channel panels 7–8 of 11, top to bottom

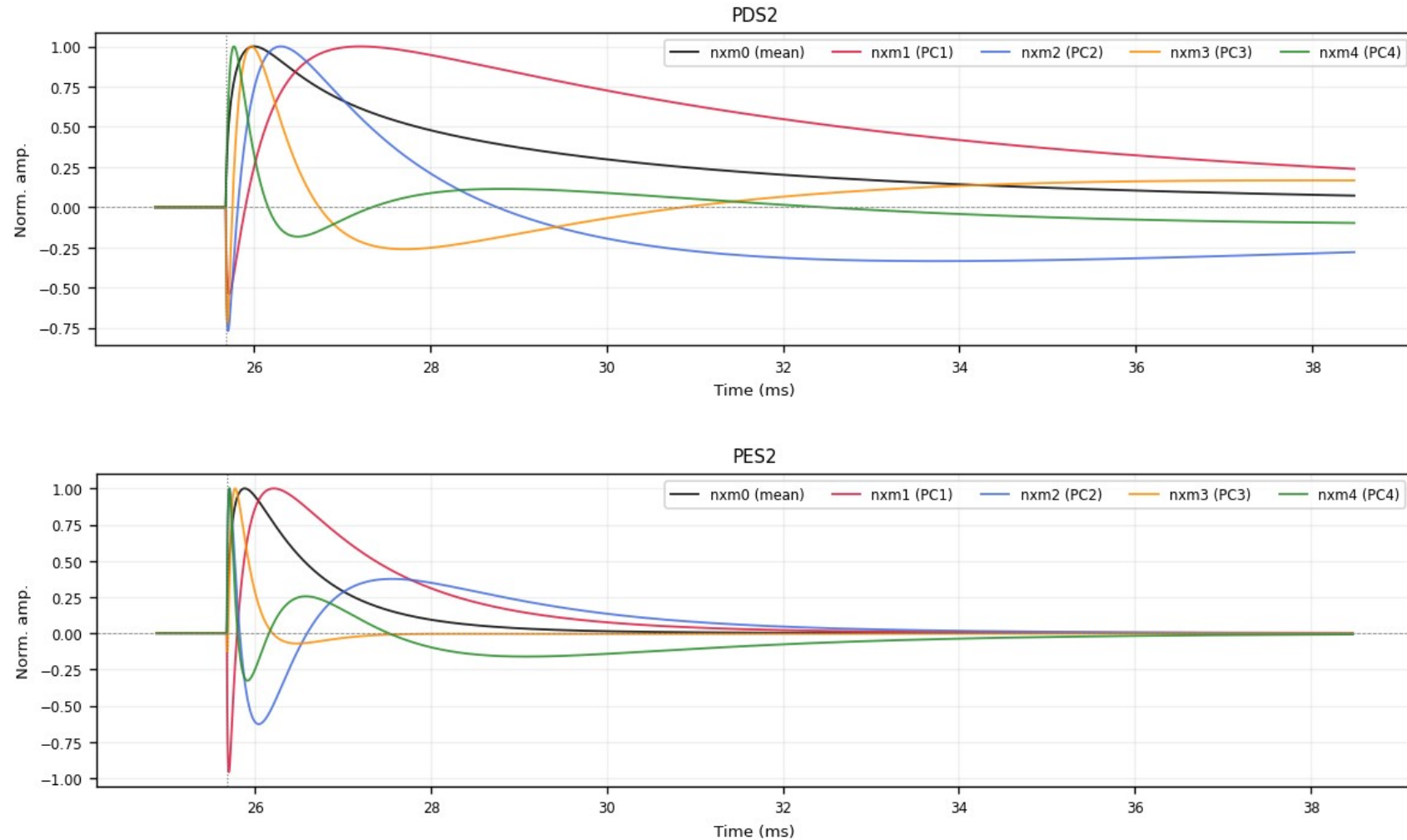
nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized; population = $\text{fit_ok} + \text{NRMSE} \leq 0.4 + \tau_{\text{rise}} \leq 0.3 \text{ ms}$.



Z4 · deliverables/nxm: PCA templates nxm0–4 (5/6)

figure: zip4_pca_templates.png, channel panels 9–10 of 11, top to bottom

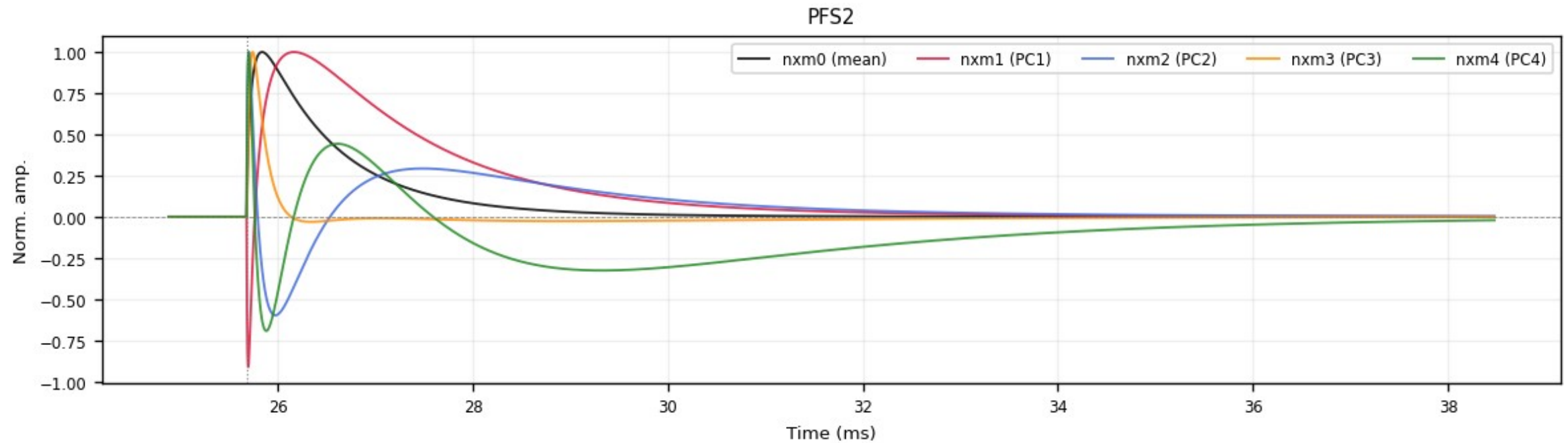
nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized; population = $\text{fit_ok} + \text{NRMSE} \leq 0.4 + \tau_{\text{rise}} \leq 0.3 \text{ ms}$.



Z4 · deliverables/nxm: PCA templates nxm0–4 (6/6)

figure: zip4_pca_templates.png, channel panels 11–11 of 11, top to bottom

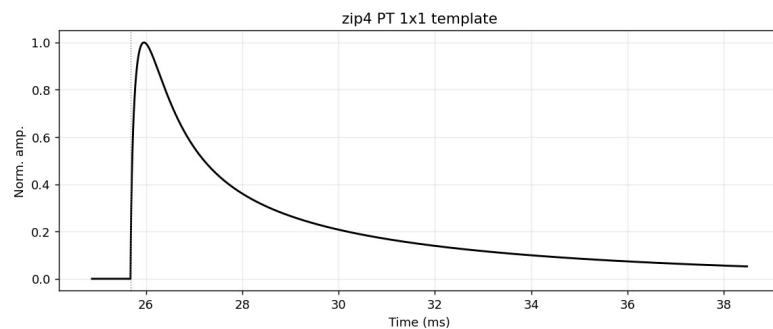
nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized; population = fit_ok + NRMSE $\leq$ 0.4 + $\tau_{\text{rise}} \leq$ 0.3 ms.



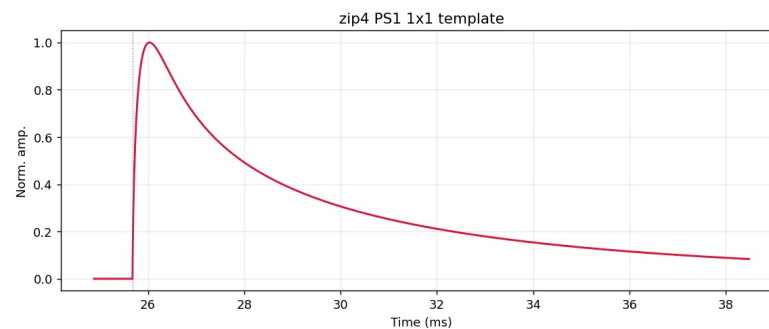
Z4 · 1x1 summed templates PT / PS1 / PS2

figures: deliverables/1x1/plots/{PT,PS1,PS2}/zip4_*.png

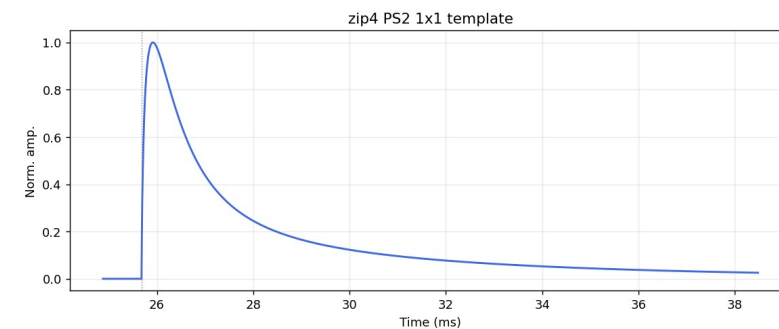
Single-template (1x1) summed curves: peak-normalized average of the per-channel nxm0 templates: PT = all channels, PS1 / PS2 = side-1 / side-2 only.



Z4 PT



Z4 PS1



Z4 PS2